

BINCOM ICT — CYBER SECURITY INTERNSHIP

ASSESSMENT 3: ATTACK AND DEFEND WEB APPLICATION

Complete Technical Assessment, Security Hardening, WAF Defense Validation & Attack Simulation

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16 September 2026

Academic / Internship Assessment Evidence

Document Control

Field	Value
Assignment	BINCOM Assessment 3 — Attack and Defend Web Application
Objective	Perform and block web-based attacks and assess a deployed PrestaShop application.
Assessment date	16 September 2026
Primary platforms	Kali Linux, DVWA, Apache/ModSecurity/OWASP CRS, AWS EC2, PrestaShop
Target application	PrestaShop 9.1.5
Public test endpoint	http://13.61.143.227

1. Executive Summary

This report documents the execution of BINCOM Assessment 3, which required controlled web-application attacks, defensive WAF configuration and validation, and a security assessment of a deployed PrestaShop instance. The work was performed within an authorized educational laboratory using Kali Linux, DVWA, Apache/ModSecurity with OWASP CRS, and an AWS-hosted PrestaShop 9.1.5 deployment. The assessment demonstrates both attack execution and defensive validation: SQL Injection and reflected XSS were exercised against DVWA; ModSecurity rules were configured and validated; a public PrestaShop instance was hardened against the checklist supplied in the assignment; and a controlled SQL injection probe against the deployed PrestaShop instance was blocked with HTTP 403 by OWASP CRS rule 942100.

2. Assignment Requirements

- Execute a SQL Injection and XSS attack against DVWA.
- Configure ModSecurity WAF controls to block the attacks and re-test them.
- Deploy/use PrestaShop and complete an application-specific security checklist.
- Simulate an attack against the deployed PrestaShop instance and document the vector, evidence, logs and mitigation.
- Produce screenshots, WAF configuration evidence, defense-validation findings and a structured final report.

3. Scope, Authorization and Safety

Testing was restricted to the authorized educational systems used for the BINCOM internship. The PrestaShop attack simulation was intentionally non-destructive. No unauthorized third-party system was targeted, and no database extraction, modification, deletion, credential theft or persistence was attempted.

4. Assessment Environment

Component	Value
Attacker / test workstation	Kali Linux
DVWA target	192.168.56.103
PrestaShop public IPv4	13.61.143.227
PrestaShop private IPv4	172.31.31.174
AWS region	eu-north-1 — Europe (Stockholm)
EC2 instance	PrestaShop-Web-Server — t3.micro
Operating system	Ubuntu 26.04 LTS
Web server	Apache HTTP Server 2.4.66
PHP	8.5.4
Application	PrestaShop 9.1.5
Theme	Hummingbird 2.0.0
Database	MariaDB 11.8.6 on a separate database server

WAF	ModSecurity 2.9.12 + OWASP CRS 3.3.8
Transport	HTTP on port 80; HTTPS/TLS not completed in this iteration

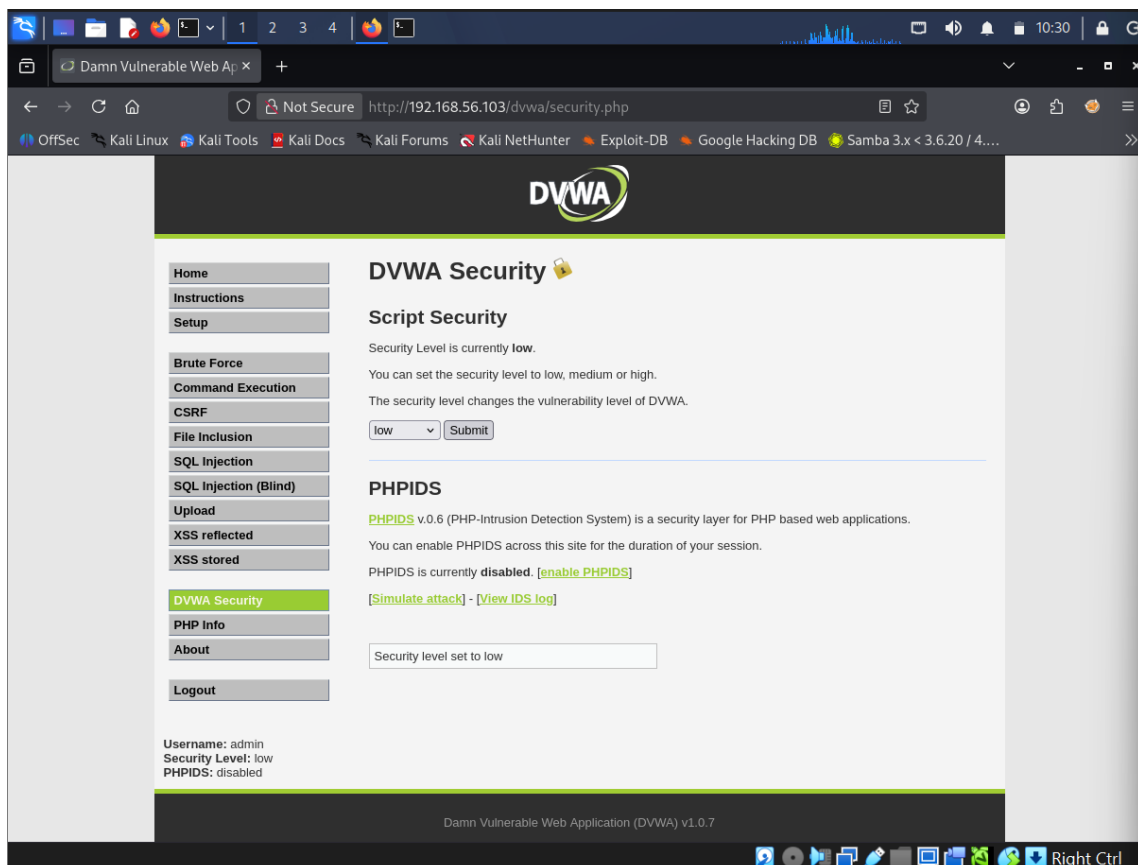
5. Methodology

1. Establish the DVWA attack baseline and capture evidence of controlled SQL Injection and XSS behavior.
2. Install/enable ModSecurity and OWASP CRS, create custom rules where required, validate Apache syntax and service health.
3. Re-test DVWA attacks to confirm defensive behavior.
4. Review the AWS/PrestaShop architecture, network controls, host firewall, application configuration and public attack surface.
5. Complete the ten-item PrestaShop security checklist using direct configuration, HTTP, database and log evidence.
6. Validate backup recovery in a temporary database without overwriting the live database.
7. Perform a controlled SQL Injection simulation against the deployed PrestaShop and correlate Kali response with ModSecurity audit evidence.

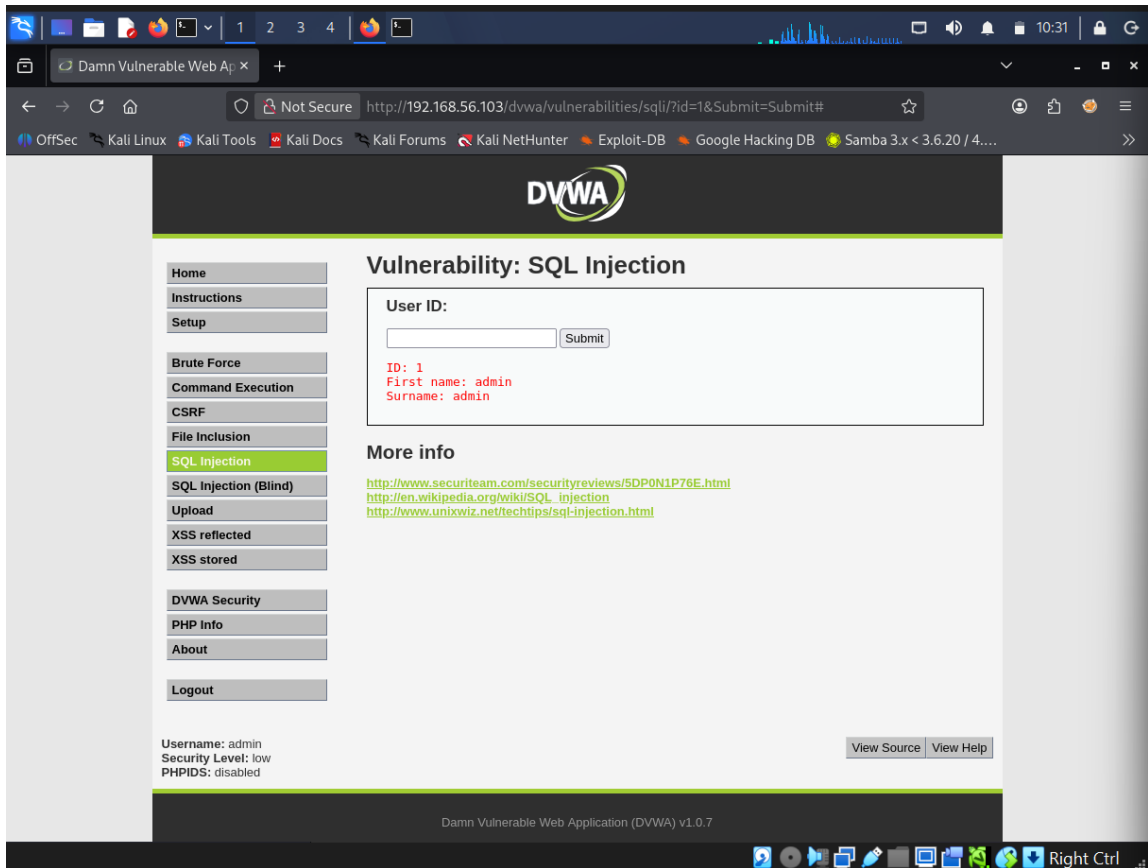
6. Part A — DVWA Attack and Defense

The DVWA portion followed the assignment's required attack/defense sequence. A baseline SQL Injection test and reflected XSS test were executed against DVWA before/while the WAF controls were being established. The captured evidence shows the SQLi and XSS test pages, controlled payload tests, the WAF rule file, and Apache/ModSecurity operation. The re-test phase was treated as successful when the configured WAF controls prevented the malicious test traffic from completing as an application attack.

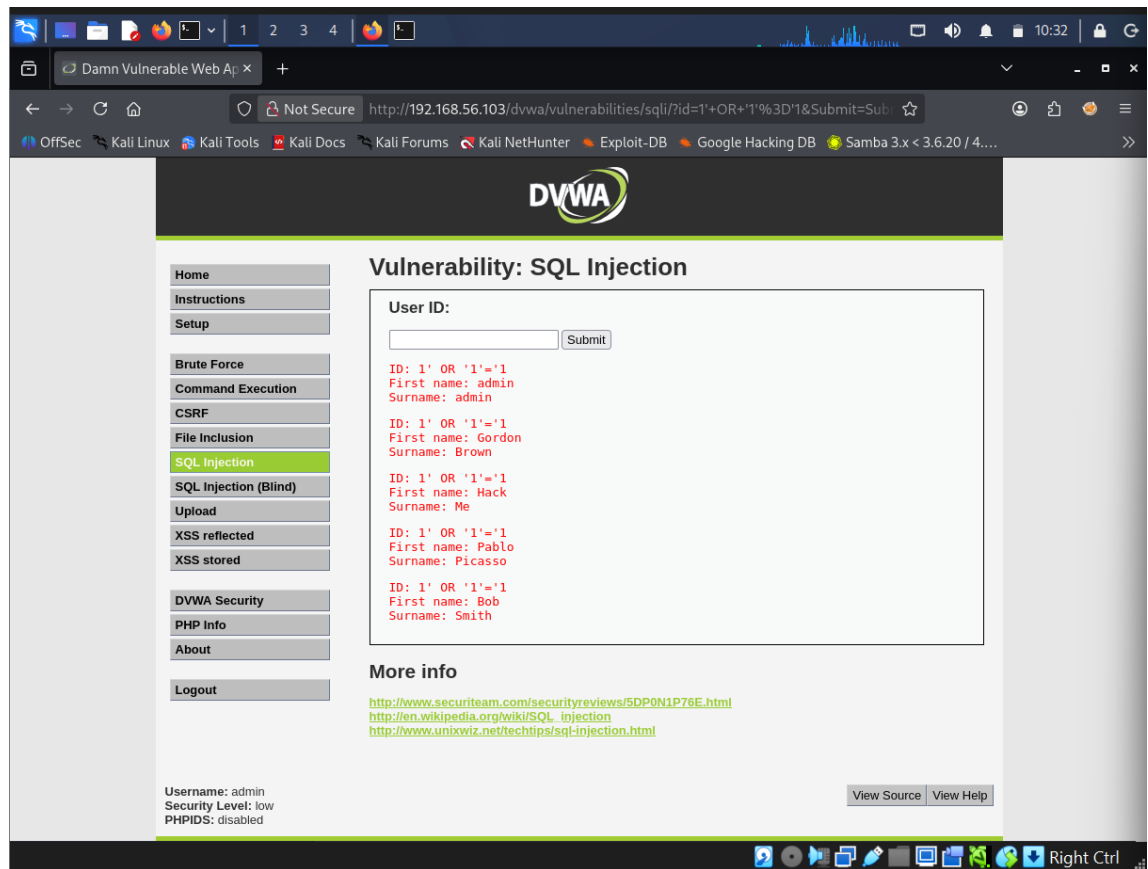
6.1 DVWA Evidence



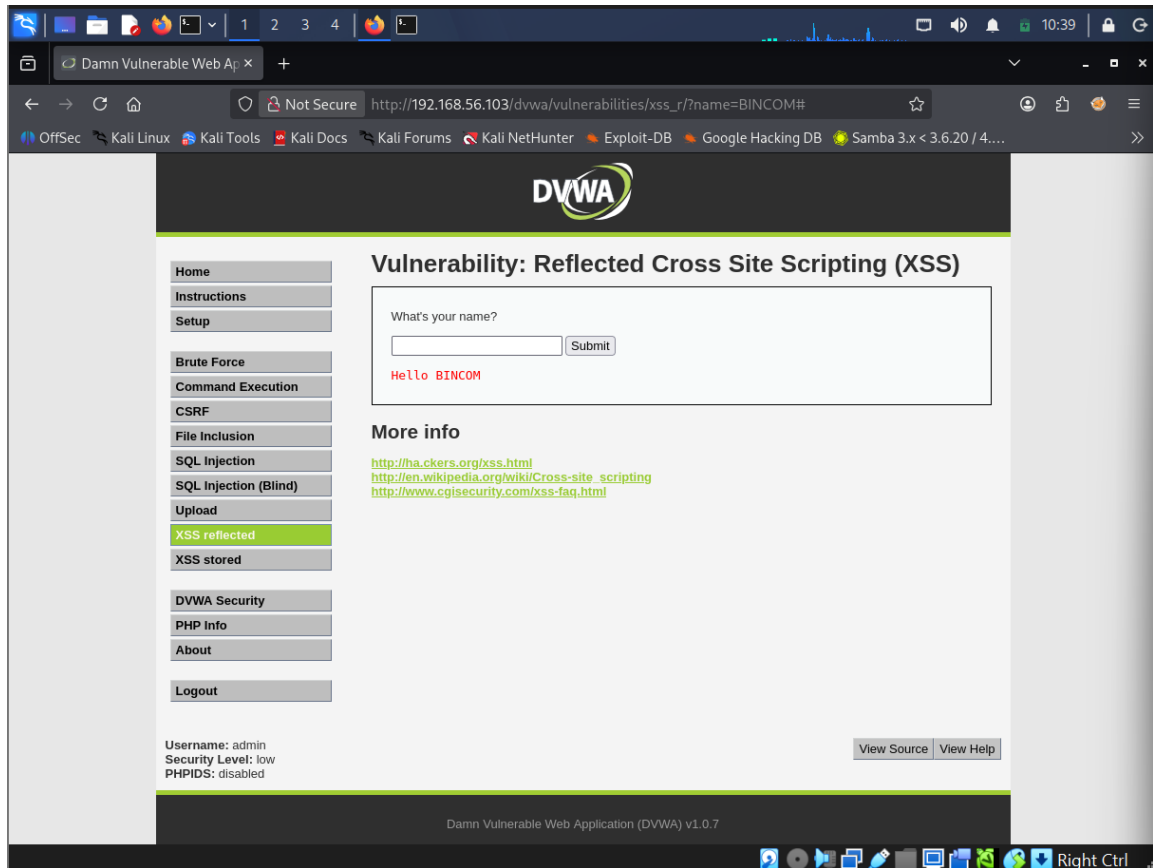
Evidence: DVWA SECURITY SETTING.png



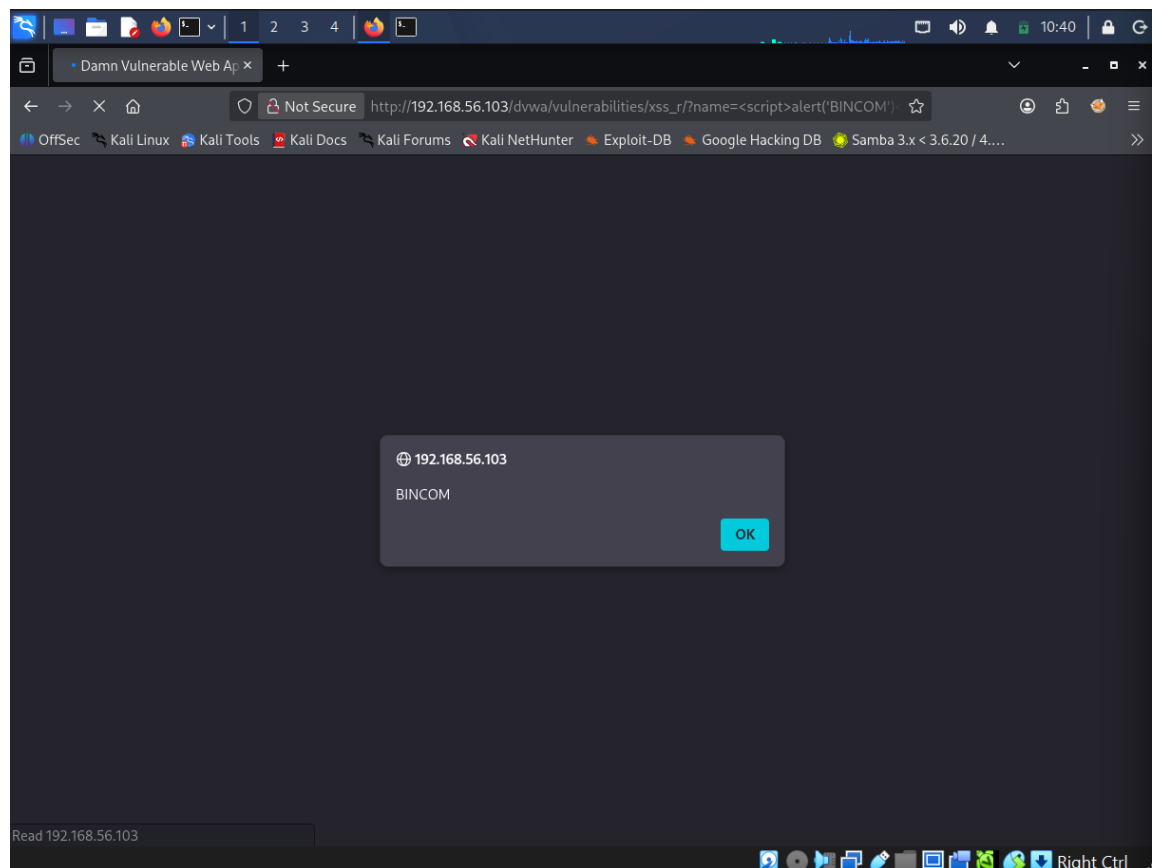
Evidence: DVWA SQL INJECTION NORMAL TEST.png



Evidence: DVWA SQLI PAYLOAD TEST.png



Evidence: NORMAL XSS TEST ON DVWA.png



Evidence: Controlled XSS TEST ON DVWA.png

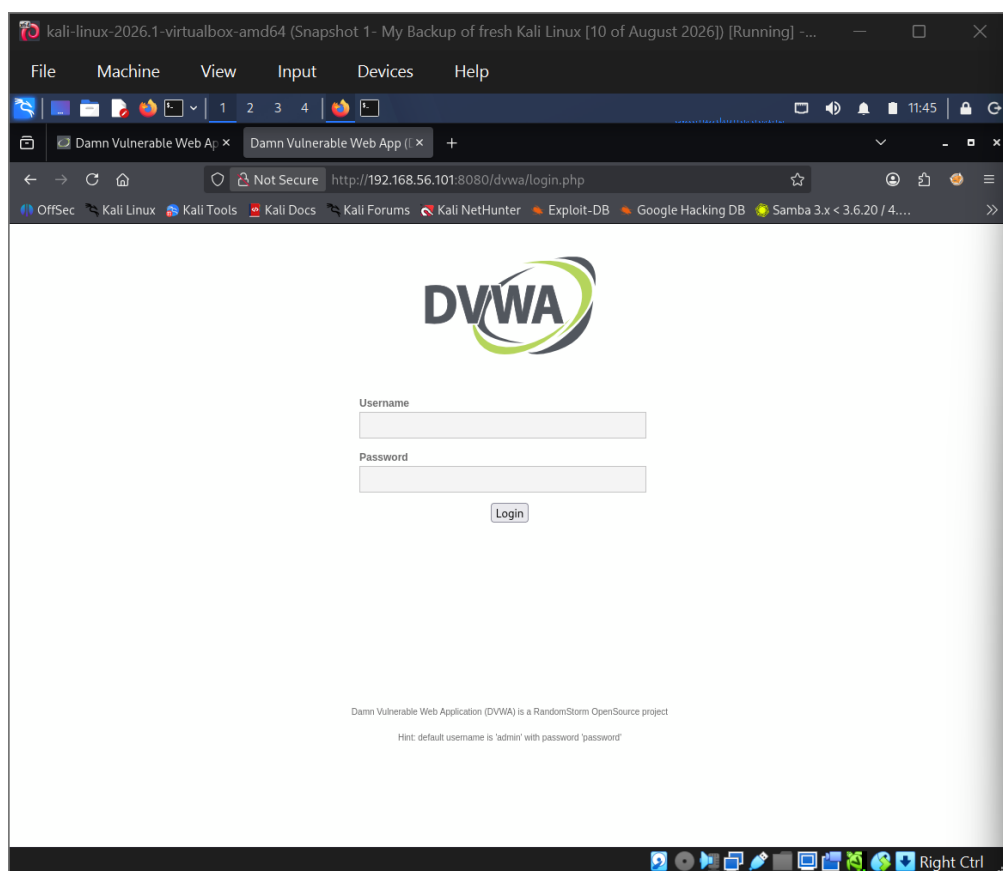
```
(kali@kali)-[~]
└─$ sudo nano /etc/modsecurity/bincom-assignment3.conf
(kali@kali)-[~]
└─$ sudo cat /etc/modsecurity/bincom-assignment3.conf
# BINCOM Assignment 3 - Custom ModSecurity Rules
# Purpose: Block SQL Injection and Cross-Site Scripting test payloads
# Environment: Authorized isolated DVWA lab

#
# SQL Injection protection
#
SecRule ARGS "@rx (?:'|"%27)s*(?:or|and)s*(?:'|"%27)?\w*(?:'|"%27)?\s*=\s*(?:'|"%27)?\w*(?:'|"%27)?" \
  "id:1000001,phase:2,deny,status:403,log,msg:'BINCOM A3 - SQL Injection attempt detected',severity:CRITICAL"

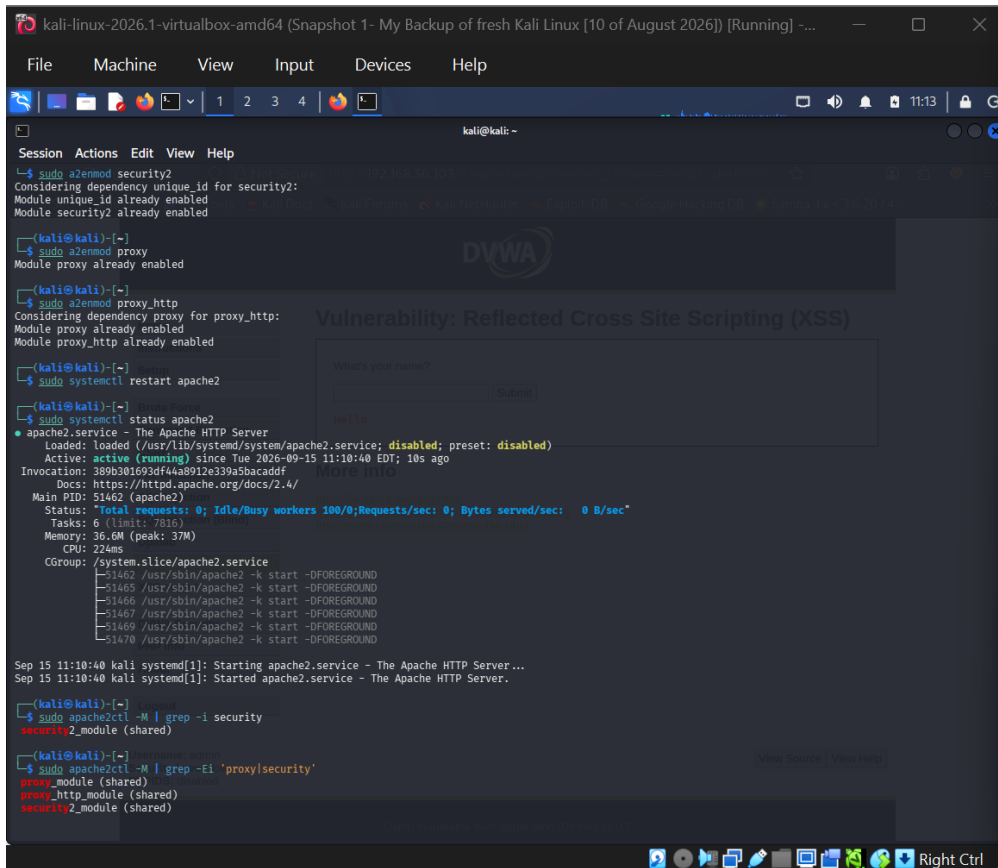
#
# Cross-Site Scripting protection
#
SecRule ARGS "@rx (?:i)<s*script\b[\">]*>.*?<s*/s*script\s*>" \
  "id:1000002,phase:2,deny,status:403,log,msg:'BINCOM A3 - XSS attempt detected',severity:CRITICAL"

(kali@kali)-[~]
└─$ sudo ls -lh /etc/modsecurity/bincom-assignment3.conf
-rw-r--r-- 1 root root 823 Sep 15 12:10 /etc/modsecurity/bincom-assignment3.conf
(kali@kali)-[~]
└─$
```

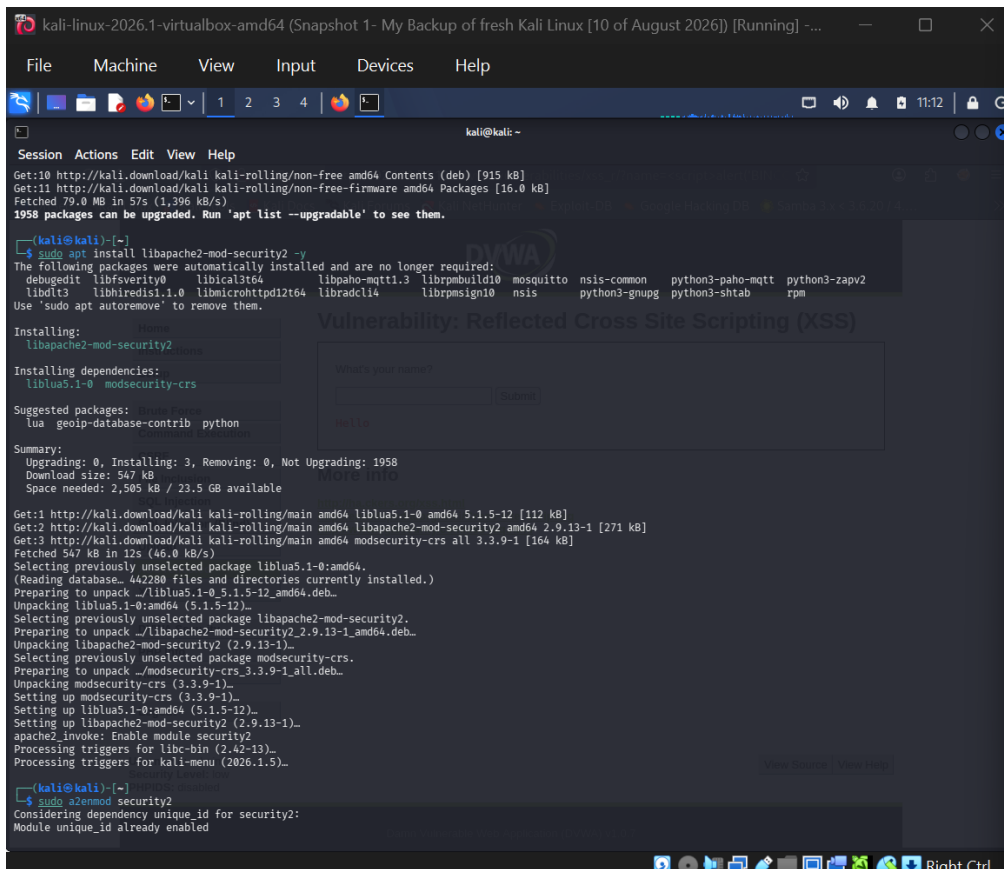
Evidence: Configured WAF RULE file.png



Evidence: DVWA through KALI APACHE MODSECURITY.png



Evidence: MODSECURITY Apache PROXY & MODULES ENABLED (1).png



Evidence: MODSECURITY Apache PROXY & MODULES ENABLED (2).png

```
(kali@kali)-[~]
└─$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 11:10:40 EDT; 1h 21min ago
  Invocation: 389b301693df44a8912e339a5bacaddf
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 83489 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
    Main PID: 51462 (apache2)
      Status: "Total requests: 13; Idle/Busy workers 100/0;Requests/sec: 0.00266; Bytes served/sec: 7 B/sec"
        Tasks: 6 (limit: 7816)
       Memory: 42.1M (peak: 78.1M)
          CPU: 3.609s
    CGroup: /system.slice/apache2.service
            └─51462 /usr/sbin/apache2 -k start -DFOREGROUND
              └─83495 /usr/sbin/apache2 -k start -DFOREGROUND
                └─83496 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─83497 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─83498 /usr/sbin/apache2 -k start -DFOREGROUND
                      └─83499 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 15 11:10:40 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Sep 15 11:10:40 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 15 11:28:37 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 11:28:37 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 11:40:13 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 11:40:14 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 12:07:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 12:07:28 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 12:14:06 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 12:14:06 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.

(kali@kali)-[~]
└─$ sudo apache2ctl -M | grep -i security
security2_module (shared)

(kali@kali)-[~]
└─$ sudo ss -ltnp | grep ':8080'
LISTEN 0      511          *:*          users:((("apache2",pid=83499,fd=8),("apache2",pid=83498,fd=8),("apache2",pid=83497,fd=8),("apache2",pid=83496,fd=8),("apache2",pid=83495,fd=8),("apache2",pid=51462,fd=8)))

(kali@kali)-[~]
└─$
```

Evidence: WAF Configuration Evidence.png

```
(kali@kali)-[~]
└─$ curl -I http://192.168.56.101:8080/dvwa/
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 15:42:25 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Pragma: no-cache
Location: login.php
Content-Type: text/html
Set-Cookie: PHPSESSID=baae3cdfca90cd032d3002d48e361fef; path=/
Set-Cookie: security=high

(kali@kali)-[~]
└─$ curl -s http://192.168.56.101:8080/dvwa/ | head -20
PHPIDS: disabled

(kali@kali)-[~]
└─$
```

Evidence: Traffic passes through WAF.png

```
(kali@kali)-[~]
└─$ curl -I http://192.168.56.101:8080/dvwa/
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 15:42:25 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Pragma: no-cache
Location: login.php
Content-Type: text/html
Set-Cookie: PHPSESSID=baae3cdfca90cd032d3002d48e361fef; path=/
Set-Cookie: security=high

(kali@kali)-[~]
└─$ curl -s http://192.168.56.101:8080/dvwa/ | head -20
PHPIDS: disabled

(kali@kali)-[~]
└─$
```

Evidence: Traffic passes through WAF(1).png

6.2 WAF Configuration

ModSecurity was installed with the Ubuntu package and OWASP ModSecurity CRS. The engine was changed from DetectionOnly to On. Apache configuration was repeatedly validated with `apache2ctl -t`, and the enabled `security2` module and CRS include chain were verified.

```
# BINCOM Assessment 3 - PrestaShop admin path protection
SecRule REQUEST_URI "@rx ^/admin113pn8yfwfb8jhihird" "id:1001001,phase:1,deny,status:403,log,msg:'BINCOM:"
```

```
PrestaShop admin path access blocked'"

# Effective ModSecurity configuration
SecRuleEngine On
SecAuditEngine RelevantOnly
SecAuditLogRelevantStatus "^(?:5|4(?:?!04))"
SecAuditLog /var/log/apache2/modsec_audit.log

# Apache module configuration
IncludeOptional /etc/modsecurity/*.conf
IncludeOptional /usr/share/modsecurity-crs/*.load
```

7. Part B — PrestaShop AWS Deployment and Hardening

The PrestaShop deployment was hosted on an AWS t3.micro web server with a separate database tier. The deployment report records layered AWS Security Groups, UFW host firewalling, Apache/PHP resource tuning, MariaDB isolation, production mode, disabled debug output and cache troubleshooting. The application was publicly reachable at the EC2 public IPv4 address.

7.1 Architecture and Baseline Controls

- Application server: Ubuntu 26.04 LTS / Apache / PHP 8.5.4 / PrestaShop 9.1.5.
- Database hosted separately from the application server as required by the assignment.
- MariaDB 11.8.6 bound to 127.0.0.1:3306 on the database host, with a dedicated database security group.
- UFW configured with default deny incoming and allow outgoing; required web/administration ports explicitly allowed.
- Apache prefork worker settings and web PHP memory were constrained for the t3.micro resource ceiling; a 1 GiB swap file was added.
- PrestaShop verified in production mode with debug disabled.
- Apache security headers added for content type sniffing, framing, referrer policy, CSP frame ancestors and permissions policy.

7.2 Application Cleanup and Configuration Review

- PrestaShop 9.1.5 verified from application metadata.
- Default/demo CMS pages disabled.
- Demo-content modules ps_imageslider 3.2.2, ps_customtext 4.2.2 and ps_banner 2.1.2 disabled through the Back Office.
- One active administrator; password policy verified at minimum length 8, maximum length 72 and minimum score 3.
- No world-writable entries identified in the permissions audit.
- Public storefront re-tested after cleanup.

8. PrestaShop Security Checklist

Checklist item	Status	Evidence / observation
1. Application/modules up to date	PASS / documented	PrestaShop 9.1.5 verified; module inventory reviewed; current deployment state documented.
2. Default/demo accounts & sample data	PASS	Default CMS/demo content was disabled; demo storefront modules were disabled; one active administrator remained.
3. Strong admin password & MFA	PARTIAL	Password policy verified: minimum 8, maximum 72, minimum score 3. MFA was not evidenced in the available Back Office.
4. Permissions & directory listing	PASS	Web-root permissions reviewed; no world-writable entries found; /modules and /themes redirected to the application directory.
5. Secure session cookies	PARTIAL	HttpOnly and SameSite=Lax observed. Secure was not present because the deployment remained HTTP-only.
6. Unnecessary modules/plugins	PASS for identified demo modules	ps_imageslider, ps_customtext and ps_banner were disabled after identifying demo/sample content.
7. Payment/API security	PARTIAL / evidence documented	Payment methods and API configuration were reviewed; Admin API routes were not exposed by unauthenticated GET tests.
8. TLS configured and enforced	NOT IMPLEMENTED	Port 443 was not active and HTTPS was not completed because no domain/certificate was available for the exercise.
9. Exposed admin path / WAF protection	PASS	Randomized admin path was identified and protected by custom ModSecurity rule 1001001; direct access returned 404.
10. Backup & recovery	PASS	978 KB SQL dump verified; 301 table definitions restored into a temporary database; key tables/data verified; temporary database removed.

9. TLS, Cookies, Payment and API Validation

HTTP response testing confirmed HttpOnly and SameSite=Lax on session cookies. Secure was not present because the deployment remained HTTP-only. An HTTPS request to the public IP on port 443 failed to connect, and Apache virtual-host listing showed only a port-80 web configuration. Payment methods were reviewed in the Back Office. Unauthenticated GET tests of /api/, /admin-api/, /admin-api/docs.json and /admin-api/api-client/info did not expose the API. Full payment/API assurance remains limited without HTTPS and provider-specific configuration validation.

10. Admin Path Protection and WAF Validation

The actual randomized administration path was /admin113pn8yfwfb8jihird. Custom ModSecurity rule 1001001 denies requests beginning with that path. After correcting a temporary configuration comment typo and restarting

Apache, a Kali request returned HTTP 403 Forbidden and the ModSecurity audit log recorded the denial.

```
shunt@kali:~$ sudo tail -n 80 /var/log/apache2/modsec_audit.log
...
-0442e01-F-
HTTP/1.1 403 Forbidden
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Length: 239
Content-Type: text/html; charset=iso-8859-1

-0442e01-E-
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<div>Forbidden</div>
<p>You don't have permission to access this resource.</p>
</body></html>

-0442e01-H-
Message: Access denied with code 403 (phase 3). Pattern match "A/admin1pgs9yvwfb8jshird" at REQUEST_URI. [file "/etc/modsecurity/prestashop-admin.conf" (line "2") [id "1001001"] [msg "BINLOCK: Prestashop admin path access blocked"]
Apache Error: [file "/etc/modsecurity/prestashop-admin.conf" (line "2") [id "1001001"] [msg "BINLOCK: Prestashop admin path access blocked"] [hostname "13.61.143.227"] [uri "/admin1pgs9yvwfb8jshird"] [unique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Action: Intercepted (phase 3)
Stopwatch: 178954575188834 538 (- - -)
Stopwatch: 178954575188834 538; combined=212, p1=76, p2=0, p3=0, p4=0, p5=136, sr=0, sw=0, l=0, gc=0
Response-Body-Transformed: Dechunked
Producer: ModSecurity for Apache/2.9.12 (http://www.modsecurity.org/) ; OWASP_CRS/3.3.8
Server: Apache
Engine Mode: "ENABLED"

-0442e01-Z-

-0442e01-A-
[16/Sep/2024:14:00:49.17981+0000] aqq6m6q9m7Vyr9Gduh2wAaAAA 102.89.41.103 28091 172.31.31.174 80
-0442e01-B-
GET /16-1a27a2008a29a271a27a2a271 HTTP/1.1
Host: 13.61.143.227
User-Agent: curl/8.10.0
Accept: */*

-0442e01-F-
HTTP/1.1 403 Forbidden
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Length: 239
Content-Type: text/html; charset=iso-8859-1

-0442e01-E-
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<div>Forbidden</div>
<p>You don't have permission to access this resource.</p>
</body></html>

-0442e01-H-
Message: Warning. Pattern match "A[\\d.]+.*" at REQUEST_HEADERS:Host. [file "/usr/share/modsecurity-crs/rules/REQUEST-920-PROTOCOL-ENFORCEMENT.conf" (line "735") [id "920310"] [msg "Host header is a numeric IP address"] (data "13.61.143.227") [sev
erity "WARNING"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-protocol"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes/1000/210/272"] [tag "PCI/6.5.10"]
Message: Warning. detected SQLi using libinjection with fingerprint 'sasos' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf" (line "66") [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] (data "M
atched data: sasos found within AHRS:Id: 1' OR '1'='1") [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes
/1000/152/248/66"] [tag "PCI/6.5.2"]
Message: Access denied with code 403 (phase 3). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf" (line "94") [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Score: 8
)"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-generic"]
Message: Warning. Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/RESPONSE-980-CORRELATION.conf" (line "92") [id "980110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,
LFI=0,RFID=0,HPP=0,HPP=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf" (line "66") [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] (data "Matched data: sasos found within AHRS:Id: 1' OR '1'='1") [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack
-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes/1000/152/248/66"] [tag "PCI/6.5.2"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf" (line "94") [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Score: 8)"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-generic"] [hostname "13.61.143.227"] [uri "/" ] [un
ique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/RESPONSE-980-CORRELATION.conf" (line "92") [id "980110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,LFI=0,RFID=0,HPP=0,HPP=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq
6m6q9m7Vyr9Gduh2wAaAAA"]
Action: Intercepted (phase 2)
Stopwatch: 178954724914175 7845 (- - -)
Stopwatch: 178954724914175 7847; combined=2372, p1=698, p2=1513, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body-Transformed: Dechunked
Producer: ModSecurity for Apache/2.9.12 (http://www.modsecurity.org/) ; OWASP_CRS/3.3.8
Server: Apache
Engine Mode: "ENABLED"

-0442e01-Z-

shunt@kali:~$ sudo grep -E "SQL[042]injection[id=1]" /var/log/apache2/modsec_audit.log | tail -n 30
GET /16-1a27a2008a29a271a27a2a271 HTTP/1.1
Message: Warning. detected SQLi using libinjection with fingerprint 'sasos' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf" (line "66") [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] (data "M
atched data: sasos found within AHRS:Id: 1' OR '1'='1") [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes
/1000/152/248/66"] [tag "PCI/6.5.2"]
Message: Warning. Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/RESPONSE-980-CORRELATION.conf" (line "92") [id "980110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,
LFI=0,RFID=0,HPP=0,HPP=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf" (line "66") [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] (data "Matched data: sasos found within AHRS:Id: 1' OR '1'='1") [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack
-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes/1000/152/248/66"] [tag "PCI/6.5.2"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/RESPONSE-980-CORRELATION.conf" (line "92") [id "980110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,LFI=0,RFID=0,HPP=0,HPP=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq
6m6q9m7Vyr9Gduh2wAaAAA"]
shunt@kali:~$
```

Kali — protected admin path returned HTTP 403.

```
...
shunt@kali:~$ sudo grep -E "SQL[042]injection[id=1]" /var/log/apache2/modsec_audit.log | tail -n 30
GET /16-1a27a2008a29a271a27a2a271 HTTP/1.1
Message: Warning. detected SQLi using libinjection with fingerprint 'sasos' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf" (line "66") [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] (data "M
atched data: sasos found within AHRS:Id: 1' OR '1'='1") [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes
/1000/152/248/66"] [tag "PCI/6.5.2"]
Message: Warning. Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/RESPONSE-980-CORRELATION.conf" (line "92") [id "980110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,
LFI=0,RFID=0,HPP=0,HPP=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf" (line "66") [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] (data "Matched data: sasos found within AHRS:Id: 1' OR '1'='1") [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack
-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "capes/1000/152/248/66"] [tag "PCI/6.5.2"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq6m6q9m7Vyr9Gduh2wAaAAA"]
Apache Error: [file "/usr/share/modsecurity-crs/rules/RESPONSE-980-CORRELATION.conf" (line "92") [id "980110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,LFI=0,RFID=0,HPP=0,HPP=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/" ] [unique_id "aqq
6m6q9m7Vyr9Gduh2wAaAAA"]
shunt@kali:~$
```

Server — ModSecurity audit/debug evidence.

```
(kali@kali)~$ curl -i http://13.61.143.227/?id=1%27%200R%20%271%27%3D%271
HTTP/1.1 403 Forbidden
Date: Wed, 16 Sep 2026 14:00:49 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Length: 239
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<div>Forbidden</div>
<p>You don't have permission to access this resource.</p>
</body></html>

(kali@kali)~$
```

Server — supporting WAF/log evidence.

11. Backup and Recovery Validation

A pre-cleanup SQL backup of approximately 978 KB was verified and contained 301 unique CREATE TABLE definitions. The dump had no top-level CREATE DATABASE, DROP DATABASE or USE statements, so it was safely imported into a temporary prestashop_recovery_test database. The recovery database contained 301 BASE TABLE objects; ps_configuration contained 492 rows, ps_employee contained 1 row and ps_shop contained 1 row. The temporary database was dropped and the live prestashop database remained present.

12. PrestaShop Attack Simulation

A non-destructive SQL injection probe was sent from Kali to the deployed PrestaShop public endpoint. The test request used the classic boolean SQLi pattern in a query parameter. The application returned HTTP 403 Forbidden. The ModSecurity audit log identified OWASP CRS rule 942100 in REQUEST-942-APPLICATION-ATTACK-SQLi.conf, detected SQL injection using libinjection, recorded CRITICAL severity/tags including attack-sqli, and showed that the inbound anomaly score exceeded the blocking threshold. No database extraction, modification or destructive action was attempted.

12.1 Attack Evidence

```
abuntu@ip:172.31.31.174-# sudo tail -n 80 /var/log/apache2/modsec_audit.log
Accept: */*

...-d64d2e01-F-
HTTP/1.1 403 Forbidden
Content-Type: Options: none
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: Frame-Ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Length: 229
Content-Type: text/html; charset=iso-8859-1

...-d64d2e01-E-
...DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<div>Forbidden</div>
<p>You don't have permission to access this resource.</p>
</body></html>

...-d64d2e01-H-
Message: Access denied with code 403 (phase 1). Pattern match "/^adminlllpwyfwb8ihird" at REQUEST_URI. [file "/etc/modsecurity/prestashop-admin.conf"] [line "2"] [id "1001001"] [msg "BINCOM: Prestashop admin path access blocked"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 1). Pattern match "/^adminlllpwyfwb8ihird" at REQUEST_URI. [file "/etc/modsecurity/prestashop-admin.conf"] [line "2"] [id "1001001"] [msg "BINCOM: Prestashop admin path access blocked"] [hostname "13.61.143.227"] [uri "/adminlllpwyfwb8ihird/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Action: Intercepted (phase 1)
Stopwatch: 1789543718884 538 (- - -)
Stopwatch2: 1789543718884 538; combined=212, p1=76, p2=0, p3=0, p4=0, p5=136, sr=0, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-

...-d64d2e01-A-
[16/Sep/2026:14:09:49.117891 +0000] aqg6kqpmv9yrgdihuzwAAAA 102.86.41.193 28051 172.31.31.174 80
...-d64d2e01-B-
GET /16-1427A20082A9271A27A2D271 HTTP/1.1
Host: 13.61.143.227
User-Agent: curl/7.19.0
Accept: */*

...-d64d2e01-F-
HTTP/1.1 403 Forbidden
Content-Type: Options: none
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: Frame-Ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Length: 229
Content-Type: text/html; charset=iso-8859-1

...-d64d2e01-E-
...DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<div>Forbidden</div>
<p>You don't have permission to access this resource.</p>
</body></html>

...-d64d2e01-H-
Message: Warning. Pattern match "A[\\d.]+@" at REQUEST_HEADERS:Host. [file "/usr/share/modsecurity-crs/rules/REQUEST-920-PROTOCOL-ENFORCEMENT.conf"] [line "735"] [id "920350"] [msg "Host header is a numeric IP address"] [data "13.61.143.227"] [severity "WARNING"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-protocol"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Message: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"] [tag "PCI/6.5.10"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Pattern match "A[\\d.]+@" at REQUEST_HEADERS:Host. [file "/usr/share/modsecurity-crs/rules/REQUEST-920-PROTOCOL-ENFORCEMENT.conf"] [line "735"] [id "920350"] [msg "Host header is a numeric IP address"] [data "13.61.143.227"] [severity "WARNING"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-protocol"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Actions: Intercepted (phase 2)
Stopwatch: 17895724911475 1847 (- - -)
Stopwatch2: 17895724911475 1847; combined=2372, p1=698, p2=1531, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-

abuntu@ip:172.31.31.174-# sudo grep -Rl "942[injection]id=1" /var/log/apache2/modsec_audit.log | tail -n 80
GET /16-1427A20082A9271A27A2D271 HTTP/1.1
Message: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"] [tag "PCI/6.5.10"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Actions: Intercepted (phase 2)
Stopwatch: 17895724911475 1847 (- - -)
Stopwatch2: 17895724911475 1847; combined=2372, p1=698, p2=1531, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-

abuntu@ip:172.31.31.174-# sudo grep -Rl "942[injection]id=1" /var/log/apache2/modsec_audit.log | tail -n 80
GET /16-1427A20082A9271A27A2D271 HTTP/1.1
Message: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"] [tag "PCI/6.5.10"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Actions: Intercepted (phase 2)
Stopwatch: 17895724911475 1847 (- - -)
Stopwatch2: 17895724911475 1847; combined=2372, p1=698, p2=1531, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-
```

Kali — controlled SQL Injection probe received HTTP 403 Forbidden.

```
abuntu@ip:172.31.31.174-# sudo grep -Rl "942[injection]id=1" /var/log/apache2/modsec_audit.log | tail -n 80
GET /16-1427A20082A9271A27A2D271 HTTP/1.1
Message: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"] [tag "PCI/6.5.10"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Actions: Intercepted (phase 2)
Stopwatch: 17895724911475 1847 (- - -)
Stopwatch2: 17895724911475 1847; combined=2372, p1=698, p2=1531, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-

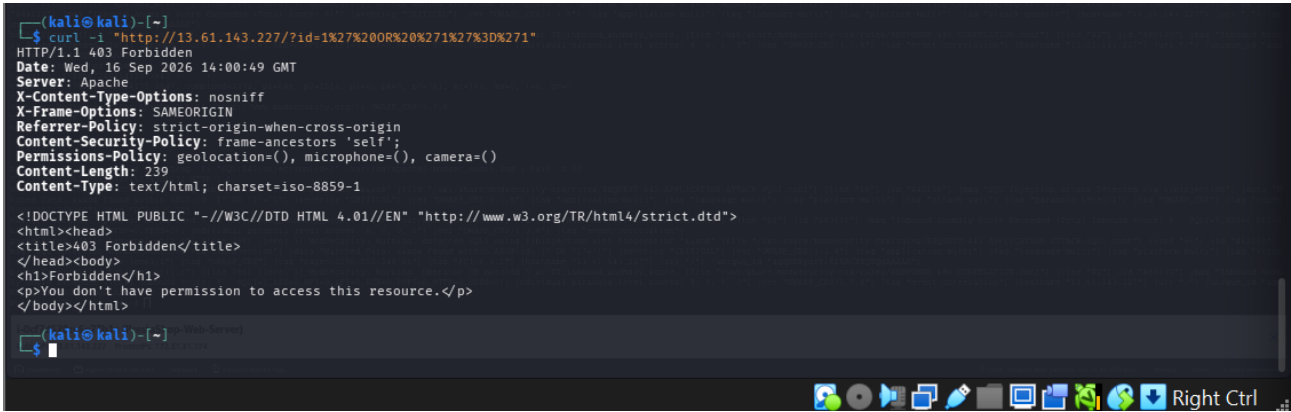
abuntu@ip:172.31.31.174-# sudo grep -Rl "942[injection]id=1" /var/log/apache2/modsec_audit.log | tail -n 80
GET /16-1427A20082A9271A27A2D271 HTTP/1.1
Message: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"] [tag "PCI/6.5.10"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Actions: Intercepted (phase 2)
Stopwatch: 17895724911475 1847 (- - -)
Stopwatch2: 17895724911475 1847; combined=2372, p1=698, p2=1531, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-

abuntu@ip:172.31.31.174-# sudo grep -Rl "942[injection]id=1" /var/log/apache2/modsec_audit.log | tail -n 80
GET /16-1427A20082A9271A27A2D271 HTTP/1.1
Message: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"] [tag "PCI/6.5.10"]
Message: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Warning. Detected SQLi using libinjection with fingerprint 'sasso' [file "/usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLi.conf"] [line "66"] [id "942100"] [msg "SQL Injection Attack Detected via libinjection"] [data "Matched Data: sasso found within ARGS:id: 1' OR '1'='1'"] [severity "CRITICAL"] [ver "OWASP_CRS/3.3.8"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-sqli"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "cpe:/a:apache:2.4.18"]
Apache Error: [file "apache2_util.c"] [line 284] [level 3] ModSecurity: Access denied with code 403 (phase 2). Operator GE matched 5 at TX:inbound_anomaly_score. [file "/usr/share/modsecurity-crs/rules/REQUEST-940-BLOCKING-EVALUATION.conf"] [line "94"] [id "940110"] [msg "Inbound Anomaly Score Exceeded (Total Inbound Score: 8 - SQLI=5,XSS=0,RFI=0,RFI=0,RC=0,RFI=0,HTTP=0,SESS=0); individual paranoia level scores: 8, 0, 0, 0"] [ver "OWASP_CRS/3.3.8"] [tag "event-correlation"] [hostname "13.61.143.227"] [uri "/"] [unique_id "aqg6kqpmv9yrgdihuzwAAAA"]
Actions: Intercepted (phase 2)
Stopwatch: 17895724911475 1847 (- - -)
Stopwatch2: 17895724911475 1847; combined=2372, p1=698, p2=1531, p3=0, p4=0, p5=161, sr=100, sw=0, l=0, gc=0
Response-Body Transformed: Dechunked
Producer: ModSecurity for Apache/2.5.12 (http://www.modsecurity.org/); OWASP_CRS/3.3.8.
Server: Apache
Engine-Mode: "ENABLED"

...-d64d2e01-E-
```

Server — ModSecurity audit evidence for the blocked request.



Server — supporting WAF/log evidence.

12.2 Technical Interpretation

- OWASP CRS rule 942100 from REQUEST-942-APPLICATION-ATTACK-SQLI.conf identified SQL injection using libinjection.
- The event was recorded at CRITICAL severity with attack-sqli / OWASP CRS tags.
- The inbound anomaly score exceeded the blocking threshold and the request was intercepted in phase 2.
- The web server returned HTTP 403 Forbidden.
- No successful database extraction or modification was demonstrated.

13. Troubleshooting and Resolution Record

Memory pressure on t3.micro: A 1 GiB swap file was created and made persistent; Apache prefork worker concurrency and web PHP memory were constrained. Subsequent verification reported no recent OOM events.

PrestaShop/Symfony cache permissions: Apache logs showed permission-denied/cache-write failures. Ownership and write access were checked, a direct www-data write test passed, and production cache was cleared successfully.

AWS EC2 Instance Connect failure: Connect Assist eventually showed network/IAM checks passing, while CloudTrail showed SendSSHPublicKey success. Direct SSH from Kali returned connection refused. The troubleshooting established that the issue was after AWS key delivery rather than an EC2 launch/network-state failure.

SSM Session Manager network failure: The initial SSM attempt displayed Network Failure; the work later continued through server-side terminal access. The investigation considered SSM Agent, IAM instance profile and endpoint connectivity.

Disk pressure: Unused /var/www/prestashop.unused content was removed, recovering approximately 527 MB and reducing root usage from about 86% to 79%. Further cleanup was deliberately stopped to avoid unnecessary changes.

ModSecurity debug configuration typo: A mistaken '-' '#' comment prefix caused Apache syntax failure. It was corrected to '#', apache2ctl -t returned Syntax OK, and Apache was restarted successfully.

14. Findings and Mitigations

ID	Finding	Status / risk	Impact	Mitigation
F-01	HTTP-only deployment / missing TLS	High	HTTP is used for the public application; session data and the certificate, both HTTP and TLS, are sent over the network.	Deployed domain and the certificate, both HTTP and TLS, are sent over the network.
F-02	MFA not evidenced for administrator	Medium	The reviewed employee security interface did not have MFA enabled for administrator.	Established MFA for administrator and supported by the user factor.
F-03	Publicly reachable randomized administrator endpoint	Medium	The randomized path reduces casual discovery of the endpoint.	Randomized path is not a security issue as it is not a public endpoint.
F-04	Demo/sample content initially exposed	Medium	Initial storefront/CMS content contained default demo content.	Disable demo content and modules before exposure; this was done.
F-05	Controlled SQLi probe detected by WAF	Low	OWASP CRS rule 942100 identified SQL injection attempt.	Keep ModSecurity CRS enabled; request was denied with HTTP 403.
F-06	Automated internet scanning noise	Low / contextual	Apache logs showed requests for WordPress.	Consistent logging of WAF and other ongoing scanning against the public IP. These were blocked.

15. Limitations

- HTTPS/TLS was not completed because the exercise did not use a purchased domain/certificate; the public application remained HTTP-only.
- MFA was not evidenced in the reviewed PrestaShop Back Office employee configuration.
- Payment-provider and production API configuration could not be fully assured from unauthenticated HTTP tests alone.

- The deployment is an educational/lab environment, not a fully hardened enterprise e-commerce platform.
- The public EC2 IP received automated internet scanning traffic; this was treated as background exposure and not proof of compromise.

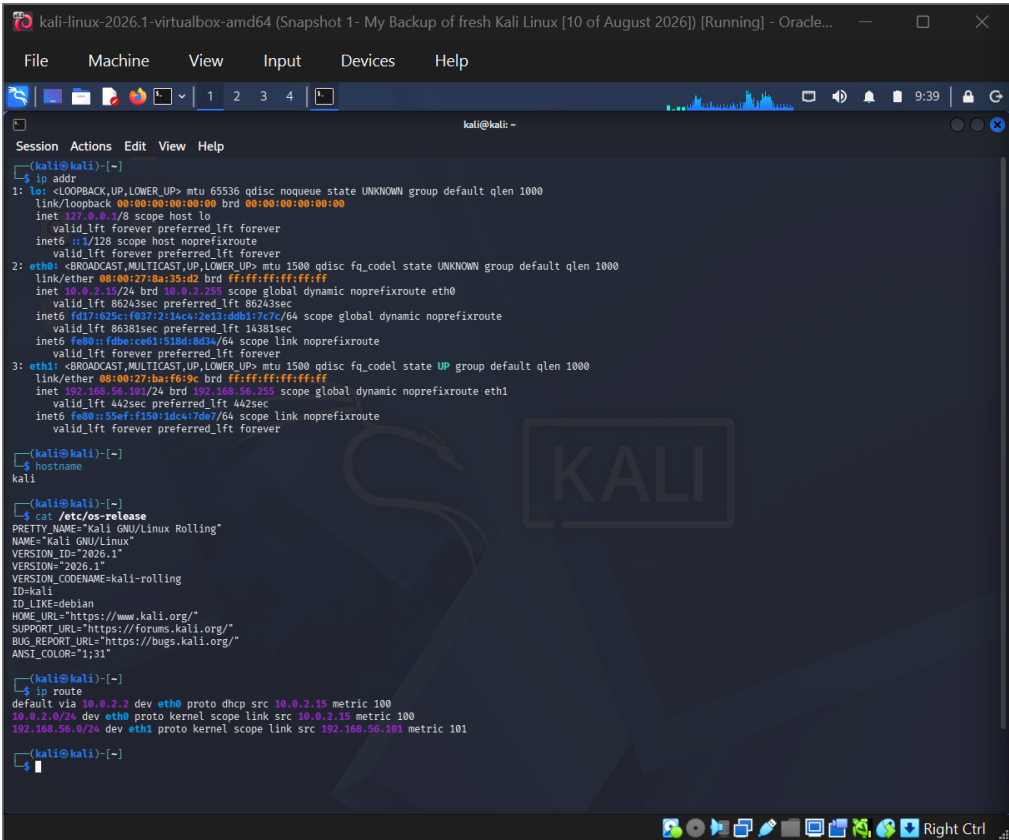
16. Conclusion

The assessment met the practical objectives of the BINCOM Assessment 3 brief. The work demonstrates a complete attack-and-defense workflow: attack execution against DVWA, WAF deployment and re-testing, application-specific PrestaShop hardening, controlled attack simulation, logging and recovery validation. The strongest evidence is the successful defensive chain in which a SQL injection probe against the public PrestaShop deployment was identified by OWASP CRS rule 942100 and blocked with HTTP 403, while the custom rule 1001001 separately protected the randomized administration path. The final posture remains a laboratory/educational deployment rather than a production-ready enterprise e-commerce platform. The principal outstanding limitations are HTTPS/TLS, the lack of evidenced MFA, and incomplete assurance around production payment/API configuration. These are explicitly recorded rather than being presented as completed controls.

17. Evidence Register

All PNG screenshots available in the project workspace at report-generation time are retained below. Named evidence is presented first; timestamped screenshots form the complete archive.

17.1 Key Evidence Screenshots



```

kali-linux-2026.1-virtualbox-amd64 (Snapshot 1- My Backup of fresh Kali Linux [10 of August 2026]) [Running] - Oracle...
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

kali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86243sec preferred_lft 86243sec
    inet6 fd17:625c:f037:214c:a2e13:dd01:7c7c/64 scope global dynamic noprefixroute
        valid_lft 86381sec preferred_lft 14381sec
    inet6 fe80::f081:ce61:518d:8d34/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ba:f6:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 442sec preferred_lft 442sec
    inet6 fe80::55ef:f150:1dca:7d07/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali:~$ hostname
kali

kali@kali:~$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.1"
VERSION="2026.1"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

kali@kali:~$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
192.168.56.0/24 dev eth1 proto kernel scope link src 192.168.56.101 metric 101

kali@kali:~$

```

Key evidence: Kali Lab Reconnaissance.png

```
(kali@kali)-[~]
$ curl -i http://13.61.143.227/ | head -30
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           0         0     0    0          0      0      0     0
HTTP/1.0 302 Moved
Date: Tue, 15 Sep 2026 17:02:38 GMT
Server: Apache/2.4.66 (Ubuntu)
Cache-Control: no-cache
Location: http://13.61.143.227/index.php
Content-Length: 0
Connection: close
Content-Type: text/html; charset=utf-8

(kali@kali)-[~]
$ curl -s http://13.61.143.227/ | grep -Ei 'prestashop|shop|apache|title'

(kali@kali)-[~]
$ sudo nmap -Pn -sV -p 80 13.61.143.227
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-15 13:02 -0400
Nmap scan report for ec2-13-61-143-227.eu-north-1.compute.amazonaws.com (13.61.143.227)
Host is up (0.29s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Ubuntu))

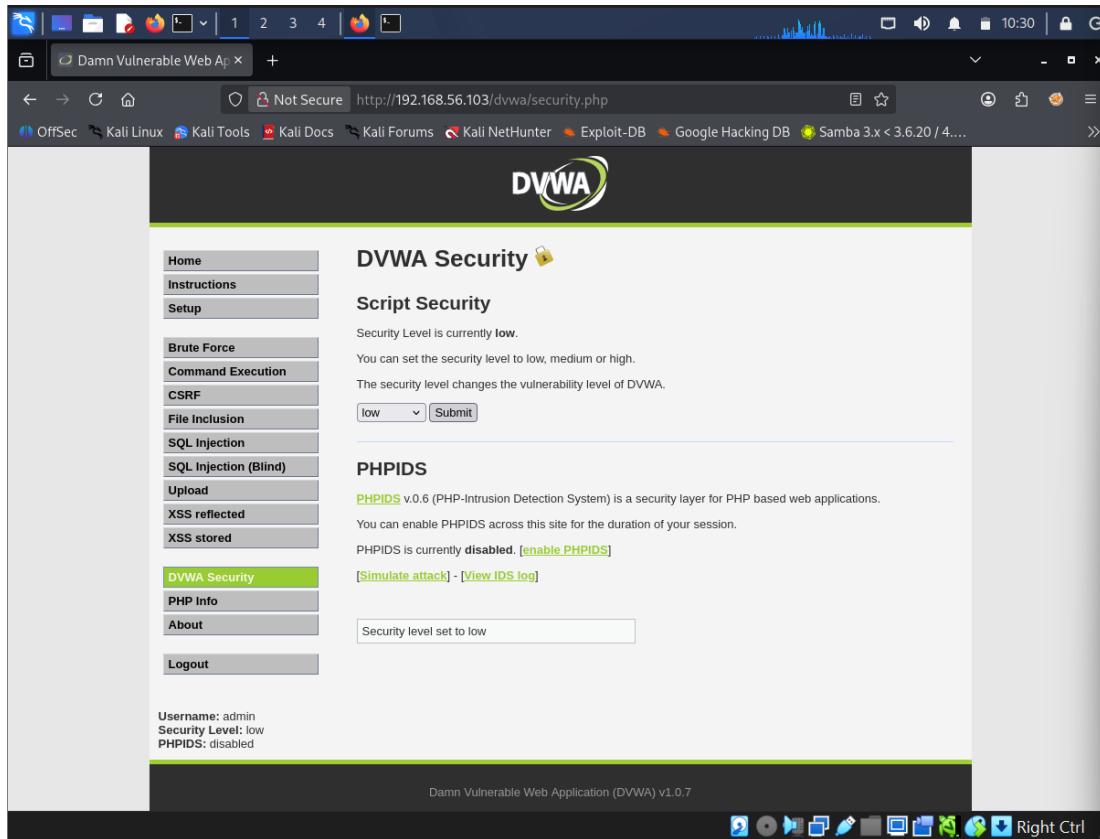
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.83 seconds
```

Key evidence: Reconnaissance State of the AWS TARGET.png

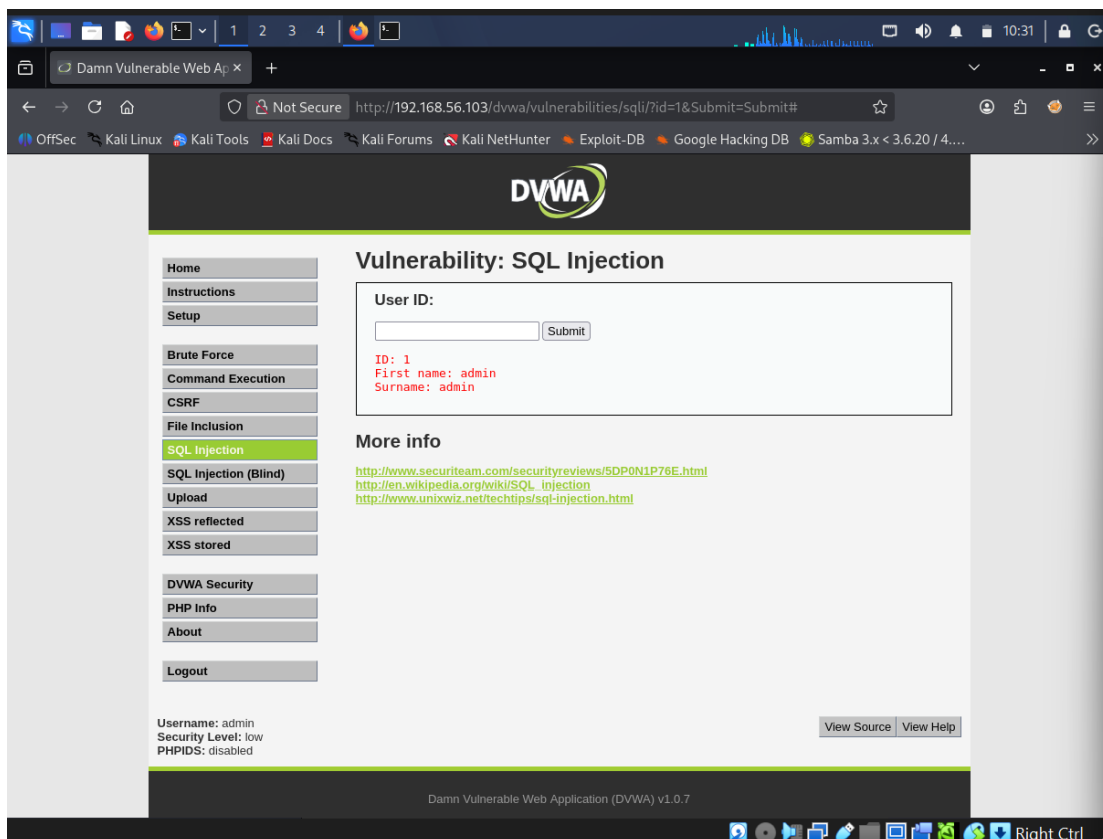
The screenshot displays the AWS Management Console interface for the 'Yusuf-CyberSecurity-Lab' account in the 'Europe (Stockholm)' region. The left-hand navigation pane shows the 'EC2' section expanded, with 'Security Groups' selected. The main content area shows a list of four security groups associated with the EC2 instance 'PrestaShop-Web-SG'. The first security group, 'sg-0cca5901c40658cd0', is selected. Below the list, the 'Inbound rules (4)' section is visible, showing four rules that restrict access to the instance. The rules are:

Name	Security group rule ID	IP version	Type	Protocol	Port
-	sgr-0d53578e0f30f4bf6	-	SSH	TCP	22
-	sgr-0a8dfa706d1beb4b	IPv4	HTTPS	TCP	443
-	sgr-01c456f05187e903f	IPv4	SSH	TCP	22
-	sgr-0930f7fc72c1f2de4	IPv4	HTTP	TCP	80

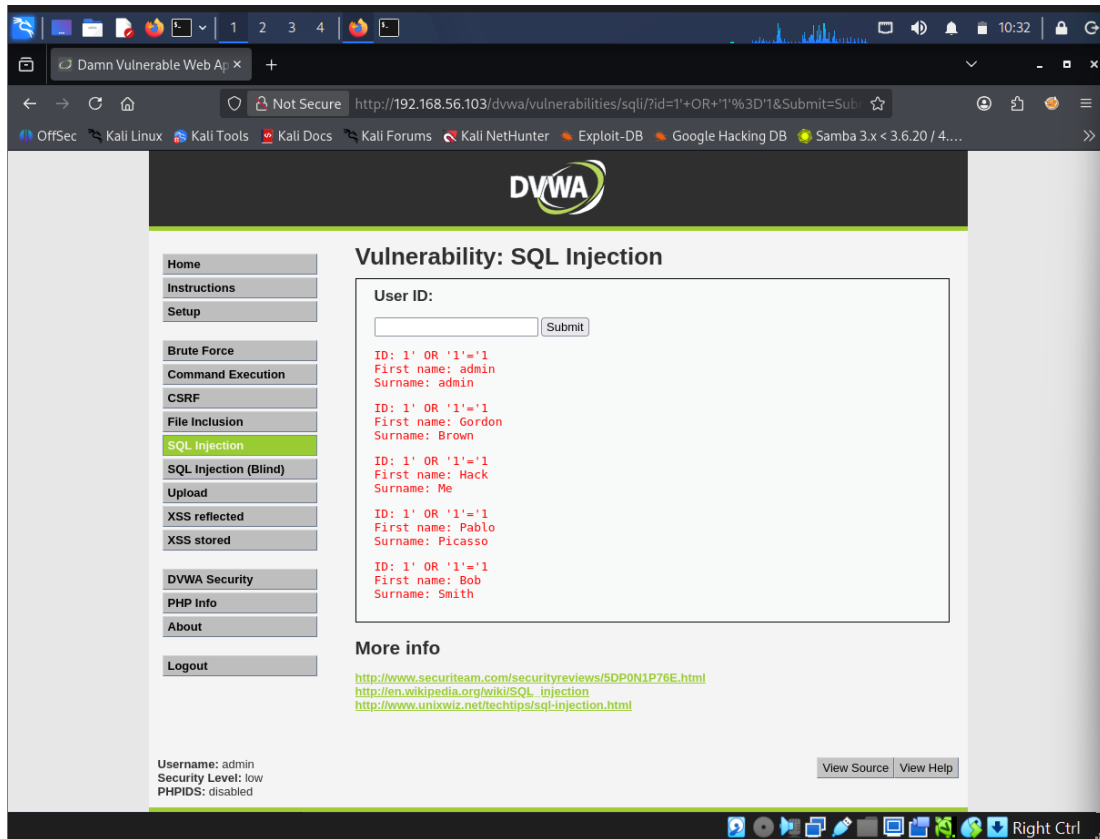
Key evidence: Restricted SSH Access.png



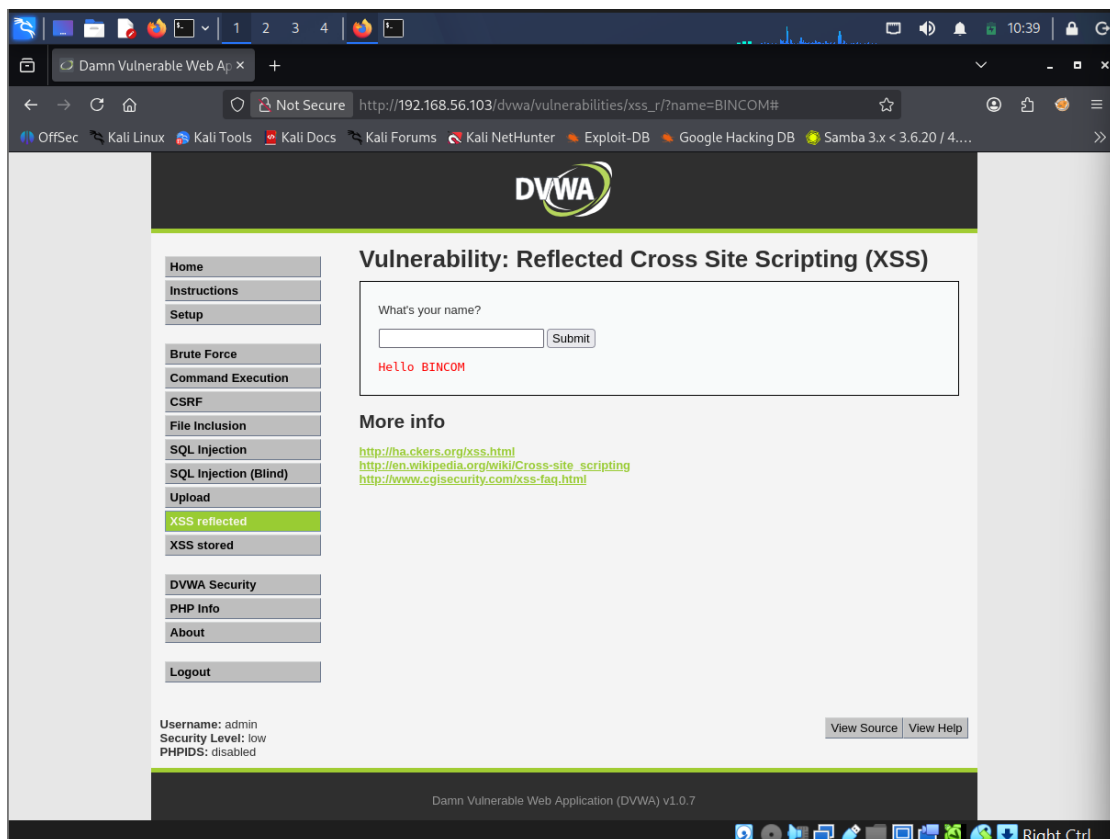
Key evidence: DVWA SECURITY SETTING.png



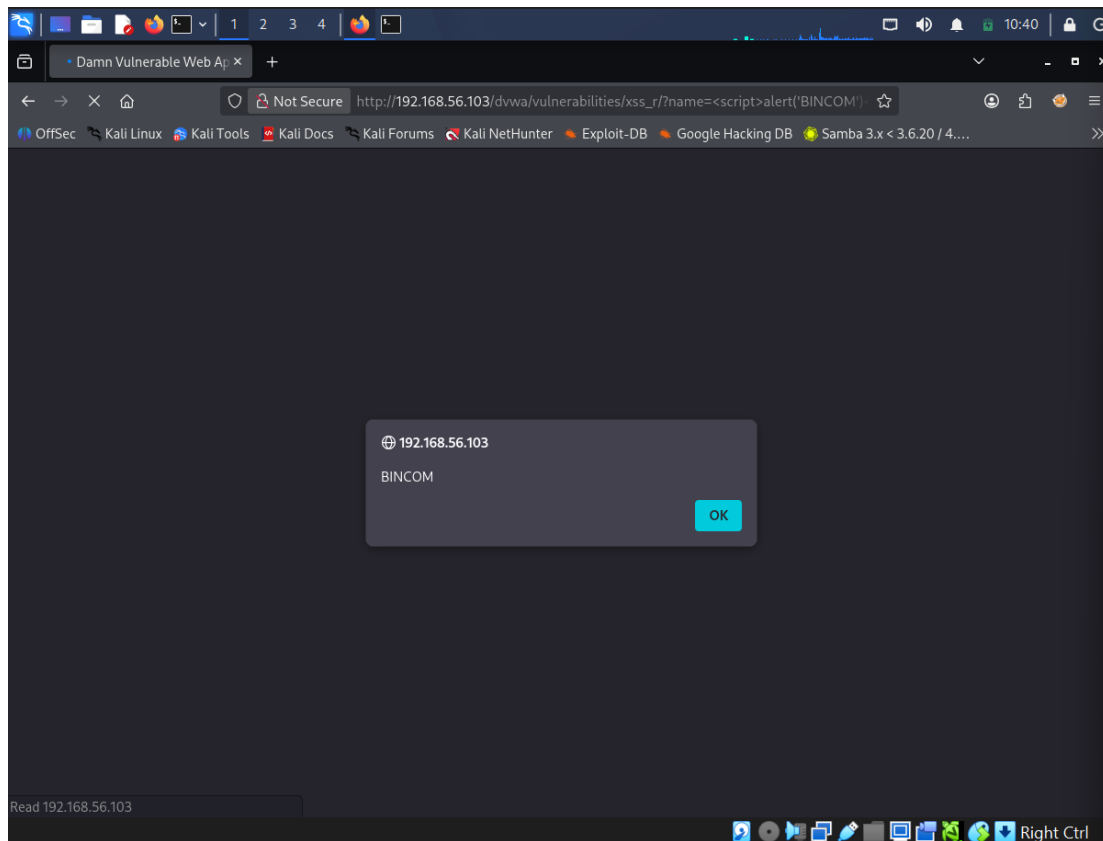
Key evidence: DVWA SQL INJECTION NORMAL TEST.png



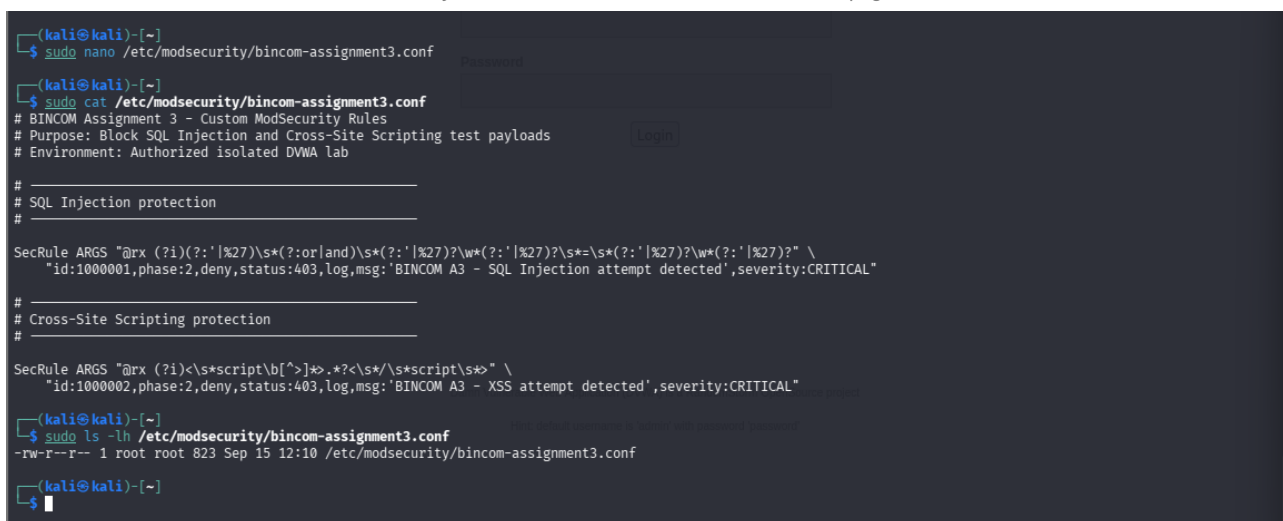
Key evidence: DVWA SQLI PAYLOAD TEST.png



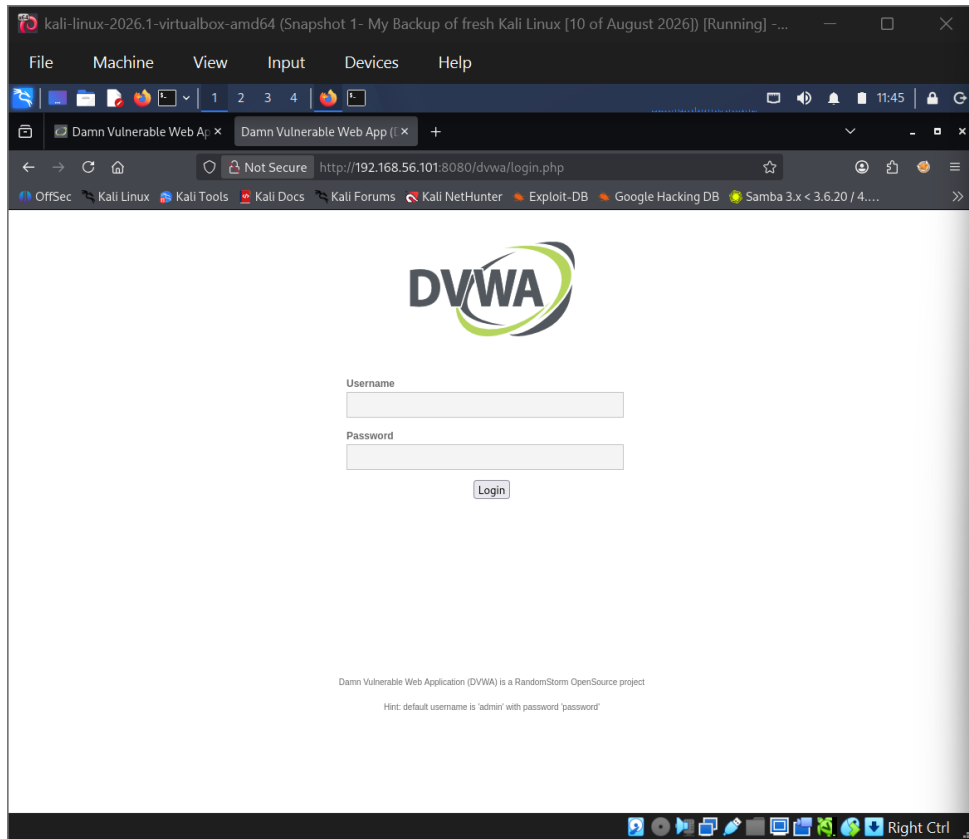
Key evidence: NORMAL XSS TEST ON DVWA.png



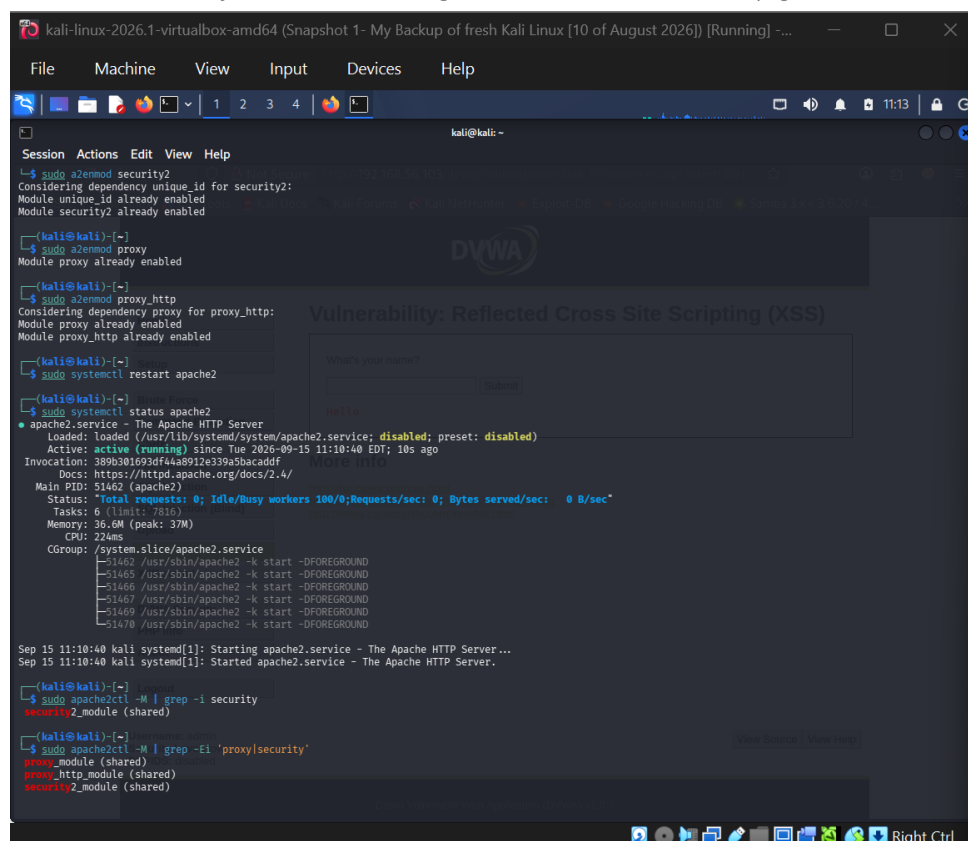
Key evidence: Controlled XSS TEST ON DVWA.png



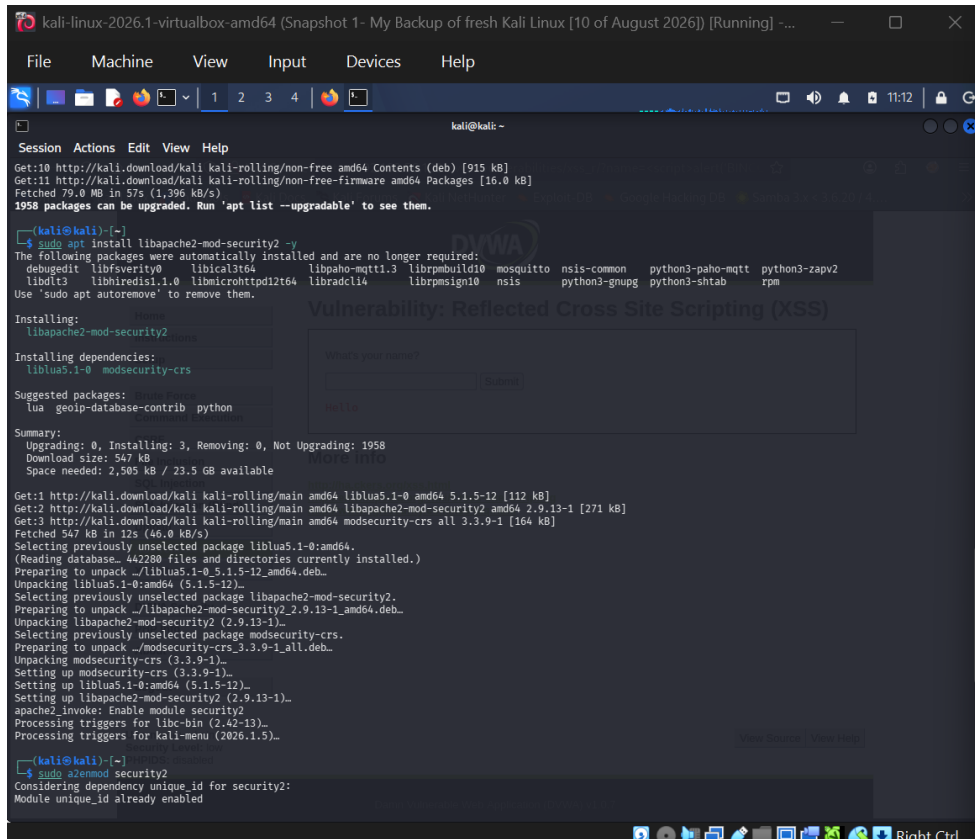
Key evidence: Configured WAF RULE file.png



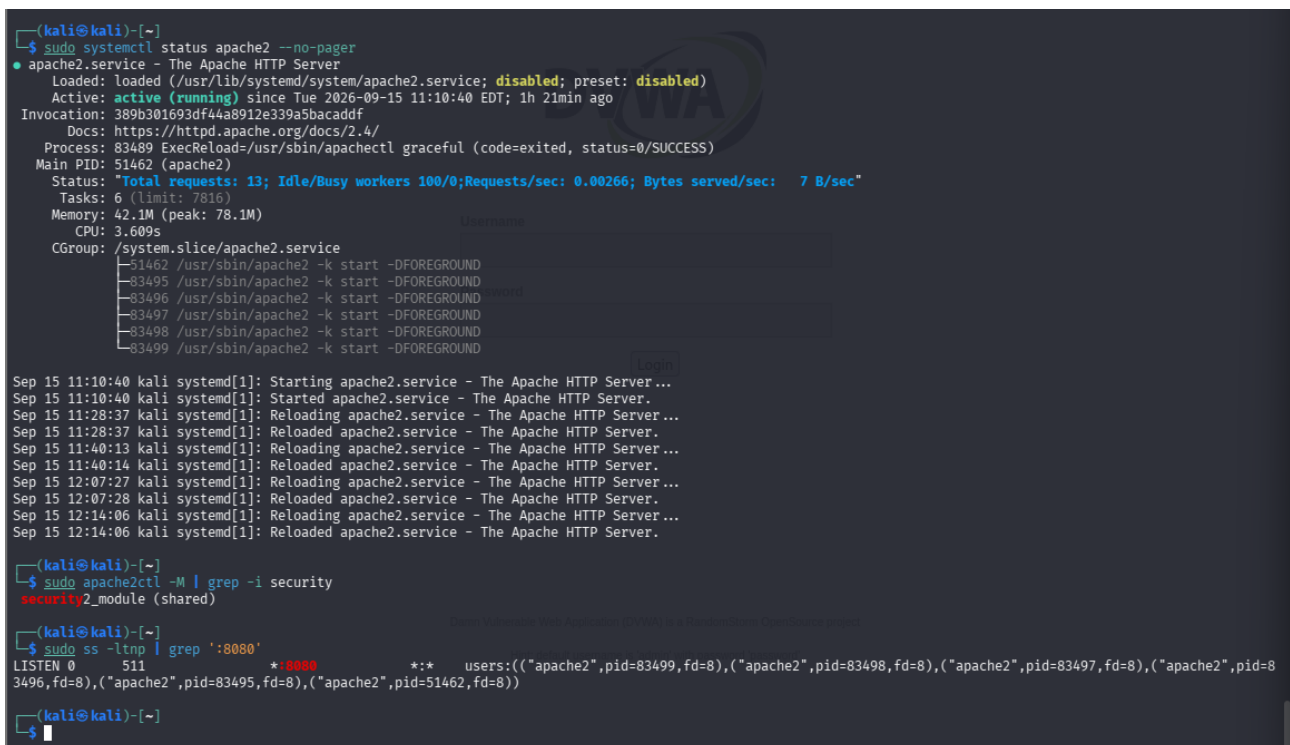
Key evidence: DVWA through KALI APACHE MODSECURITY.png



Key evidence: MODSECURITY Apache PROXY & MODULES ENABLED (1).png



Key evidence: MODSECURITY Apache PROXY & MODULES ENABLED (2).png



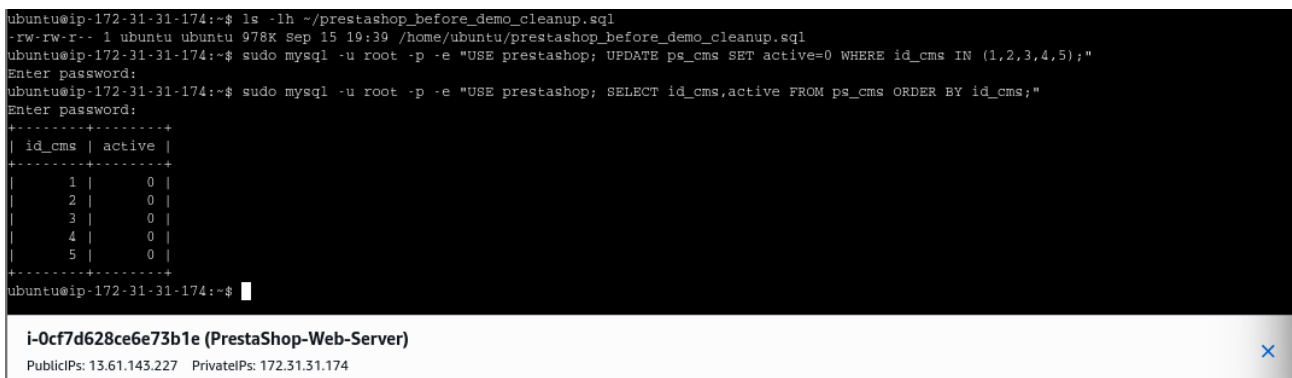
Key evidence: WAF Configuration Evidence.png



Key evidence: Traffic passes through WAF.png

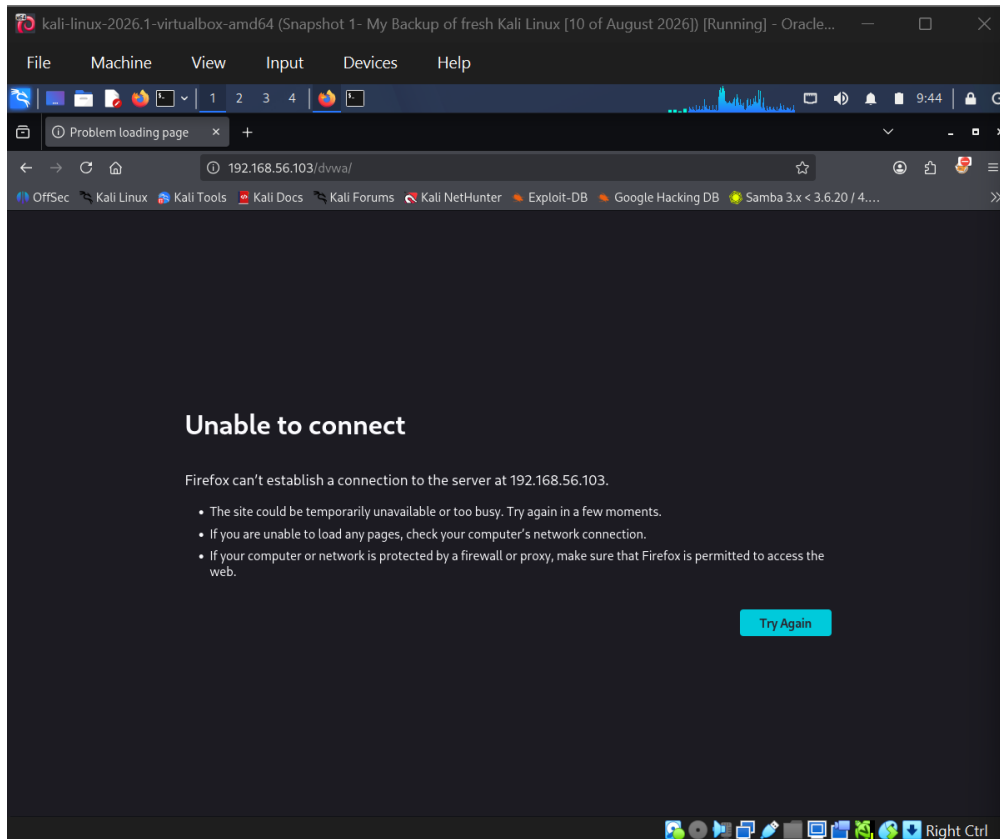


Key evidence: Traffic passes through WAF(1).png

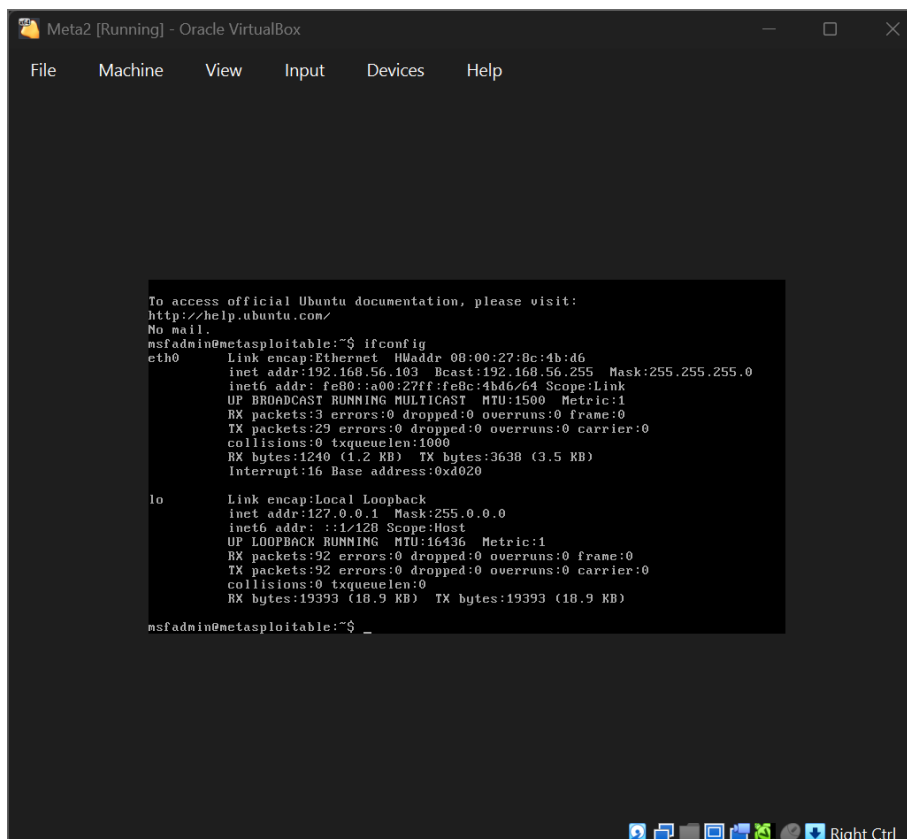


Key evidence: Defaultdemo accounts and sample data PASS after remediation..png

17.2 Complete Screenshot Archive



Archive screenshot: Screenshot 2026-09-15 144455.png



Archive screenshot: Screenshot 2026-09-15 145451.png

```
(kali@kali)-[~]
└─$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=16.5 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=3.90 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=5.10 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=4.47 ms

— 192.168.56.103 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 3.902/7.479/16.454/5.198 ms

(kali@kali)-[~]
└─$ curl -I http://192.168.56.103
HTTP/1.1 200 OK
Date: Tue, 15 Sep 2026 14:06:36 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

(kali@kali)-[~]
└─$ nmap -Pn -p 80,443 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-15 10:06 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0034s latency).
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   closed https
MAC Address: 08:00:27:8C:4B:D6 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.39 seconds

(kali@kali)-[~]
└─$ sudo service apache2 status
[sudo] password for kali:
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: https://httpd.apache.org/docs/2.4/

(kali@kali)-[~]
└─$
```

Archive screenshot: Screenshot 2026-09-15 150858.png

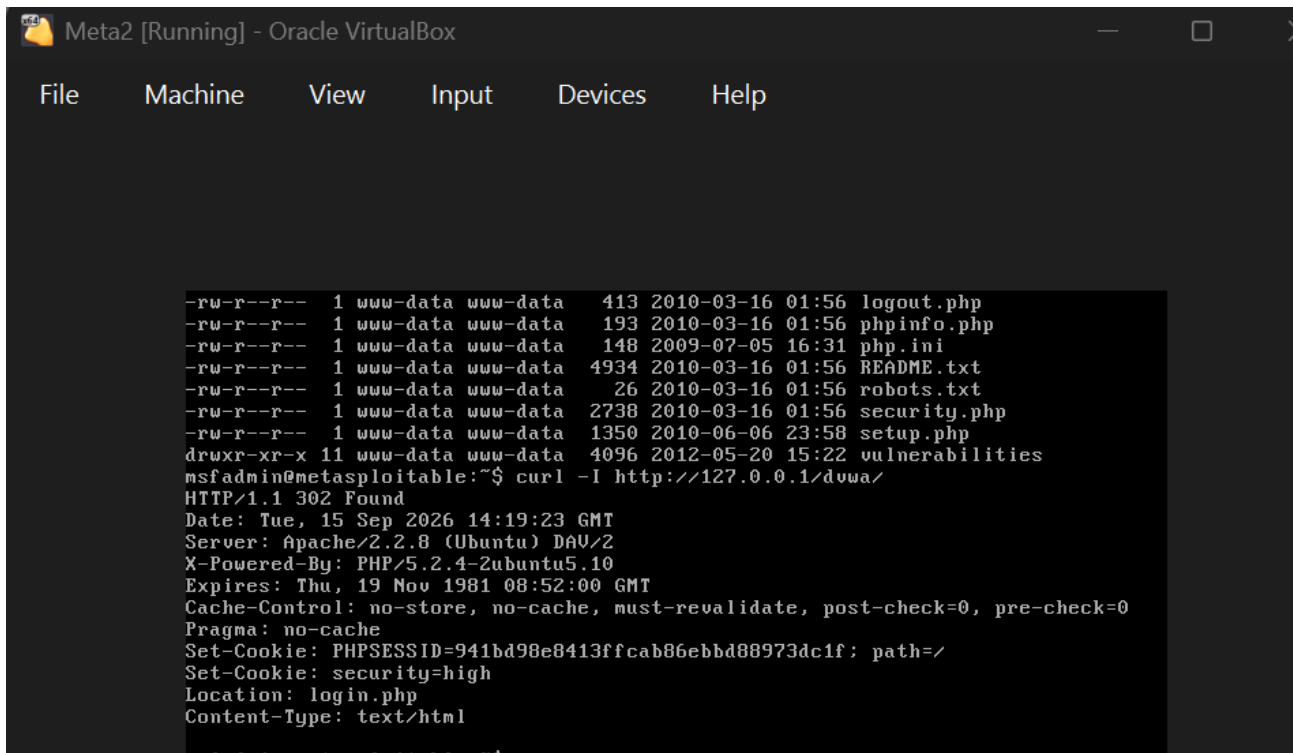
```
Meta2 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help

msfadmin@metasploitable:~$ ifconfig
eth0    Link encap:Ethernet HWaddr 08:00:27:8c:4b:d6
        inet addr:192.168.56.103 Bcast:192.168.56.255 Mask:255.255.255.0
        inet6 addr: fe80::a00:27ff:fe8c:4b46/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
        RX packets:3 errors:0 dropped:0 overruns:0 frame:0
        TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:1240 (1.2 KB) TX bytes:3638 (3.5 KB)
        Interrupt:16 Base address:0xd020

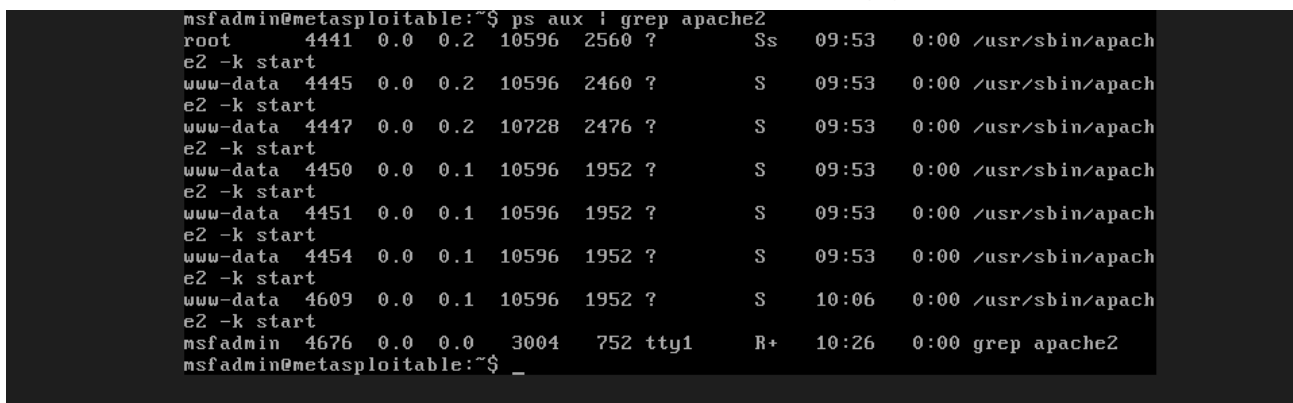
lo      Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING MTU:16436 Metric:1
        RX packets:92 errors:0 dropped:0 overruns:0 frame:0
        TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:19393 (18.9 KB) TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ sudo service apache2 status
[sudo] password for msfadmin:
sudo: service: command not found
msfadmin@metasploitable:~$
```

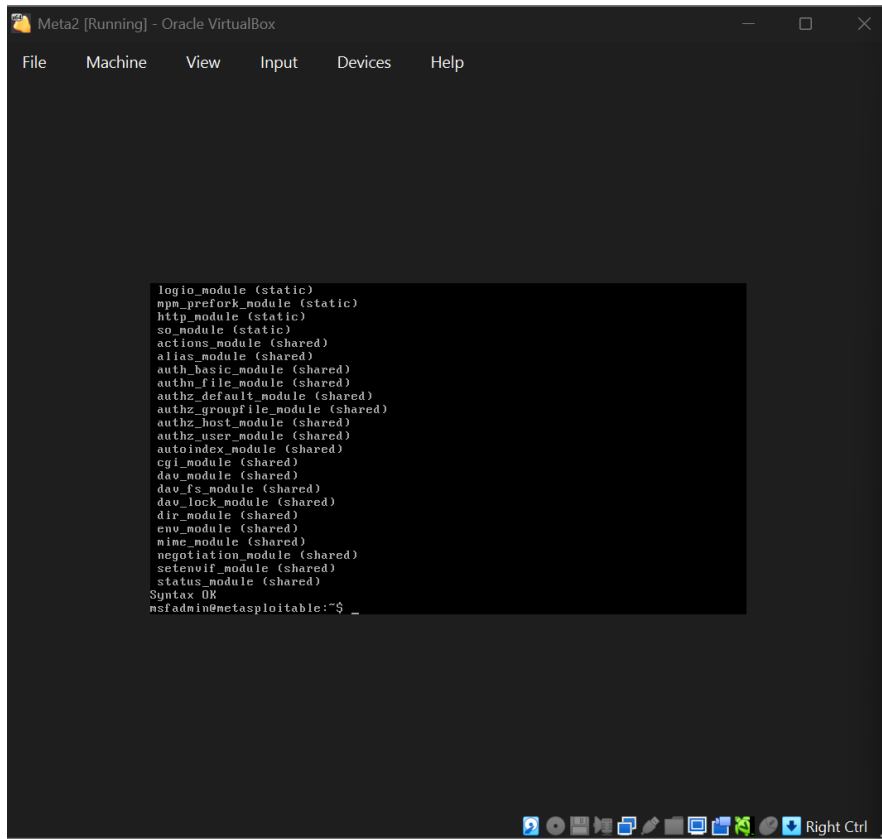
Archive screenshot: Screenshot 2026-09-15 151446.png



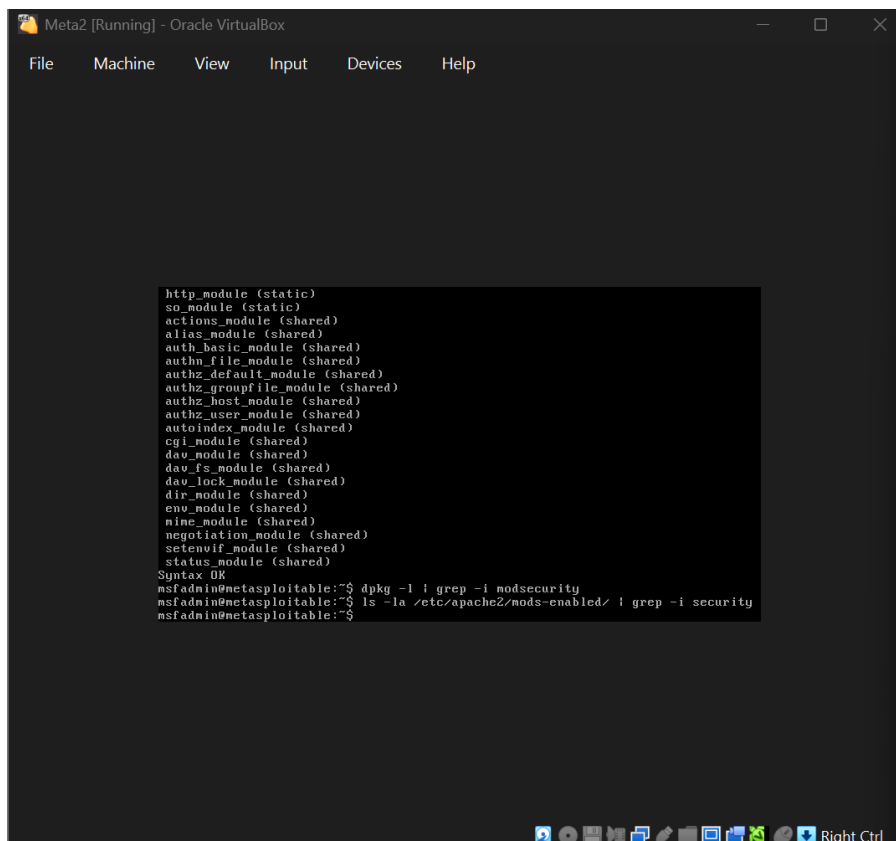
Archive screenshot: Screenshot 2026-09-15 152429.png



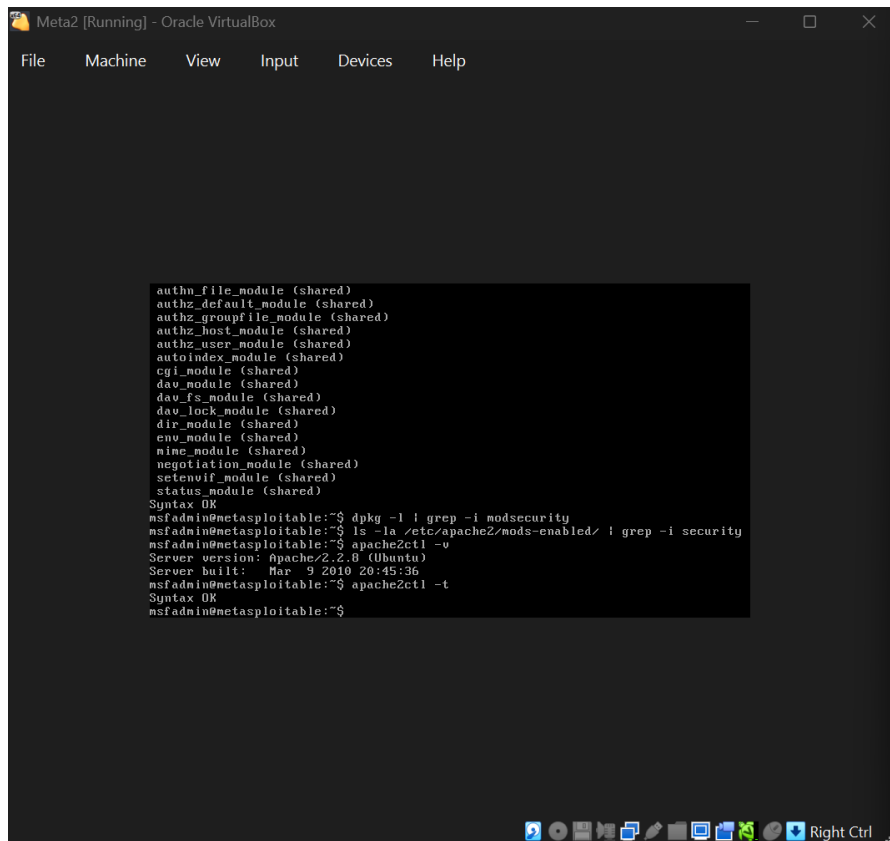
Archive screenshot: Screenshot 2026-09-15 152626.png



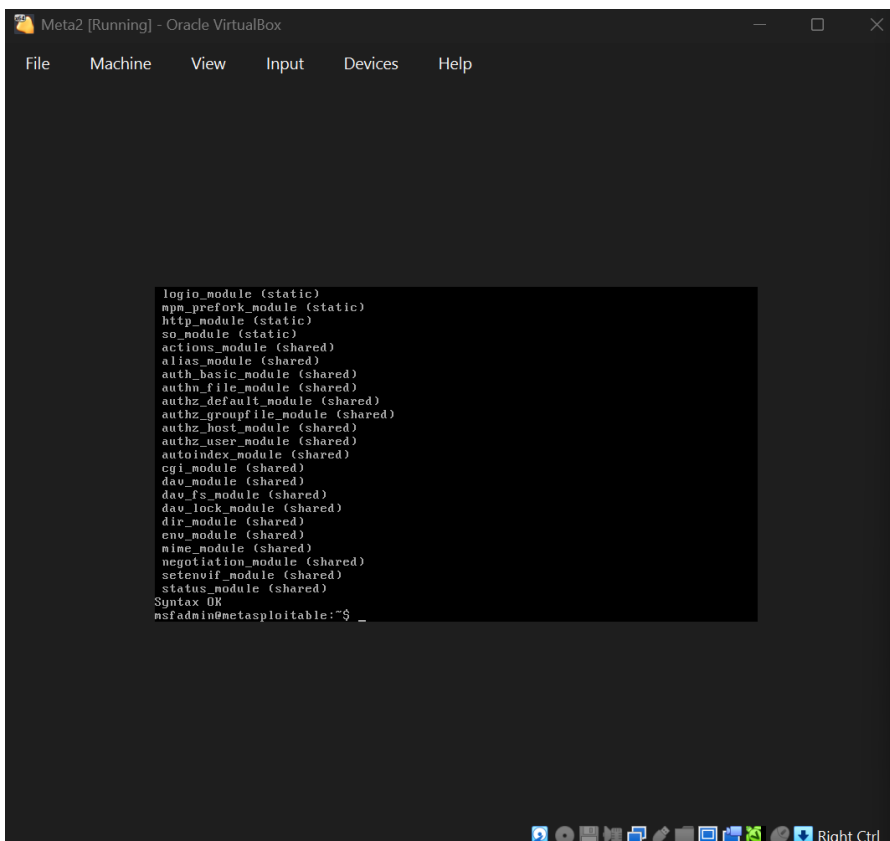
Archive screenshot: Screenshot 2026-09-15 154615.png



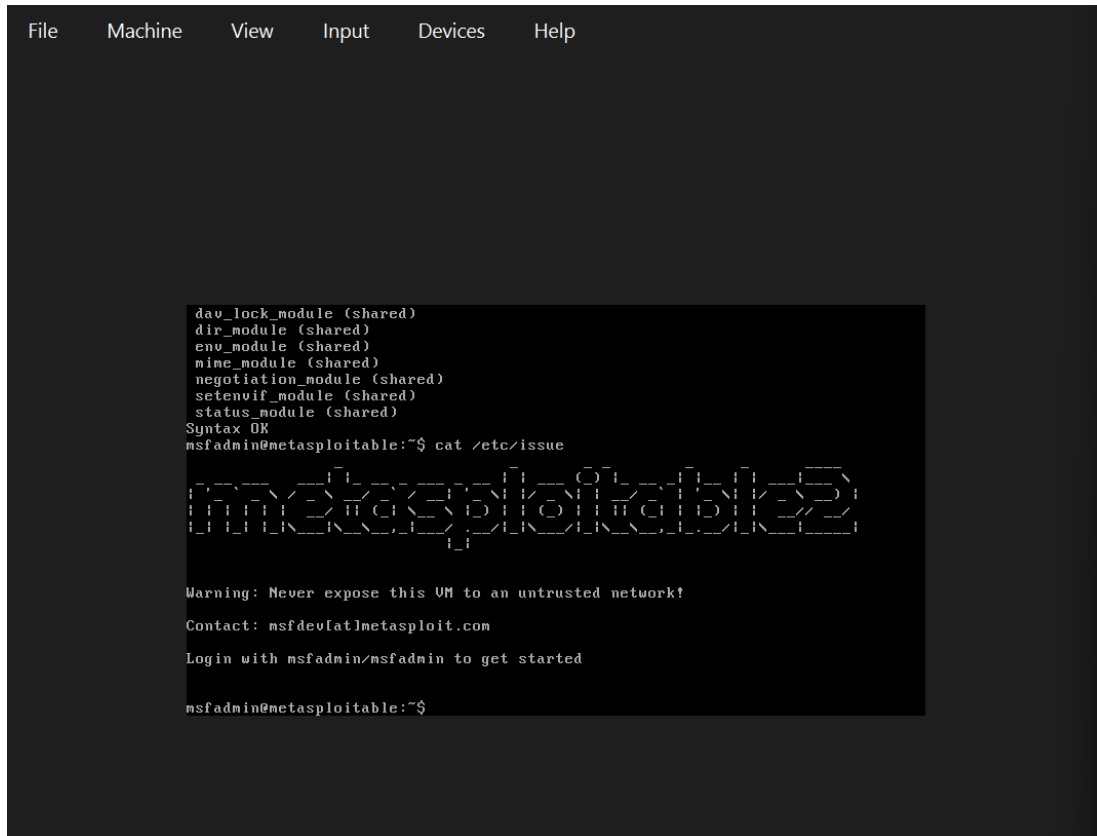
Archive screenshot: Screenshot 2026-09-15 155007.png



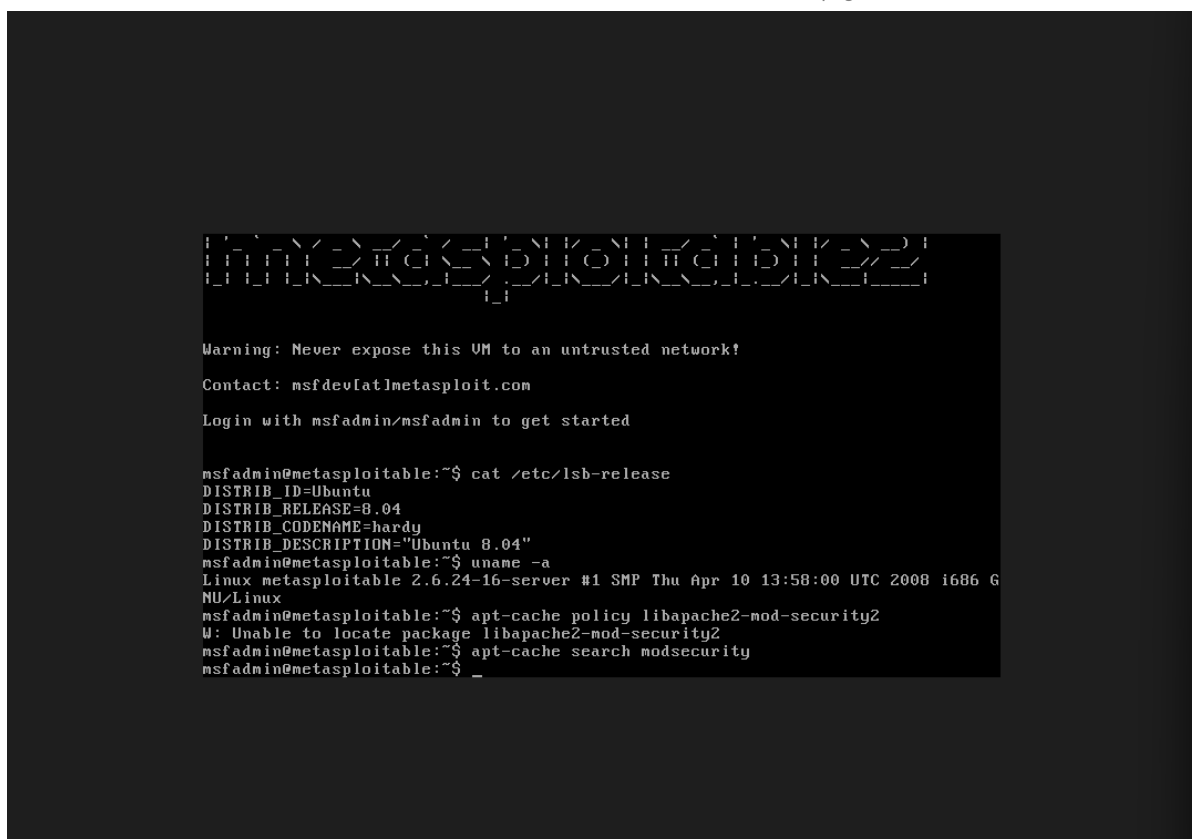
Archive screenshot: Screenshot 2026-09-15 155115.png



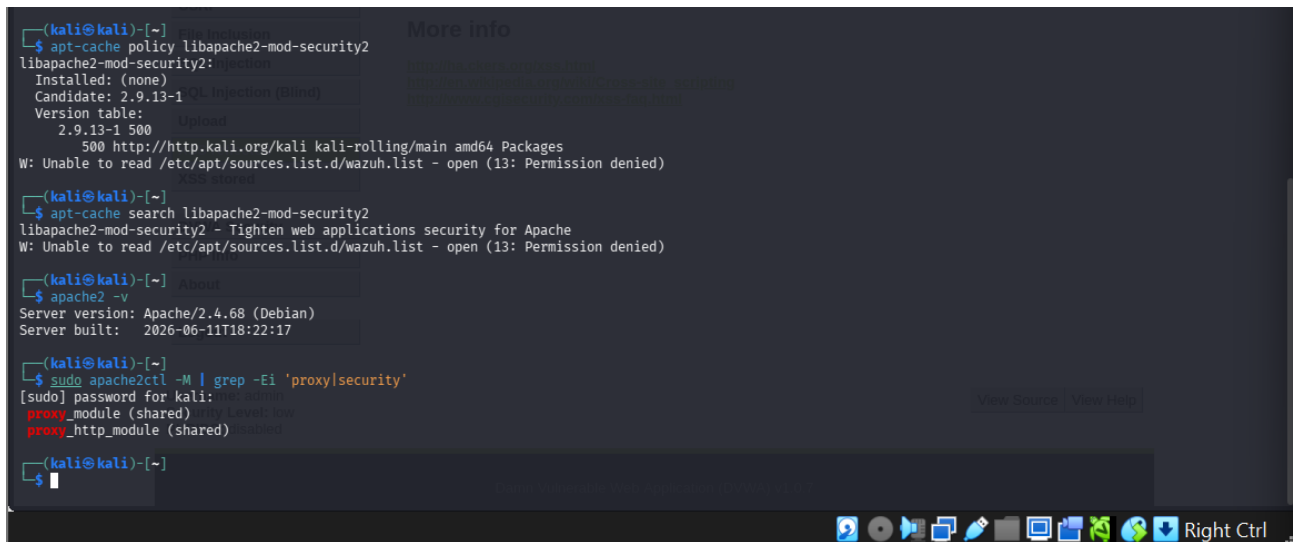
Archive screenshot: Screenshot 2026-09-15 155328.png



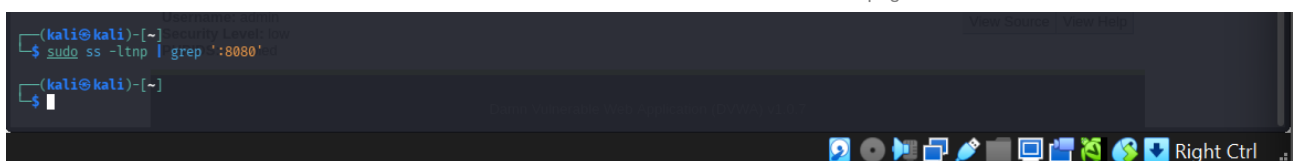
Archive screenshot: Screenshot 2026-09-15 155624.png



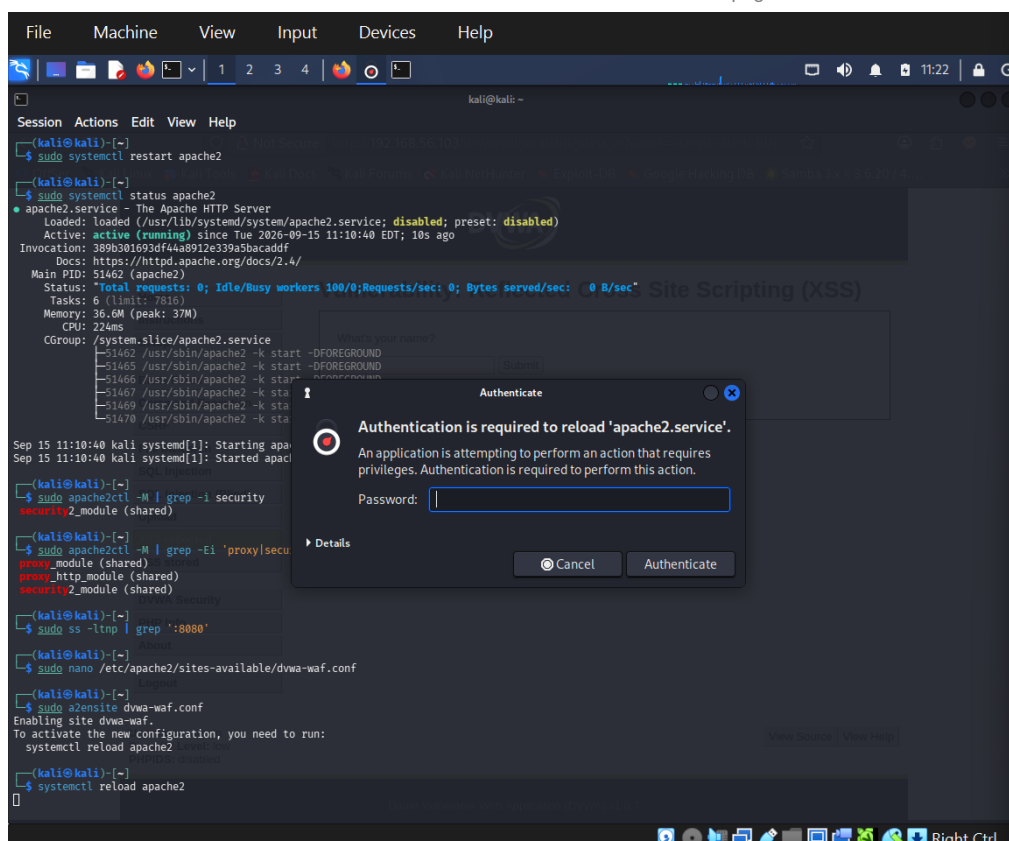
Archive screenshot: Screenshot 2026-09-15 155914.png



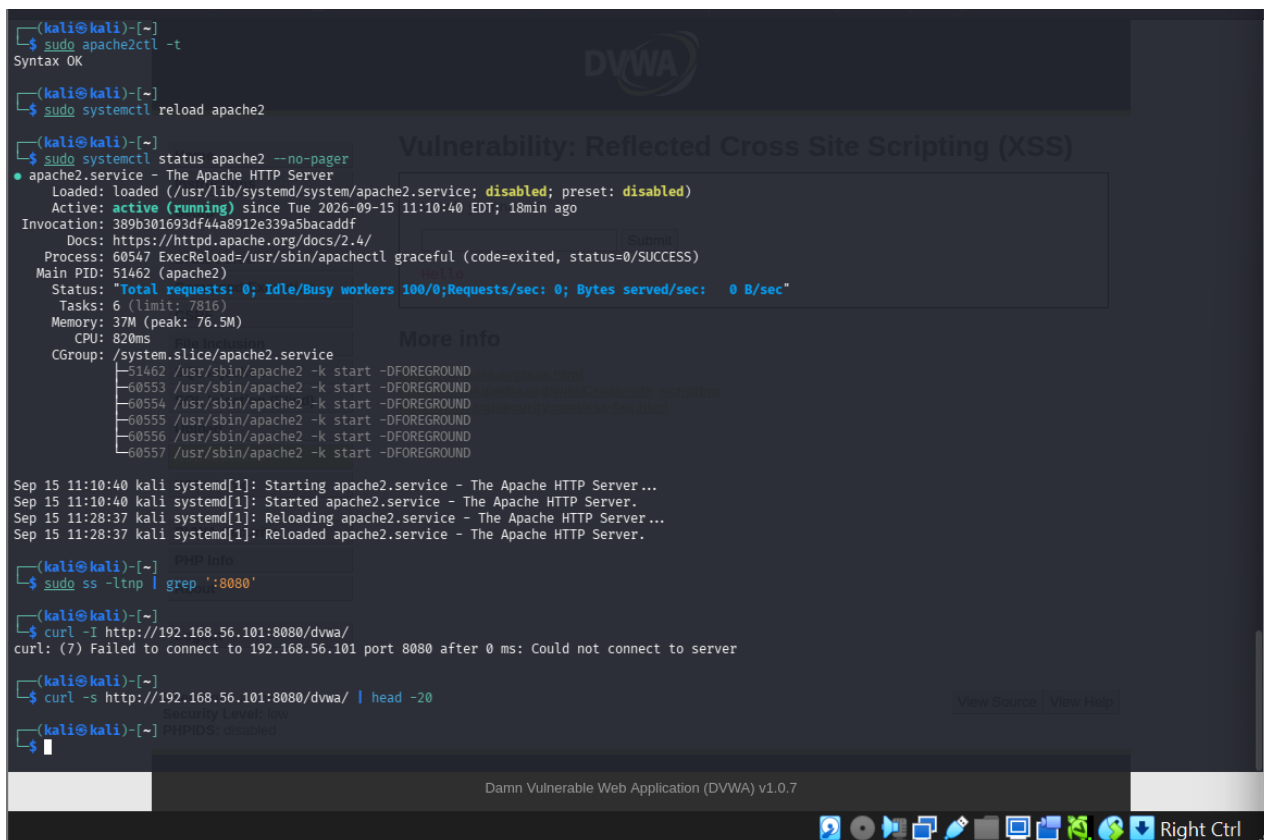
Archive screenshot: Screenshot 2026-09-15 160423.png



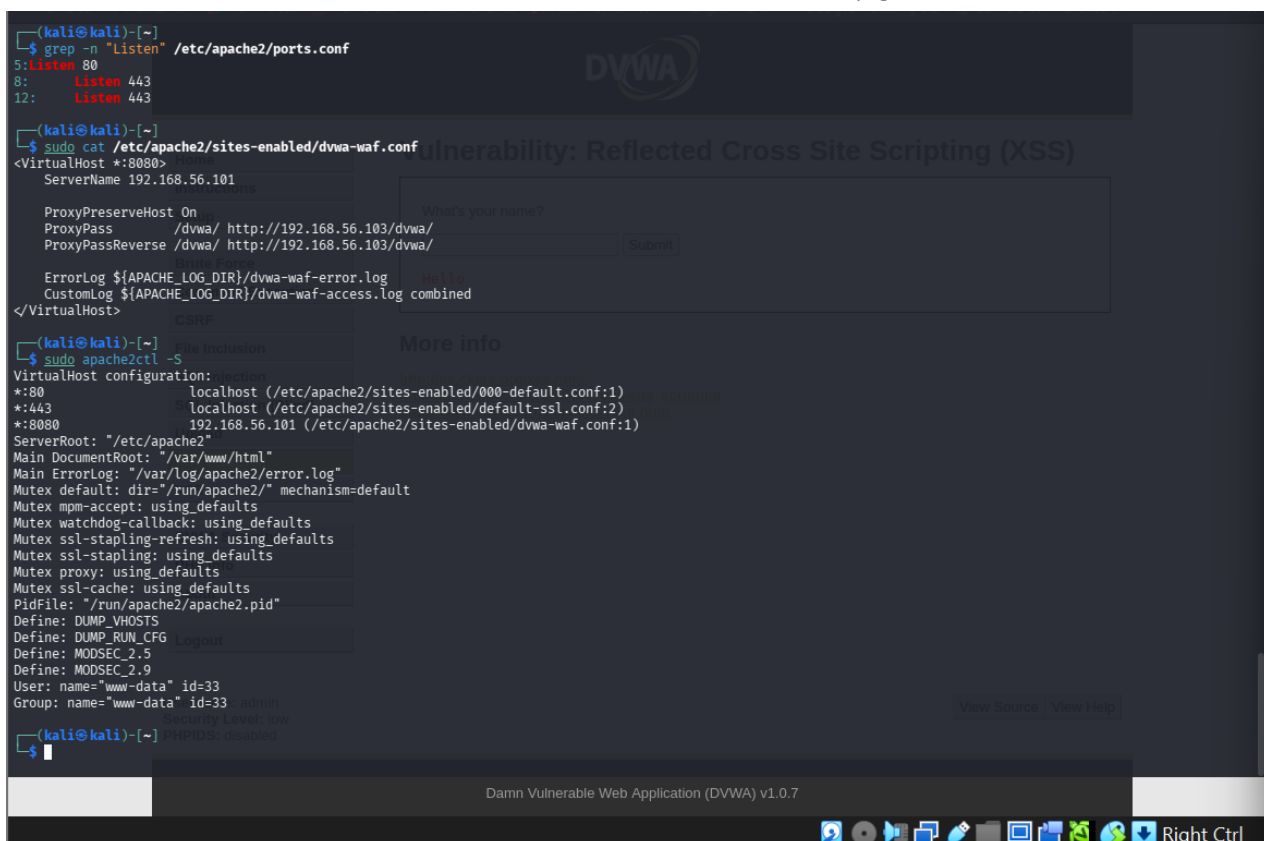
Archive screenshot: Screenshot 2026-09-15 161821.png



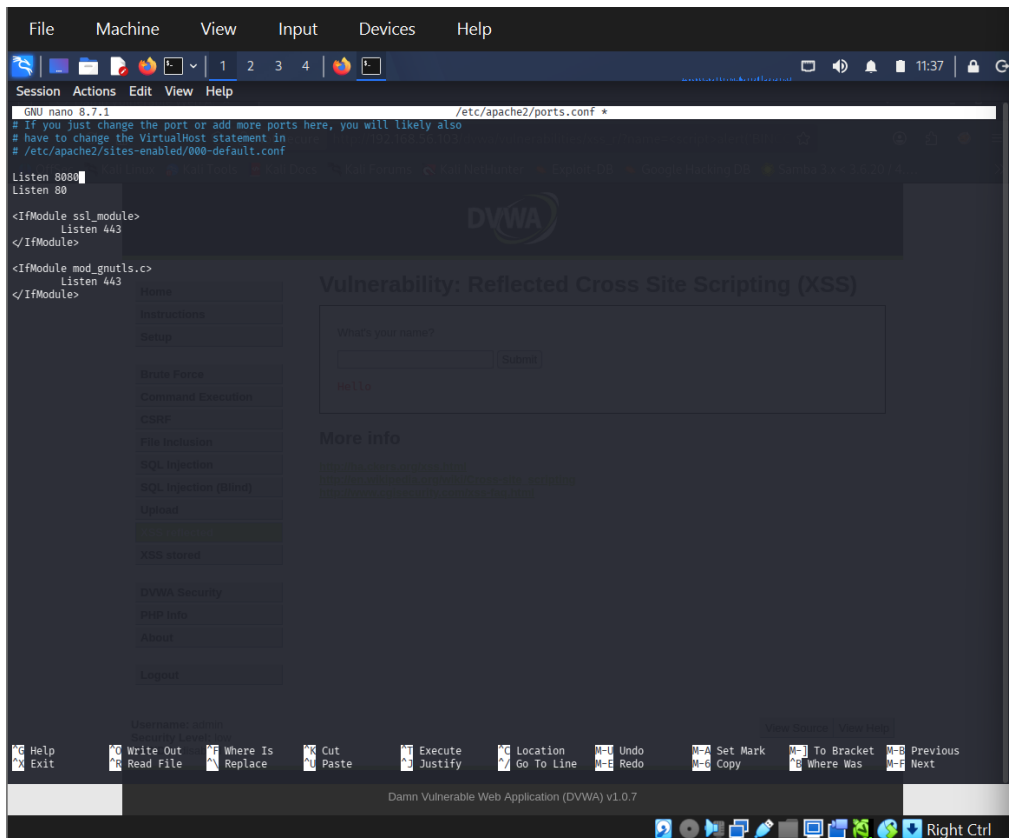
Archive screenshot: Screenshot 2026-09-15 162213.png



Archive screenshot: Screenshot 2026-09-15 163017.png



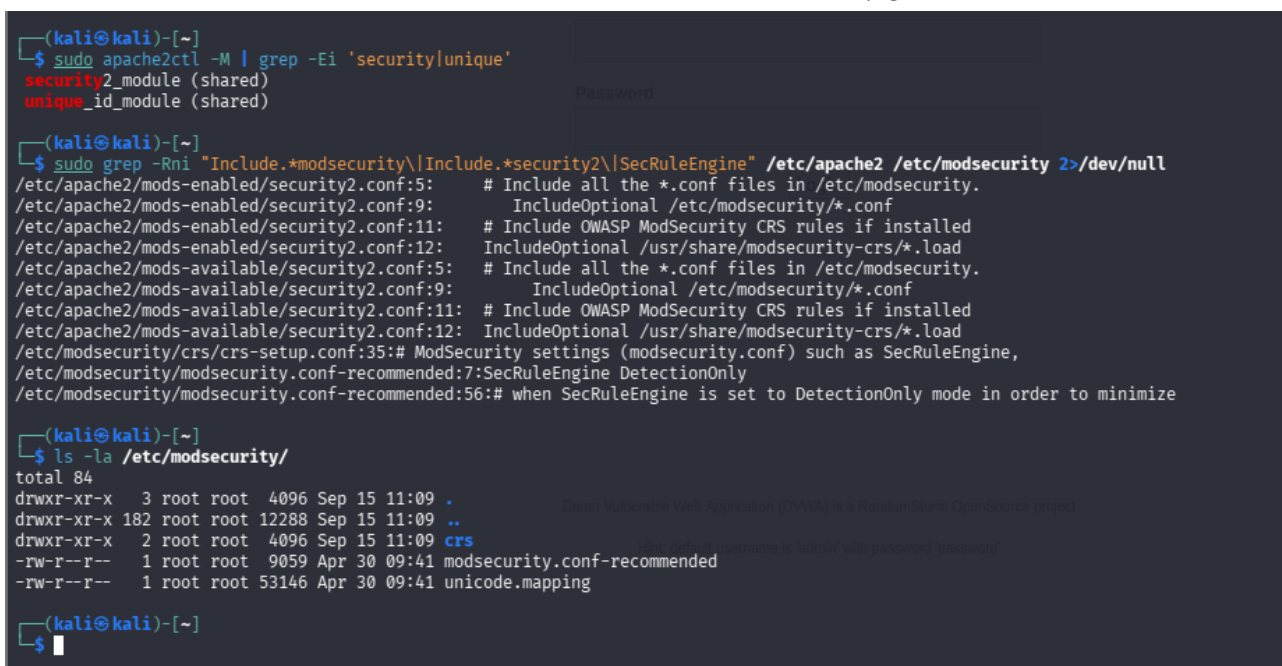
Archive screenshot: Screenshot 2026-09-15 163356.png



Archive screenshot: Screenshot 2026-09-15 163749.png



Archive screenshot: Screenshot 2026-09-15 164102.png



Archive screenshot: Screenshot 2026-09-15 165702.png

```

(kali@kali)-[~]
$ sudo cat /etc/apache2/mods-enabled/security2.conf
<IfModule security2_module>
# Default Debian dir for modsecurity's persistent data
SecDataDir /var/cache/modsecurity

# Include all the *.conf files in /etc/modsecurity.
# Keeping your local configuration in that directory
# will allow for an easy upgrade of THIS file and
# make your life easier
IncludeOptional /etc/modsecurity/*.conf

# Include OWASP ModSecurity CRS rules if installed
IncludeOptional /usr/share/modsecurity-crs/*.load
</IfModule>

(kali@kali)-[~]
$ sudo cat /usr/share/modsecurity-crs/owasp-crs.load
##
## This file loads OWASP CRS's rules when the package is installed
## It is Included by libapache2-mod-security2
##
Include /etc/modsecurity/crs/crs-setup.conf
IncludeOptional /etc/modsecurity/crs/REQUEST-932-APPLICATION-ATTACK-RCE.conf
Include /usr/share/modsecurity-crs/rules/php-config-directives.data
Include /usr/share/modsecurity-crs/rules/REQUEST-903.9004-DOKUWIKI-EXCLUSION-RULES.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-933-APPLICATION-ATTACK-PHP.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-903.9006-XENFORO-EXCLUSION-RULES.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-942-APPLICATION-ATTACK-SQLI.conf
Include /usr/share/modsecurity-crs/rules/crawlers-user-agents.data
Include /usr/share/modsecurity-crs/rules/REQUEST-903.9001-DRUPAL-EXCLUSION-RULES.conf
Include /usr/share/modsecurity-crs/rules/scanners-user-agents.data
Include /usr/share/modsecurity-crs/rules/REQUEST-930-APPLICATION-ATTACK-LFI.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-901-INITIALIZATION.conf
Include /usr/share/modsecurity-crs/rules/windows-powershell-commands.data
Include /usr/share/modsecurity-crs/rules/java-errors.data
Include /usr/share/modsecurity-crs/rules/REQUEST-903.9002-WORDPRESS-EXCLUSION-RULES.conf
Include /usr/share/modsecurity-crs/rules/scanners-urls.data
Include /usr/share/modsecurity-crs/rules/REQUEST-922-MULTIPART-ATTACK.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-921-PROTOCOL-ATTACK.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-941-APPLICATION-ATTACK-XSS.conf
Include /usr/share/modsecurity-crs/rules/RESPONSE-959-BLOCKING-EVALUATION.conf
Include /usr/share/modsecurity-crs/rules/REQUEST-903.9005-CPANEL-EXCLUSION-RULES.conf

```

Archive screenshot: Screenshot 2026-09-15 170306.png

```

(kali@kali)-[~]
$ sudo apache2ctl -t
Syntax OK

(kali@kali)-[~]
$ sudo systemctl reload apache2

(kali@kali)-[~]
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 11:10:40 EDT; 57min ago
     Invocation: 389b301693df44a8912e339a5bacaddf
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 80047 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 51462 (apache2)
    Status: "Total requests: 10; Idle/Busy workers 100/0;Requests/sec: 0.00292; Bytes served/sec: 10 B/sec"
     Tasks: 6 (limit: 7816)
    Memory: 41.8M (peak: 76.5M)
       CPU: 2.964s
    CGroup: /system.slice/apache2.service
            └─51462 /usr/sbin/apache2 -k start -DFOREGROUND
              └─80056 /usr/sbin/apache2 -k start -DFOREGROUND
                └─80057 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─80058 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─80059 /usr/sbin/apache2 -k start -DFOREGROUND
                      └─80060 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 15 11:10:40 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Sep 15 11:10:40 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 15 11:28:37 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 11:28:37 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.
Sep 15 11:40:13 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 11:40:13 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.
Sep 15 12:07:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 15 12:07:28 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.

(kali@kali)-[~]
$ sudo ss -ltnp | grep ':8080'
LISTEN 0      511          *:*          *:*        users:((("apache2",pid=80060,fd=8),("apache2",pid=80059,fd=8),("apache2",pid=80058,fd=8),("apache2",pid=80057,fd=8),("apache2",pid=80056,fd=8),("apache2",pid=51462,fd=8)))

(kali@kali)-[~]
$

```

Archive screenshot: Screenshot 2026-09-15 170825.png

```
(kali@kali)-[~]
└─$ sudo apache2ctl -t
Syntax OK

(kali@kali)-[~]
└─$ sudo systemctl reload apache2

(kali@kali)-[~]
└─$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 11:10:40 EDT; 1h 3min ago
 Invocation: 389b301693df44a8912e339a5bacaddf
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 83489 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
  Main PID: 51462 (apache2)
    Status: "Total requests: 10; Idle/Busy workers 100/0;Requests/sec: 0.00262; Bytes served/sec: 9 B/sec"
     Tasks: 6 (limit: 7816)
    Memory: 42.4M (peak: 78.1M)
       CPU: 3.398s
    CGroup: /system.slice/apache2.service
            └─51462 /usr/sbin/apache2 -k start -DFOREGROUND
              └─83495 /usr/sbin/apache2 -k start -DFOREGROUND
                └─83496 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─83497 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─83498 /usr/sbin/apache2 -k start -DFOREGROUND
                      └─83499 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 15 11:10:40 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 11:10:40 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 15 11:28:37 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 11:28:37 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 11:40:13 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 11:40:14 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 12:07:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 12:07:28 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 12:14:06 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 12:14:06 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.

(kali@kali)-[~]
└─$ sudo ss -ltnp | grep ':8080'
LISTEN 0      511          *:*          *:*    users:(("apache2",pid=83499,fd=8),("apache2",pid=83498,fd=8),("apache2",pid=83497,fd=8),("apache2",pid=83496,fd=8),("apache2",pid=83495,fd=8),("apache2",pid=51462,fd=8))

(kali@kali)-[~]
└─$
```

Archive screenshot: Screenshot 2026-09-15 171535.png

```
(kali@kali)-[~]
└─$ curl -i http://192.168.56.101:8080/dvwa/
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 16:18:50 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Pragma: no-cache
Location: login.php
Content-Length: 0
Content-Type: text/html
Set-Cookie: PHPSESSID=163d690396d7eb49c1fe9428b51ba74e; path=/
Set-Cookie: security=high

(kali@kali)-[~]
└─$ curl -i "http://192.168.56.101:8080/dvwa/vulnerabilities/sqli/?id=1%27%20OR%20%271%27=%2716Submit=Submit"
HTTP/1.1 403 Forbidden
Date: Tue, 15 Sep 2026 16:19:11 GMT
Server: Apache/2.4.68 (Debian)
Content-Length: 321
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at 192.168.56.101 Port 8080</address>
</body></html>

(kali@kali)-[~]
└─$
```

Archive screenshot: Screenshot 2026-09-15 171943.png

```

(kali@kali)-[~]
$ curl -i "http://192.168.56.101:8080/dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Ealert%28%27BINCOM%27%29%3C%2Fscript%3E6Submit=Submit"
HTTP/1.1 403 Forbidden
Date: Tue, 15 Sep 2026 16:28:09 GMT
Server: Apache/2.4.68 (Debian)
Content-Length: 321
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at 192.168.56.101 Port 8080</address>
</body></html>

(kali@kali)-[~]
$ sudo grep -rni "1000002" /var/log/apache2 /var/log/modsecurity 2>/dev/null | tail -30
/var/log/apache2/modsec_audit.log:99:Message: Access denied with code 403 (phase 2). Pattern match "(?i)<\\s*script\\b[>]*.*?<\\s*\\s*script\\s*>" at ARGS:name
. [file "/etc/modsecurity/bincom-assignment3.conf"] [line "17"] [id "1000002"] [msg "BINCOM A3 - XSS attempt detected"] [severity "CRITICAL"]
/var/log/apache2/modsec_audit.log:100:Apache-Error: [file "apache2_util.c"] [line 286] [level 3] ModSecurity: Access denied with code 403 (phase 2). Pattern match
"(?i)<\\s*script\\b[>]*.*?<\\s*\\s*script\\s*>" at ARGS:name. [file "/etc/modsecurity/bincom-assignment3.conf"] [line "17"] [id
"1000002"] [msg "BINCOM A3 - XSS attempt detected"] [severity "CRITICAL"] [hostname "192.168.56.101"] [uri "/dvwa/vulnerabilities/xss_r/"] [unique_id "aqlyGVfjNg4
K8qfLT6FboAAAAAI"]
/var/log/apache2/dvwa-waf-error.log:3:[Tue Sep 15 12:28:09.552554 2026] [security2:error] [pid 83497:tid 83497] [client 192.168.56.101:41796] ModSecurity: Access
denied with code 403 (phase 2). Pattern match "(?i)<\\s*script\\b[>]*.*?<\\s*\\s*script\\s*>" at ARGS:name. [file "/etc/modsecurity/bincom-assignment
3.conf"] [line "17"] [id "1000002"] [msg "BINCOM A3 - XSS attempt detected"] [severity "CRITICAL"] [hostname "192.168.56.101"] [uri "/dvwa/vulnerabilities/xss_r/"
] [unique_id "aqlyGVfjNg4K8qfLT6FboAAAAAI"]

(kali@kali)-[~]
$

```

Archive screenshot: Screenshot 2026-09-15 172928.png

```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 08:00:27:ba:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 75451sec preferred_lft 75451sec
    inet6 fd17:625c:f037:214c4:2e13:ddb1:7c7c/64 scope global dynamic noprefixroute
        valid_lft 86252sec preferred_lft 14252sec
    inet6 fe80::f0b0:ce61:518d:8d34/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ba:f6:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 451sec preferred_lft 451sec
    inet6 fe80::55ef:f150:1dc4:7de7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
192.168.56.0/24 dev eth1 proto kernel scope link src 192.168.56.101 metric 101

(kali@kali)-[~]
$ sudo nmap -sn 192.168.56.0/24
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-15 12:36 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).
MAC Address: 08:00:27:00:00:14 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.0081s latency).
MAC Address: 08:00:27:78:75:0C (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.103
Host is up (0.0026s latency).
MAC Address: 08:00:27:8C:48:D6 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 4.03 seconds

(kali@kali)-[~]
$

```

Archive screenshot: Screenshot 2026-09-15 173714.png

```
(kali@kali)-[~]
$ sudo nmap -sV -p 80,443,8080,8000,8443 192.168.56.100 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-15 12:39 -0400
Nmap scan report for 192.168.56.100
Host is up (0.0016s latency).

PORT      STATE SERVICE      VERSION
80/tcp    closed http
443/tcp    closed https
8000/tcp   closed http-alt
8080/tcp   closed http-proxy
8443/tcp   closed https-alt
MAC Address: 08:00:27:7B:75:0C (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.56.103
Host is up (0.0065s latency).
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 2 IP addresses (2 hosts up) scanned in 8.57 seconds

(kali@kali)-[~]
$ curl -I http://192.168.56.100/
curl: (7) Failed to connect to 192.168.56.100 port 80 after 1 ms: Could not connect to server

(kali@kali)-[~]
$ curl -k -I https://192.168.56.100/
curl: (7) Failed to connect to 192.168.56.100 port 443 after 1 ms: Could not connect to server

(kali@kali)-[~]
$
```

Archive screenshot: Screenshot 2026-09-15 174035.png

```
(kali@kali)-[~]
$ sudo nmap -p- --min-rate 1000 192.168.56.100
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-15 12:42 -0400
Nmap scan report for 192.168.56.100
Host is up (0.00057s latency).
All 65535 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 65535 closed tcp ports (reset)
MAC Address: 08:00:27:7B:75:0C (Oracle VirtualBox virtual NIC)

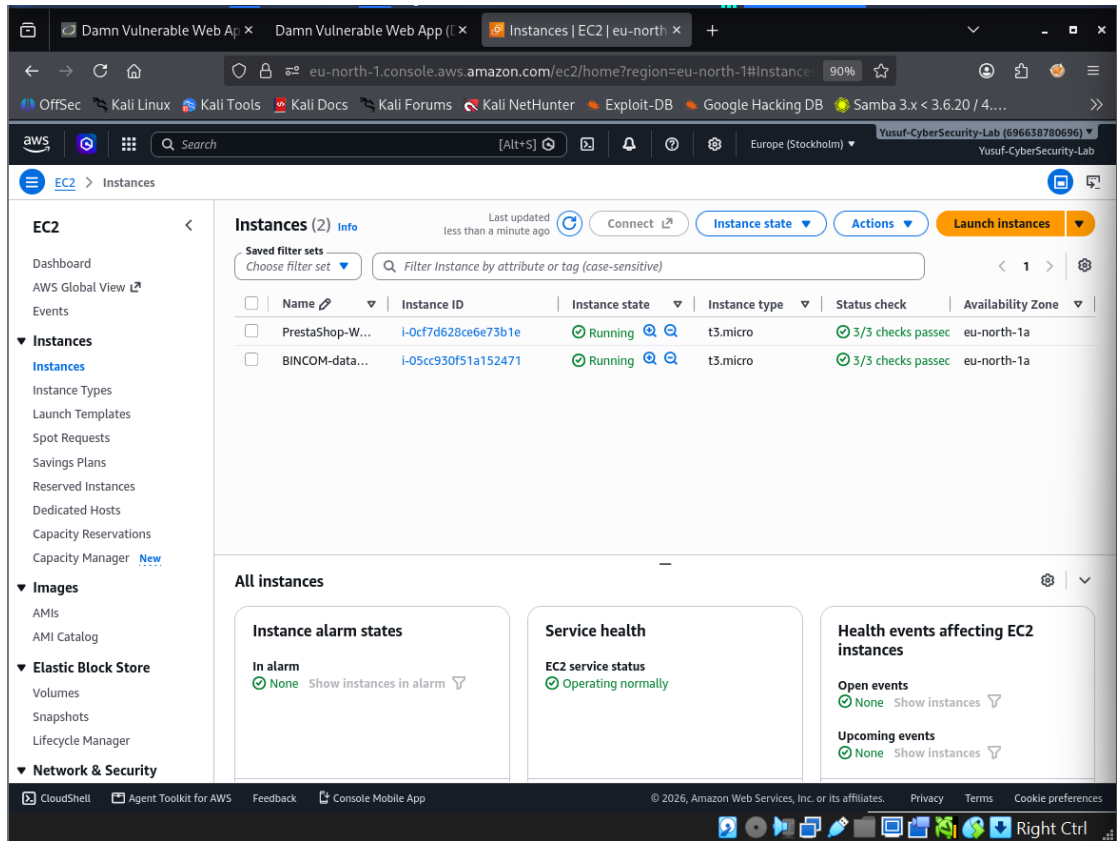
Nmap done: 1 IP address (1 host up) scanned in 26.74 seconds

(kali@kali)-[~]
$ sudo nmap -sV -O 192.168.56.100
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-15 12:43 -0400
Nmap scan report for 192.168.56.100
Host is up (0.00085s latency).
All 1000 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:7B:75:0C (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 2N Helios IP VoIP doorbell (96%), Advanced Illumination DCS-100E lighting controller (96%), AudioControl D3400 network amplifier (96%), British Gas GS-23 data logger (96%), Chamberlain myQ garage door opener (96%), Daikin DKN Cloud Wi-Fi Adaptor (96%), Daysequerra M4.2SI radio (96%), Denver Electronics AC-5000W MK2 camera (96%), Eve Cam (lwIP 2.1.0 - 2.2.0) (96%), Fatek FBS-CBEH PLC Ethernet communication board (96%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

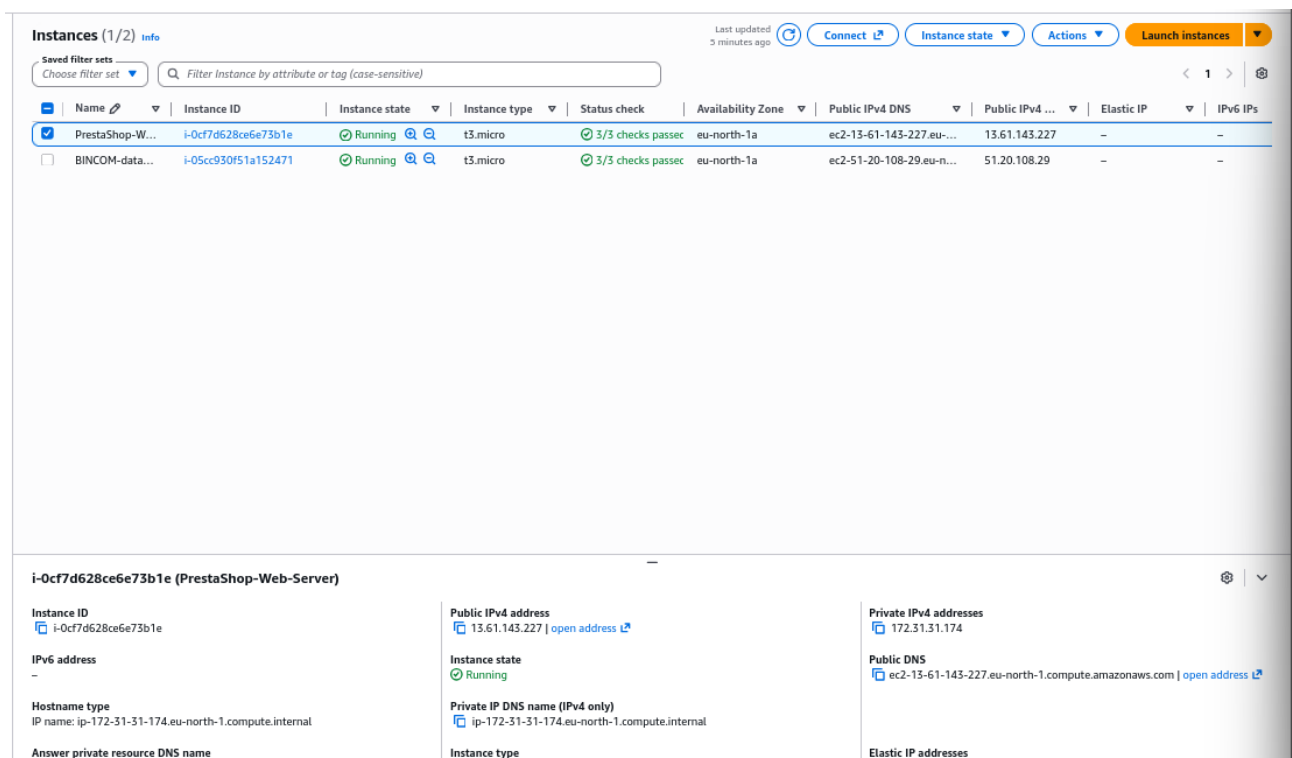
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.76 seconds

(kali@kali)-[~]
$
```

Archive screenshot: Screenshot 2026-09-15 174415.png



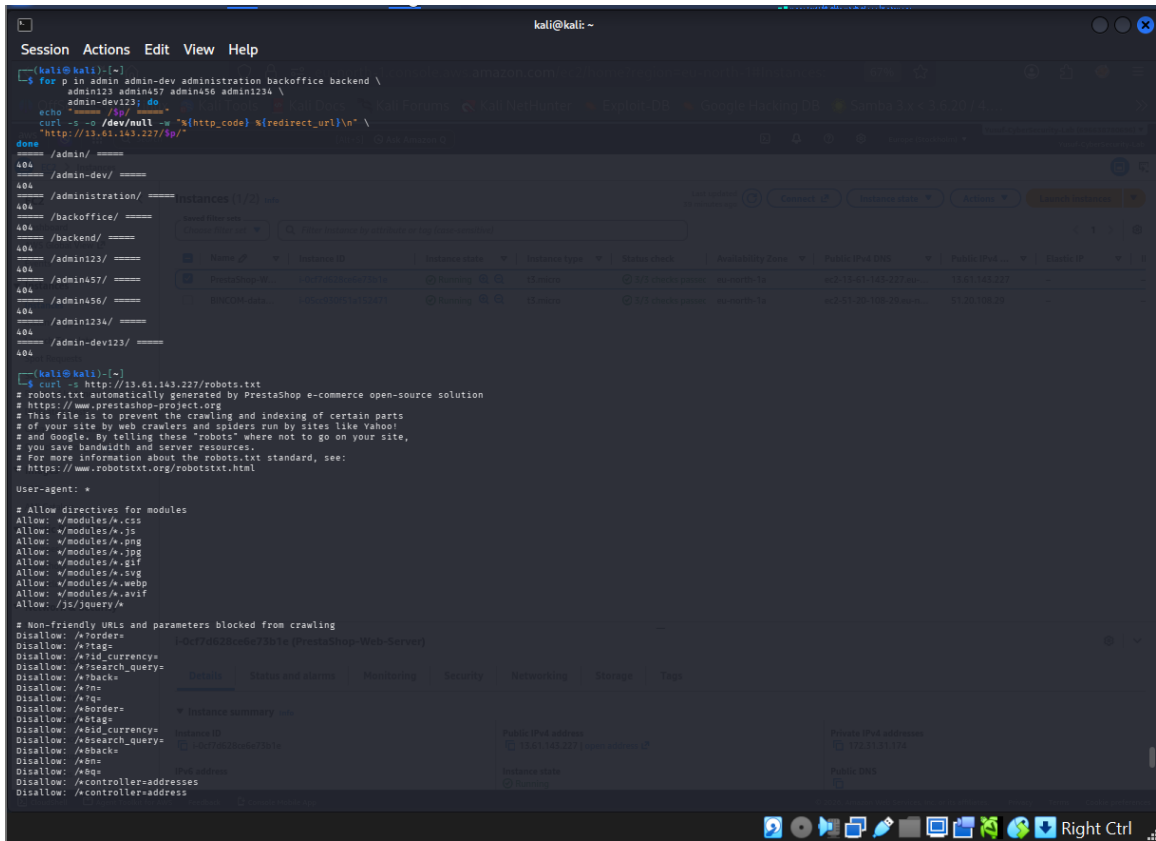
Archive screenshot: Screenshot 2026-09-15 175057.png



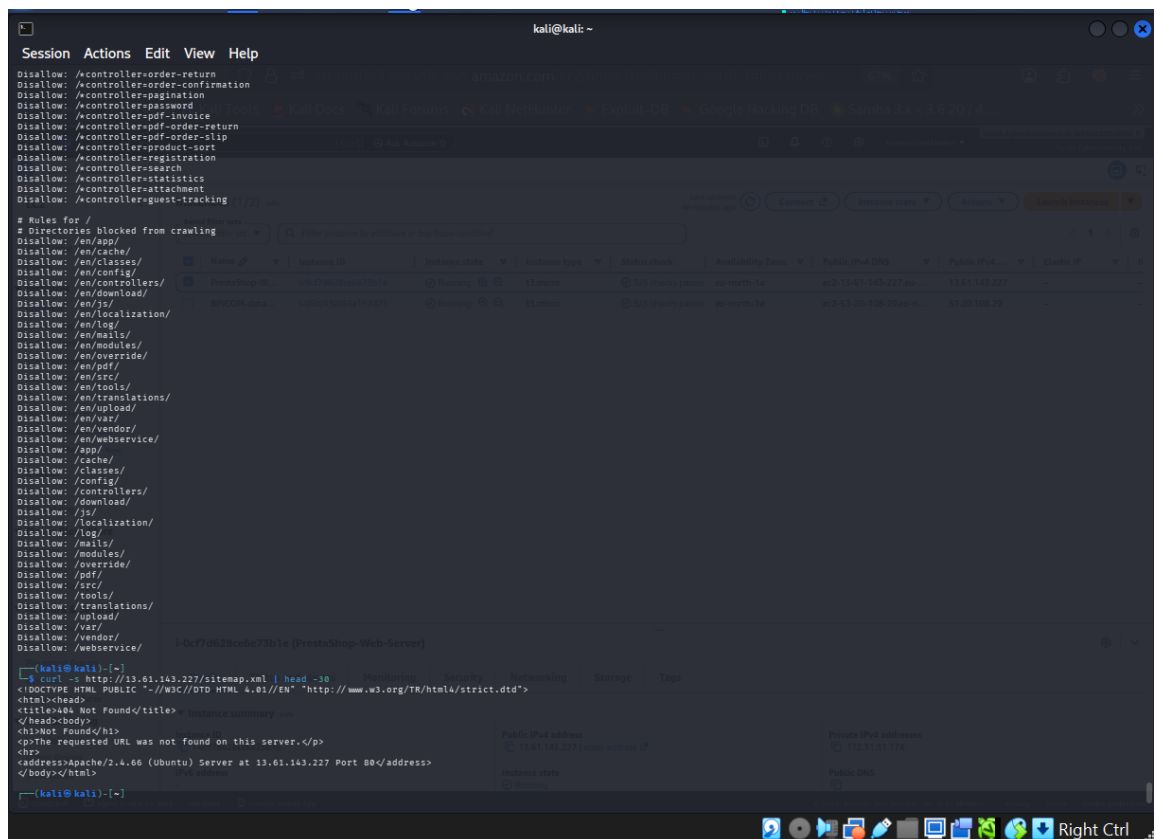
Archive screenshot: Screenshot 2026-09-15 175521.png



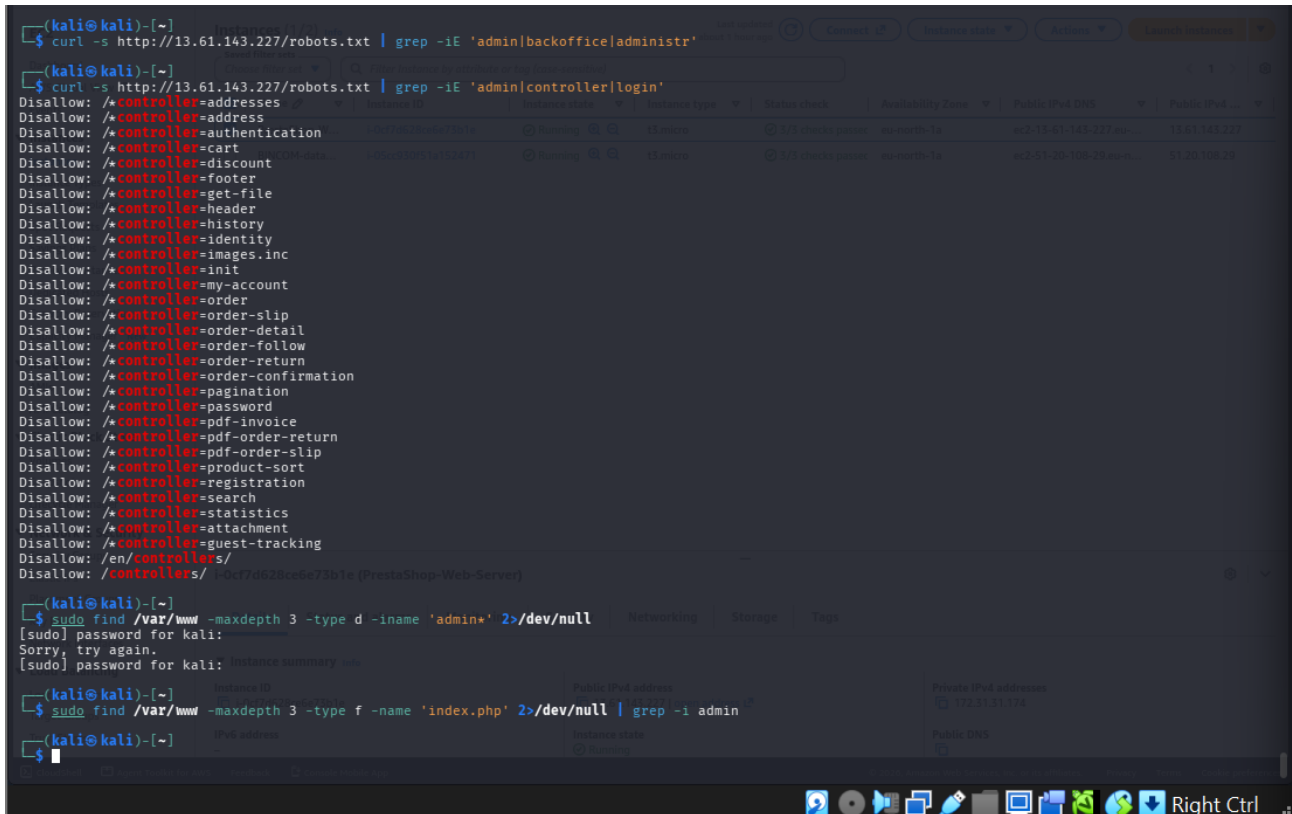
Archive screenshot: Screenshot 2026-09-15 175550.png



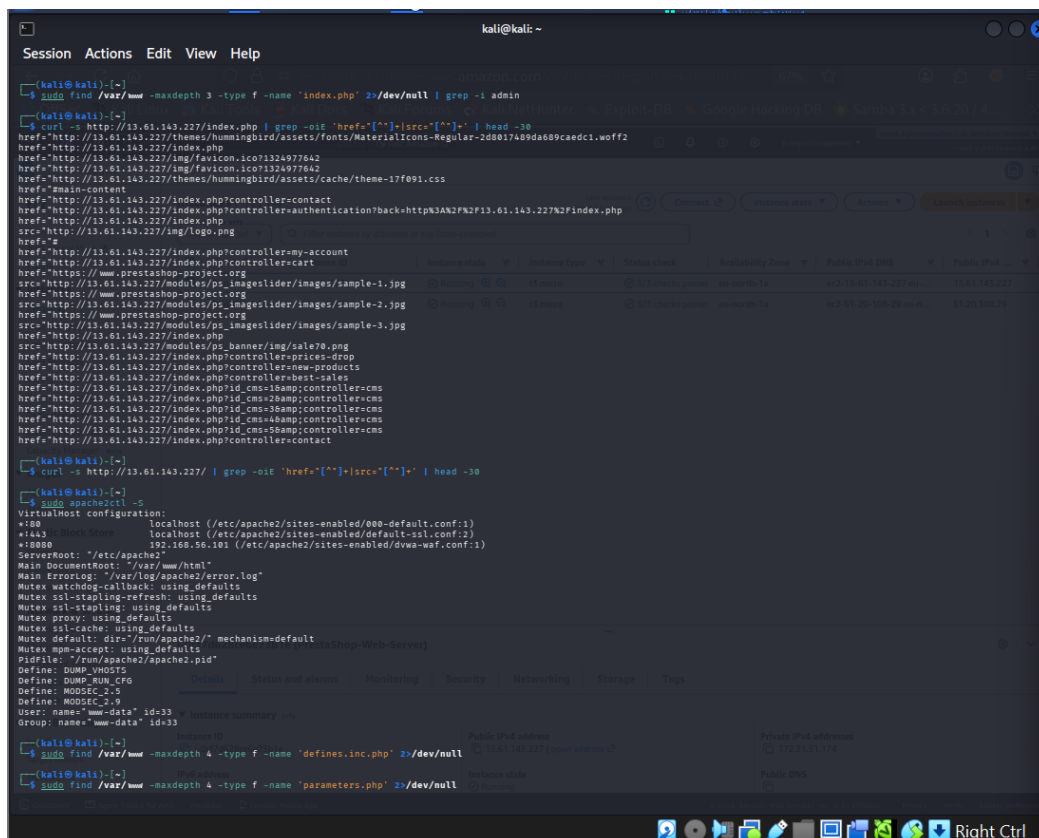
Archive screenshot: Screenshot 2026-09-15 182942.png



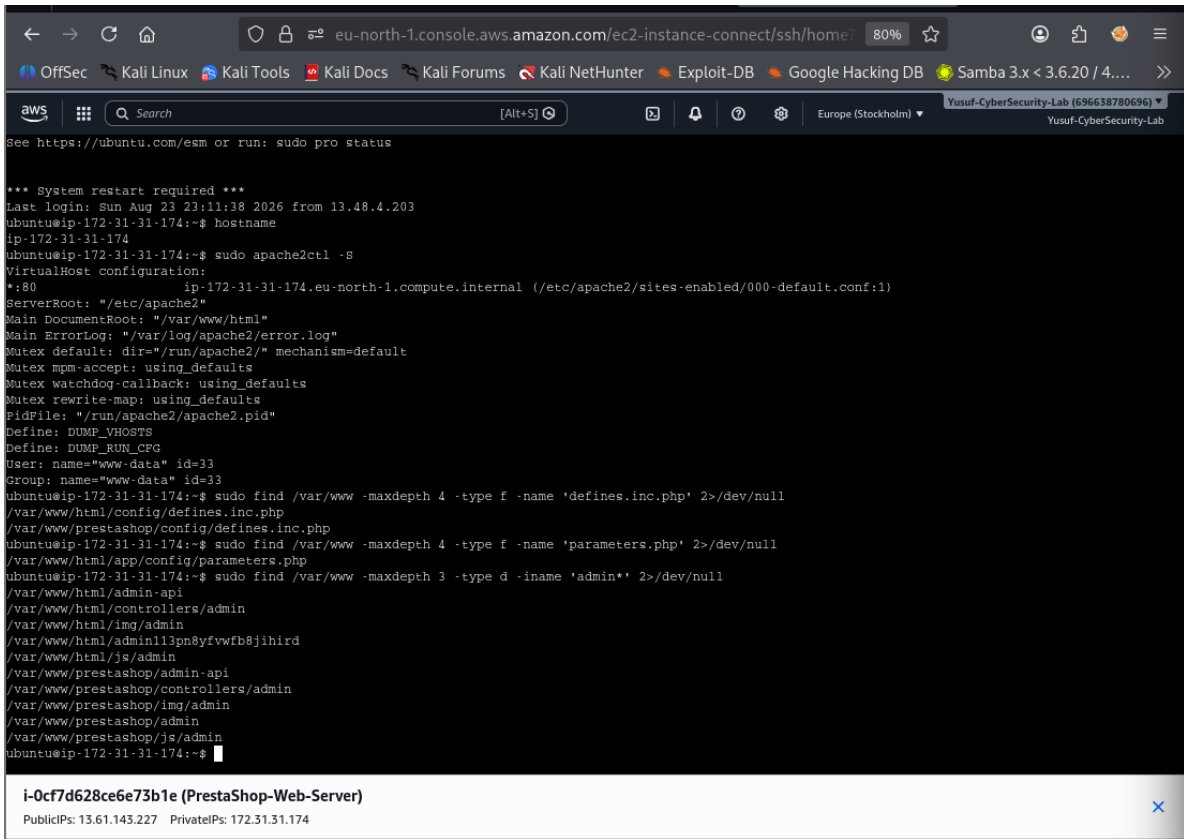
Archive screenshot: Screenshot 2026-09-15 183026.png



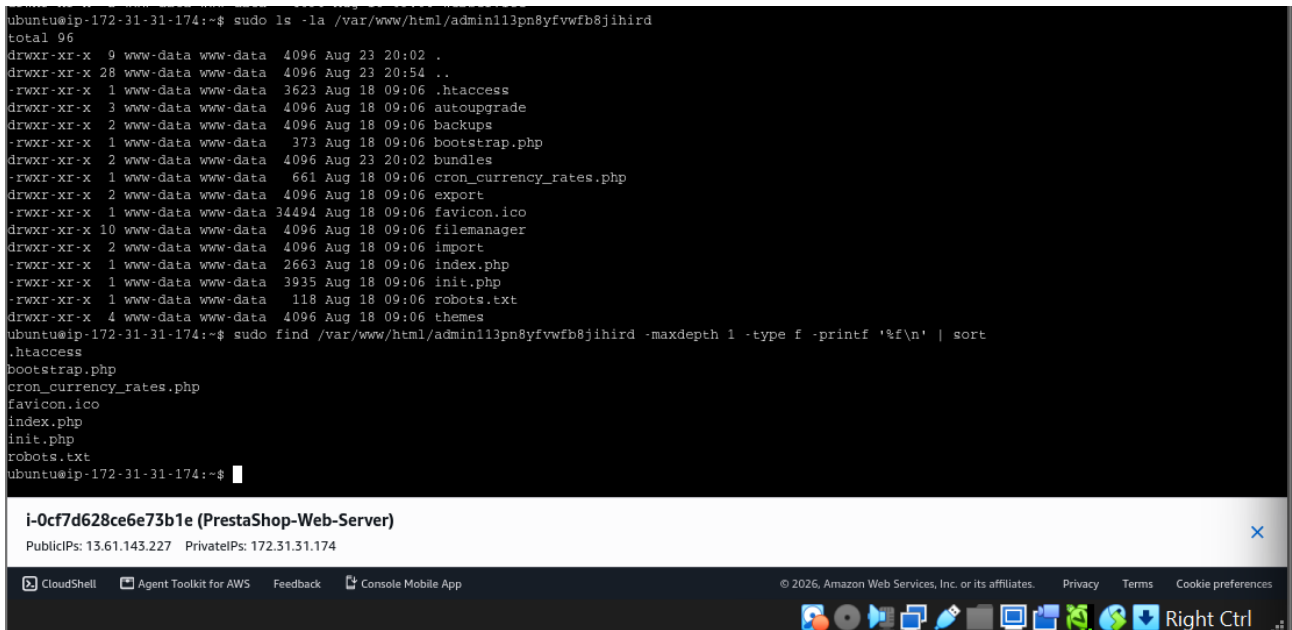
Archive screenshot: Screenshot 2026-09-15 184128.png



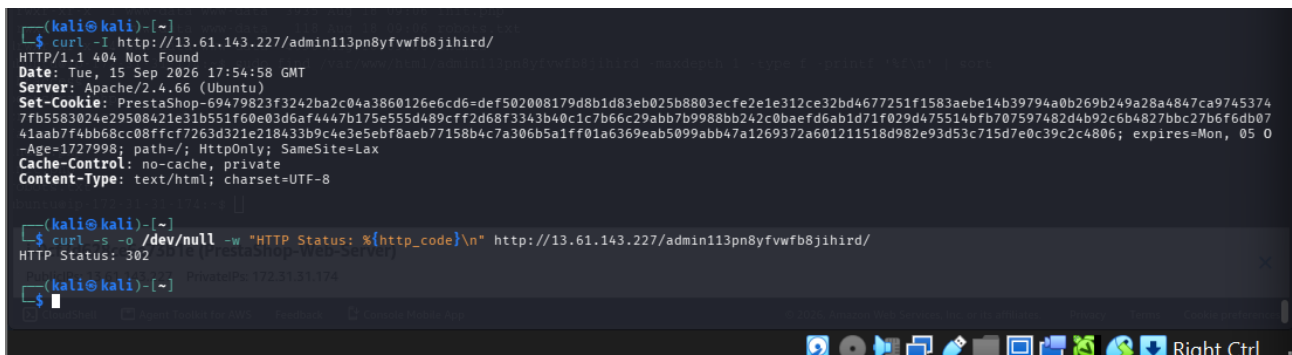
Archive screenshot: Screenshot 2026-09-15 184424.png



Archive screenshot: Screenshot 2026-09-15 185047.png



Archive screenshot: Screenshot 2026-09-15 185408.png



Archive screenshot: Screenshot 2026-09-15 185522.png

```
(kali@kali)-[~]
└─$ curl -i http://13.61.143.227/admin113pn8yfvwfb8jihird/index.php
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 17:56:59 GMT
Server: Apache/2.4.66 (Ubuntu)
Set-Cookie: PrestaShop-69479823f3242ba2c04a3860126e6cd6=def50200a201aff6bc1e24da7198a17718308a8476d4a4b0ddd779a6d0ef21d3801039bee691cc1a337d245c49f9cb1d04fbc2686b7ad7301da0525e19f0992b67428d7619dd10e565942ef9e7e8ebb1faf302bd95363ca0f587d266ceec25d7971bffb2b4b1c063102b91c25b655e79715019f4f9cfcd8828b87baa46bdf2e819c4d6dd3b429eccc3385f654737347024434797b7a788eb6c966af69a03511db021c3c3292a069010b2bfd71200a8343da6d0a67f7760252de8a06d6d700aa3dbec1490187cf49748cc599e5a2a3646e531293a27620aa55; expires=Mon, 05 Oct 2026 17:56:59 GMT; Max-Age=1728000; path=/; HttpOnly; SameSite=Lax
Cache-Control: max-age=0, must-revalidate, private
Location: /admin113pn8yfvwfb8jihird/index.php/login?_token=
Expires: Tue, 15 Sep 2026 17:56:59 GMT
Transfer-Encoding: chunked
Content-Type: text/html; charset=utf-8

<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8" />
<meta http-equiv="refresh" content="0;url='/admin113pn8yfvwfb8jihird/index.php/login?_token=' />
<title>Redirecting to /admin113pn8yfvwfb8jihird/index.php/login?_token=</title>
</head>
<body>
Redirecting to <a href="/admin113pn8yfvwfb8jihird/index.php/login?_token=">/admin113pn8yfvwfb8jihird/index.php/login?_token=</a>.
</body>
</html>

(kali@kali)-[~]
└─$ curl -s -o /dev/null -w "HTTP Status: %{http_code}\n" http://13.61.143.227/admin113pn8yfvwfb8jihird/login.php
HTTP Status: 404
└─$
```

Archive screenshot: Screenshot 2026-09-15 185857.png

```
(kali@kali)-[~]
└─$ curl -s -L -o /tmp/admin-login.html -w "Final HTTP Status: %{http_code}\nFinal URL: %{url_effective}\n" http://13.61.143.227/admin113pn8yfvwfb8jihird/index.php
Final HTTP Status: 200
Final URL: http://13.61.143.227/admin113pn8yfvwfb8jihird/index.php/login?_token=

(kali@kali)-[~]
└─$ grep -iE '<title>|<form|password|login' /tmp/admin-login.html | head -20
<title>BINCOM PrestaShop Lab</title>
<link rel="stylesheet" href="/admin113pn8yfvwfb8jihird/themes/new-theme/public/login.css?9.1.5" media="all">
<script src="/admin113pn8yfvwfb8jihird/themes/new-theme/public/login_form.bundle.js?9.1.5"></script>
<div id="login">
  <div id="login-panel">
    <div id="login-header">
      <div id="login-content-card" class="card">
        <form method="post" action="/admin113pn8yfvwfb8jihird/index.php/login?_token=" id="login_form" class="" data-alerts-success="0" data-alerts-info="0" data-alerts-warning="0" data-alerts-error="0" data-form-submitted="0" data-form-valid="1">
          <div id="login_form">
            <div class="form-group password-widget">
              <label for="passwd" class="required">
                Password
              </label>
              <input type="password" id="passwd" name="passwd" required="required" class="form-control" />
            <div class="form-group submit-widget"><button type="submit" id="submit_login" name="submit_login" class="btn-primary btn btn">Log in</button></div>
            <a id="forgot-password-link" href="#forgotten_password" class="show-forgot-password btn btn-link px-0 align-right">
              I forgot my password</a>
            <form name="request_password_reset" method="post" action="/admin113pn8yfvwfb8jihird/index.php/request-password-reset?_token=" id="forgot_password_form" class="d-none" data-alerts-success="0" data-alerts-info="0" data-alerts-warning="0" data-alerts-error="0" data-form-submitted="0" data-form-valid="1">
              Forgot your password?
            </div>
            <div id="request_password_reset" id="forgot_password_form">
              <div class="form-group email-widget">
                <label for="request_password_reset_email_forgot" class="required">
                  <input type="email" id="request_password_reset_email_forgot" name="request_password_reset[email_forgot]" required="required" class="email-input form-control" data-invalid-message="Invalid email address." />
            </div>

(kali@kali)-[~]
└─$
```

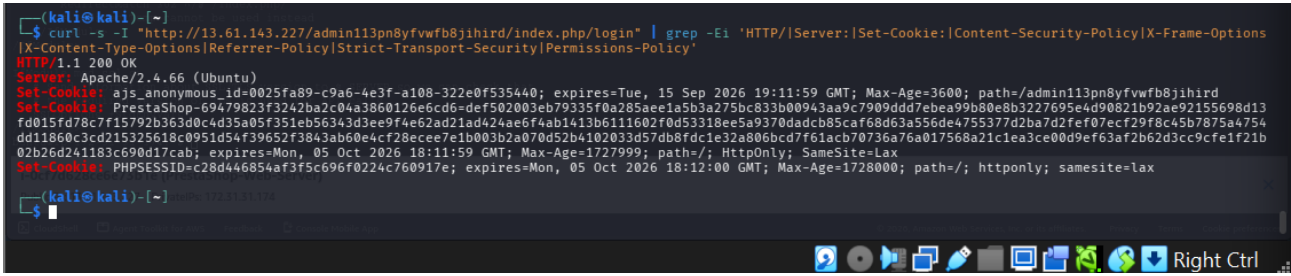
Archive screenshot: Screenshot 2026-09-15 190156.png

```
(kali@kali)-[~]
└─$ curl -s -L -o /tmp/admin-login.html -w "Final HTTP Status: %{http_code}\nFinal URL: %{url_effective}\n" http://13.61.143.227/admin113pn8yfvwfb8jihird/index.php
Final HTTP Status: 200
Final URL: http://13.61.143.227/admin113pn8yfvwfb8jihird/index.php/login?_token=

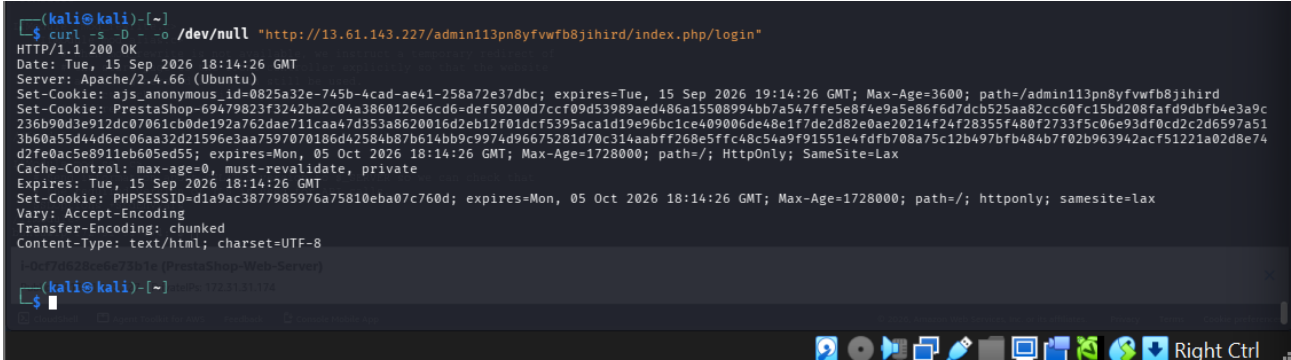
(kali@kali)-[~]
└─$ grep -iE '<title>|<form|password|login' /tmp/admin-login.html | head -20
<title>BINCOM PrestaShop Lab</title>
<link rel="stylesheet" href="/admin113pn8yfvwfb8jihird/themes/new-theme/public/login.css?9.1.5" media="all">
<script src="/admin113pn8yfvwfb8jihird/themes/new-theme/public/login_form.bundle.js?9.1.5"></script>
<div id="login">
  <div id="login-panel">
    <div id="login-header">
      <div id="login-content-card" class="card">
        <form method="post" action="/admin113pn8yfvwfb8jihird/index.php/login?_token=" id="login_form" class="" data-alerts-success="0" data-alerts-info="0" data-alerts-warning="0" data-alerts-error="0" data-form-submitted="0" data-form-valid="1">
          <div id="login_form">
            <div class="form-group password-widget">
              <label for="passwd" class="required">
                Password
              </label>
              <input type="password" id="passwd" name="passwd" required="required" class="form-control" />
            <div class="form-group submit-widget"><button type="submit" id="submit_login" name="submit_login" class="btn-primary btn btn">Log in</button></div>
            <a id="forgot-password-link" href="#forgotten_password" class="show-forgot-password btn btn-link px-0 align-right">
              I forgot my password</a>
            <form name="request_password_reset" method="post" action="/admin113pn8yfvwfb8jihird/index.php/request-password-reset?_token=" id="forgot_password_form" class="d-none" data-alerts-success="0" data-alerts-info="0" data-alerts-warning="0" data-alerts-error="0" data-form-submitted="0" data-form-valid="1">
              Forgot your password?
            </div>
            <div id="request_password_reset" id="forgot_password_form">
              <div class="form-group email-widget">
                <label for="request_password_reset_email_forgot" class="required">
                  <input type="email" id="request_password_reset_email_forgot" name="request_password_reset[email_forgot]" required="required" class="email-input form-control" data-invalid-message="Invalid email address." />
            </div>

(kali@kali)-[~]
└─$
```

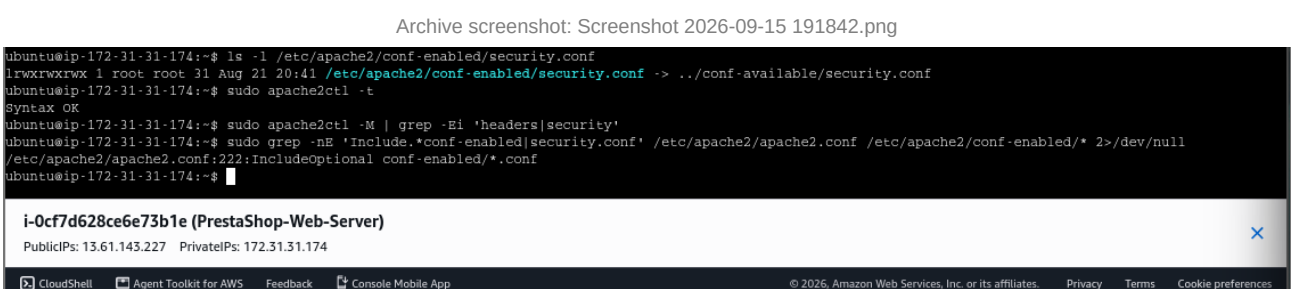
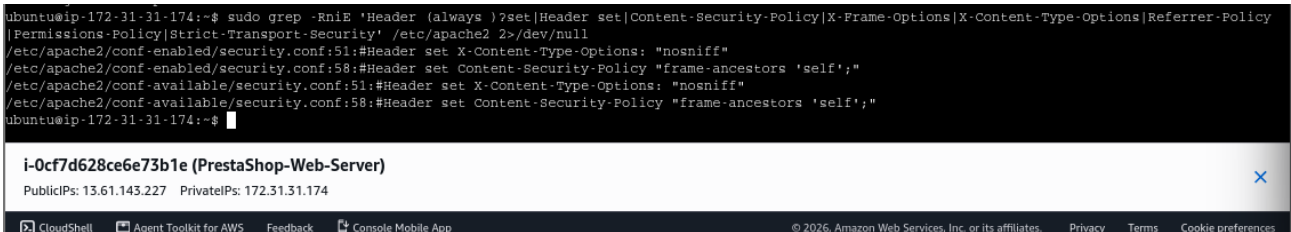
Archive screenshot: Screenshot 2026-09-15 190439(1).png



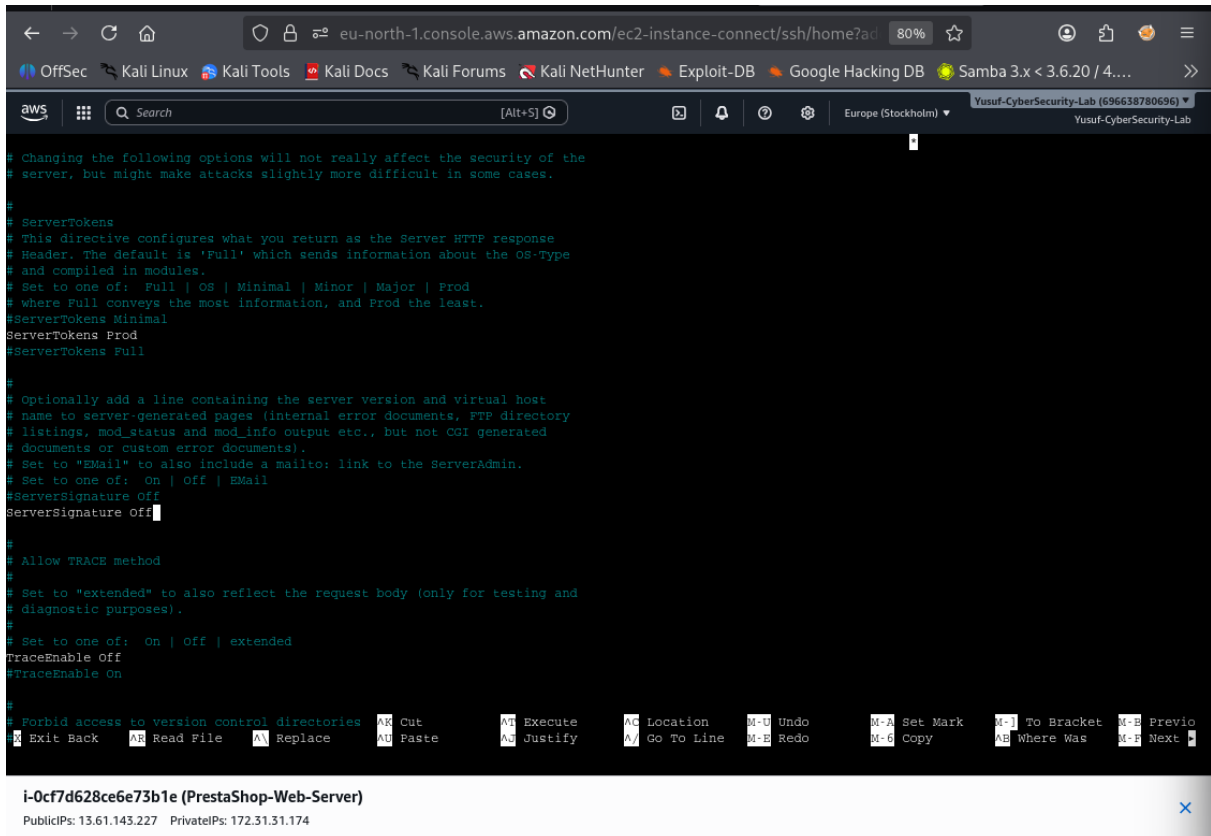
Archive screenshot: Screenshot 2026-09-15 191221.png



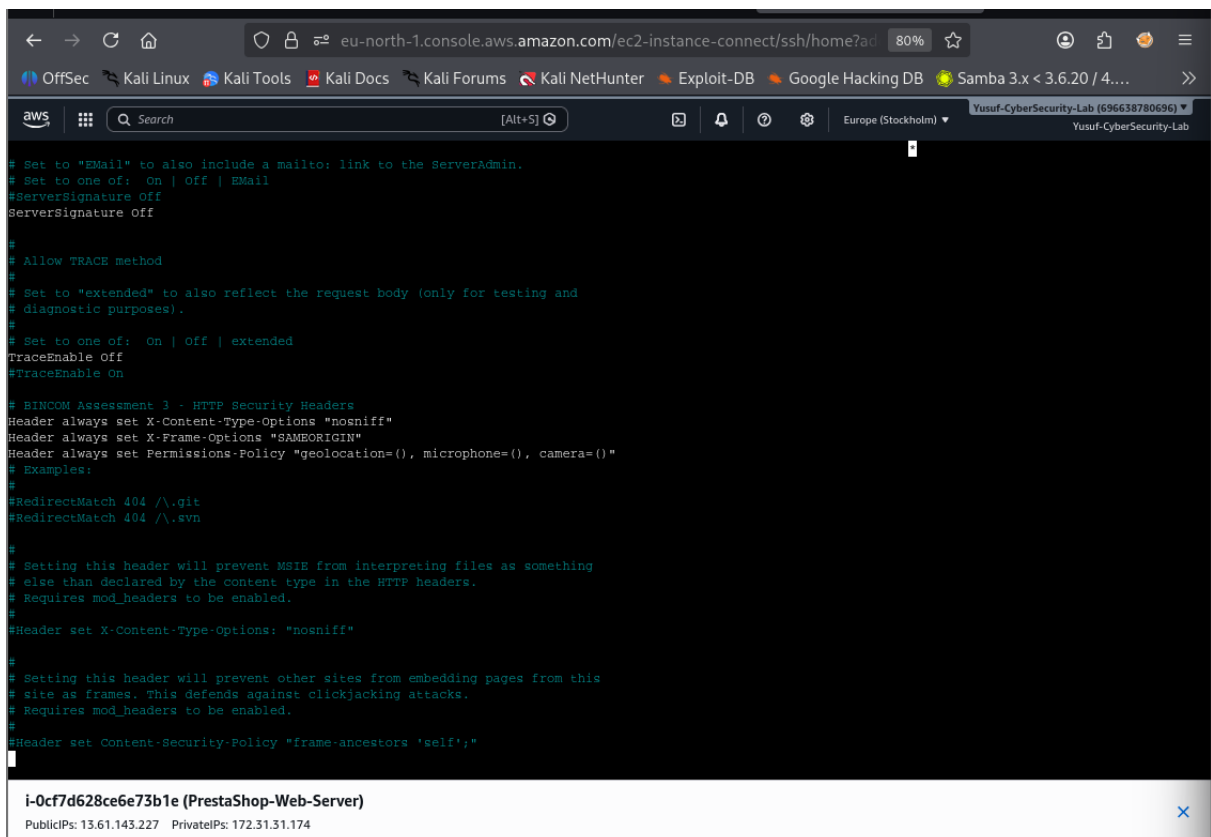
Archive screenshot: Screenshot 2026-09-15 191444.png



Archive screenshot: Screenshot 2026-09-15 192132.png



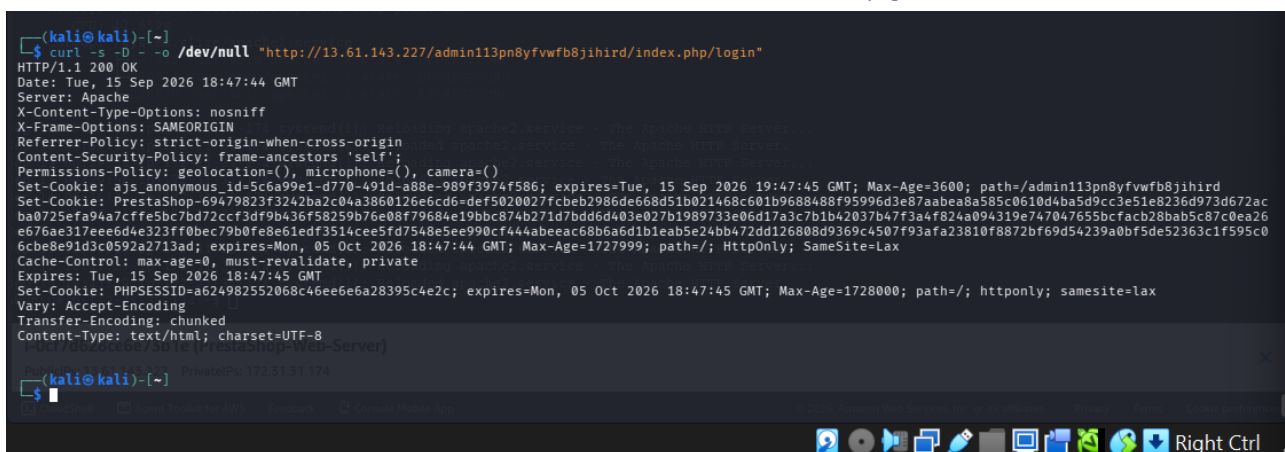
Archive screenshot: Screenshot 2026-09-15 193231.png



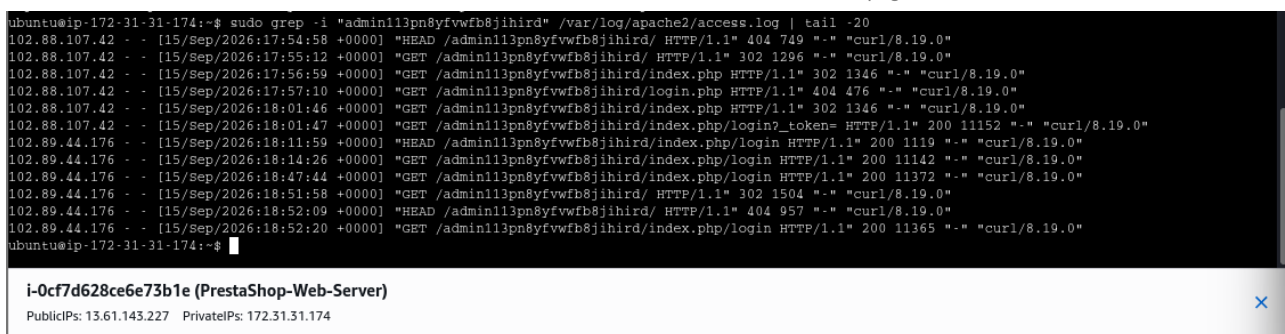
Archive screenshot: Screenshot 2026-09-15 194023.png



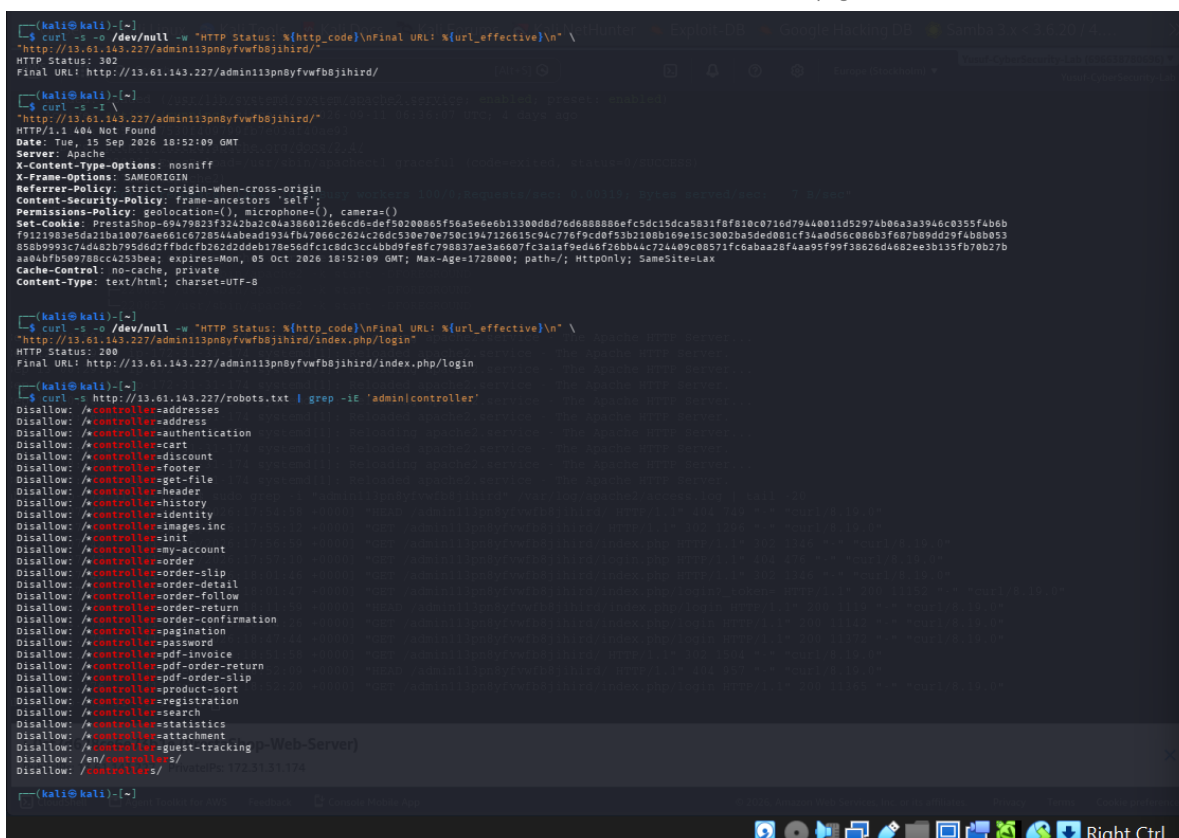
Archive screenshot: Screenshot 2026-09-15 194419.png



Archive screenshot: Screenshot 2026-09-15 194810.png



Archive screenshot: Screenshot 2026-09-15 195353.png



```

ubuntu@ip-172-31-31-174:~$ sudo grep -nE '\$_PS_VERSION_' /var/www/html/config/defines.inc.php
ubuntu@ip-172-31-31-174:~$ sudo grep -nE 'version|name' /var/www/html/config/defines.inc.php | head
57:currentDir = dirname($_FILE__);
ubuntu@ip-172-31-31-174:~$ sudo find /var/www/html/modules -maxdepth 2 -type f -name '*.php' 2>/dev/null | head -40
/var/www/html/modules/dashtrends/dashtrends.php
/var/www/html/modules/dashtrends/index.php
/var/www/html/modules/ps_facebook/ps_facebook.php
/var/www/html/modules/ps_facebook/index.php
/var/www/html/modules/ps_facebook.php.cs-fixer.dist.php
/var/www/html/modules/ps_classic_edition/ps_classic_edition.php
/var/www/html/modules/ps_classic_edition/index.php
/var/www/html/modules/statsbestcategories/statsbestcategories.php
/var/www/html/modules/statsbestcategories/index.php
/var/www/html/modules/ps_themecusto/ps_themecusto.php
/var/www/html/modules/ps_themecusto/index.php
/var/www/html/modules/ps_viewedproduct/ps_viewedproduct.php
/var/www/html/modules/ps_viewedproduct/index.php
/var/www/html/modules/statsbestvouchers/index.php
/var/www/html/modules/statsbestvouchers/statsbestvouchers.php
/var/www/html/modules/statsproduct/statsproduct.php
/var/www/html/modules/statsproduct/index.php
/var/www/html/modules/statsstock/index.php
/var/www/html/modules/statsstock/statsstock.php
/var/www/html/modules/ps_categorytree/index.php
/var/www/html/modules/ps_categorytree/ps_categorytree.php
/var/www/html/modules/contactform/contactform.php
/var/www/html/modules/contactform/index.php
/var/www/html/modules/ps_emailsubscription/ps_emailsubscription.php
/var/www/html/modules/ps_emailsubscription/index.php
/var/www/html/modules/ps_faviconnotificationbo/ps_faviconnotificationbo.php
/var/www/html/modules/ps_faviconnotificationbo/index.php
/var/www/html/modules/statscheckup/statscheckup.php
/var/www/html/modules/statscheckup/index.php
/var/www/html/modules/ps_cashondelivery/ps_cashondelivery.php
/var/www/html/modules/ps_cashondelivery/index.php
/var/www/html/modules/graphvdi/graphvdi.php
/var/www/html/modules/graphvdi/index.php
/var/www/html/modules/ps_emailalerts/ps_emailalerts.php
/var/www/html/modules/ps_emailalerts/MailAlert.php
/var/www/html/modules/ps_emailalerts/index.php
/var/www/html/modules/ps_emailalerts/.php.cs-fixer.dist.php
/var/www/html/modules/dashactivity/index.php
/var/www/html/modules/dashactivity/dashactivity.php
/var/www/html/modules/statsforecast/statsforecast.php
ubuntu@ip-172-31-31-174:~$ sudo find /var/www/html/themes -maxdepth 2 -type f -name 'theme.yml' -o -name 'config.yml' 2>/dev/null
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-15 195943.png

```

ubuntu@ip-172-31-31-174:~$ sudo sed -n '1,120p' /var/www/html/modules/ps_currencyselector/ps_currencyselector.php | grep -nE 'version|name|author'
18:* versions in the future. If you wish to customize PrestaShop for your
21:* @author PrestaShop SA <contact@prestashop.com>
30:if (!defined('_PS_VERSION_')) {
40:     $this->name = 'ps_currencyselector';
42:     $this->author = 'PrestaShop';
43:     $this->version = '2.1.1';
48:     $this->displayName = $this->trans('Currency block', [], 'Modules.Currencyselector.Admin');
51:     $this->ps_versions_compliancy = ['min' => '1.7.5.0', 'max' => _PS_VERSION_];
67:     public function getWidgetVariables($hookName, array $configuration)
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-15 200747.png

```

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?ad 80%
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB Samba 3.x < 3.6.20 / 4....
aws Search [Alt+S] Europe (Stockholm) Yusuf-CyberSecurity-Lab (59638780696) Yusuf-CyberSecurity-Lab
#page-header-desc-configuration-switch_demo: hover, :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) > *:not(#main):not(#main-div) #desc-cart_rule-sa
ve-and-stay: hover,
/var/www/html/modules/ps_classic_edition/views/css/back.css:110: :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) .table thead .filter td .btn-de
fault: hover, :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) .toolbarBox .btn-toolbar .nav-pills #page-header-desc-configuration-switch_demo: ho
ver, :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) #desc-cart_rule-save-and-stay: hover,
/var/www/html/modules/ps_classic_edition/views/css/back.css:124: :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) > *:not(#main):not(#main-div) .table
thead .filter td .btn-default: hover, dropdown-toggle .caret, :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) > *:not(#main):not(#main-div) .toolbarBox
.btn-toolbar .nav-pills #page-header-desc-configuration-switch_demo: hover, dropdown-toggle .caret, :is(body:not(.ps-bo-rebrand)):is(body.no-smb-reskin) >
*:not(#main):not(#main-div) #desc-cart_rule-save-and-stay: hover, dropdown-toggle .caret,
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT id_customer,email,firstname,lastname,active FROM ps_customer;"
Enter password:
+-----+-----+-----+-----+-----+
| id_customer | email | firstname | lastname | active |
+-----+-----+-----+-----+-----+
| 1 | anonymous@psgdpr.com | Anonymous | Anonymous | 0 |
+-----+-----+-----+-----+-----+
ubuntu@ip-172-31-31-174:~$ sudo grep -nRiE "database_name|database_user|database_host" /var/www/html/app/config /var/www/html/config 2>/dev/null | head -3
0
sudo grep -nRiE "database_name|database_user|database_host" /var/www/html/app/config /var/www/html/config 2>/dev/null | head -3
0
/var/www/html/app/config/config_test.yml:29:     dbname: "test_${database_name}"
/var/www/html/app/config/set_parameters.php:15:     'database_host' => '',
/var/www/html/app/config/set_parameters.php:17:     'database_name' => '',
/var/www/html/app/config/set_parameters.php:18:     'database_user' => '',
/var/www/html/app/config/parameters.yml.dist:5:     database_host: 127.0.0.1
/var/www/html/app/config/parameters.yml.dist:7:     database_name: prestashop
/var/www/html/app/config/parameters.yml.dist:8:     database_user: root
/var/www/html/app/config/parameters.php:4:     'database_host' => '127.0.0.1',
/var/www/html/app/config/parameters.php:6:     'database_name' => 'prestashop',
/var/www/html/app/config/parameters.php:7:     'database_user' => 'webuser',
/var/www/html/app/config/doctrine.yml:10:     host: "%database_host%"
/var/www/html/app/config/doctrine.yml:12:     dbname: "%database_name%"
/var/www/html/app/config/doctrine.yml:13:     user: "%database_user%"
/var/www/html/config/bootstrap.php:72:     $database_host = $config['parameters']['database_host'];
/var/www/html/config/bootstrap.php:75:     $database_host .= ':' . $config['parameters']['database_port'];
/var/www/html/config/bootstrap.php:78:     define('_DB_SERVER_', $database_host);
/var/www/html/config/bootstrap.php:80:     define('_DB_NAME_', 'test_' . $config['parameters']['database_name']);
/var/www/html/config/bootstrap.php:82:     define('_DB_NAME_', $config['parameters']['database_name']);
/var/www/html/config/bootstrap.php:85:     define('_DB_USER_', $config['parameters']['database_user']);
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-15 201559.png

```

ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT id_employee,id_profile,id_lang,email,firstname,lastname,active,date_add FROM ps_employee;"
mysql -u root -p -e "USE prestashop; SELECT id_employee,id_profile,id_lang,email,firstname,lastname,active,date_add FROM ps_employee;"
Enter password:
SELECT id_employee,id_profile,id_lang,email,firstname,lastname,active,date_add FROM ps_employee
ERROR 1054 (42S22) at line 1: Unknown column 'date_add' in 'SELECT'
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT id_profile,name FROM ps_profile_lang;"
Enter password:
+-----+-----+
| id_profile | name |
+-----+-----+
| 1 | SuperAdmin |
+-----+-----+
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-15 202131.png

```

ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT id_employee,id_profile,id_lang,email,firstname,lastname,active FROM ps_employee;"
mysql -u root -p -e "USE prestashop; SELECT id_employee,id_profile,id_lang,email,firstname,lastname,active FROM ps_employee;"
Enter password:
+-----+-----+-----+-----+-----+-----+
| id_employee | id_profile | id_lang | email | firstname | lastname | active |
+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 1 | Yusufvic9@gmail.com | Yusuf | Oyewumi | 1 |
+-----+-----+-----+-----+-----+-----+
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; DESCRIBE ps_employee;"
Enter password:
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id_employee | int(10) unsigned | NO | PRI | NULL | auto_increment |
| id_profile | int(10) unsigned | NO | MUL | NULL |
| id_lang | int(10) unsigned | NO | MUL | 0 |
| lastname | varchar(255) | NO | NULL |
| firstname | varchar(255) | NO | NULL |
| email | varchar(255) | NO | MUL | NULL |
| passwd | varchar(255) | NO | NULL |
| last_passwd_gen | timestamp | NO | current_timestamp() |
| stats_date_from | date | YES | NULL |
| stats_date_to | date | YES | NULL |
| stats_compare_from | date | YES | NULL |
| stats_compare_to | date | YES | NULL |
| stats_compare_option | int(1) unsigned | NO | 1 |
| preselect_date_range | varchar(32) | YES | NULL |
| bo_color | varchar(32) | YES | NULL |
| bo_theme | varchar(32) | YES | NULL |
| bo_css | varchar(64) | YES | NULL |
| default_tab | int(10) unsigned | NO | 0 |
| bo_width | int(10) unsigned | NO | 0 |
| bo_menu | tinyint(1) | NO | 1 |
| active | tinyint(1) unsigned | NO | 0 |
| optin | tinyint(1) unsigned | YES | NULL |
| id_last_order | int(10) unsigned | NO | 0 |
| id_last_customer_message | int(10) unsigned | NO | 0 |
| id_last_customer | int(10) unsigned | NO | 0 |
| last_connection_date | date | YES | NULL |
| reset_password_token | varchar(40) | YES | NULL |
| reset_password_validity | datetime | YES | NULL |
| has_enabled_gravatar | tinyint(3) unsigned | NO | 0 |
+-----+-----+-----+-----+-----+-----+
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-15 202344.png

```

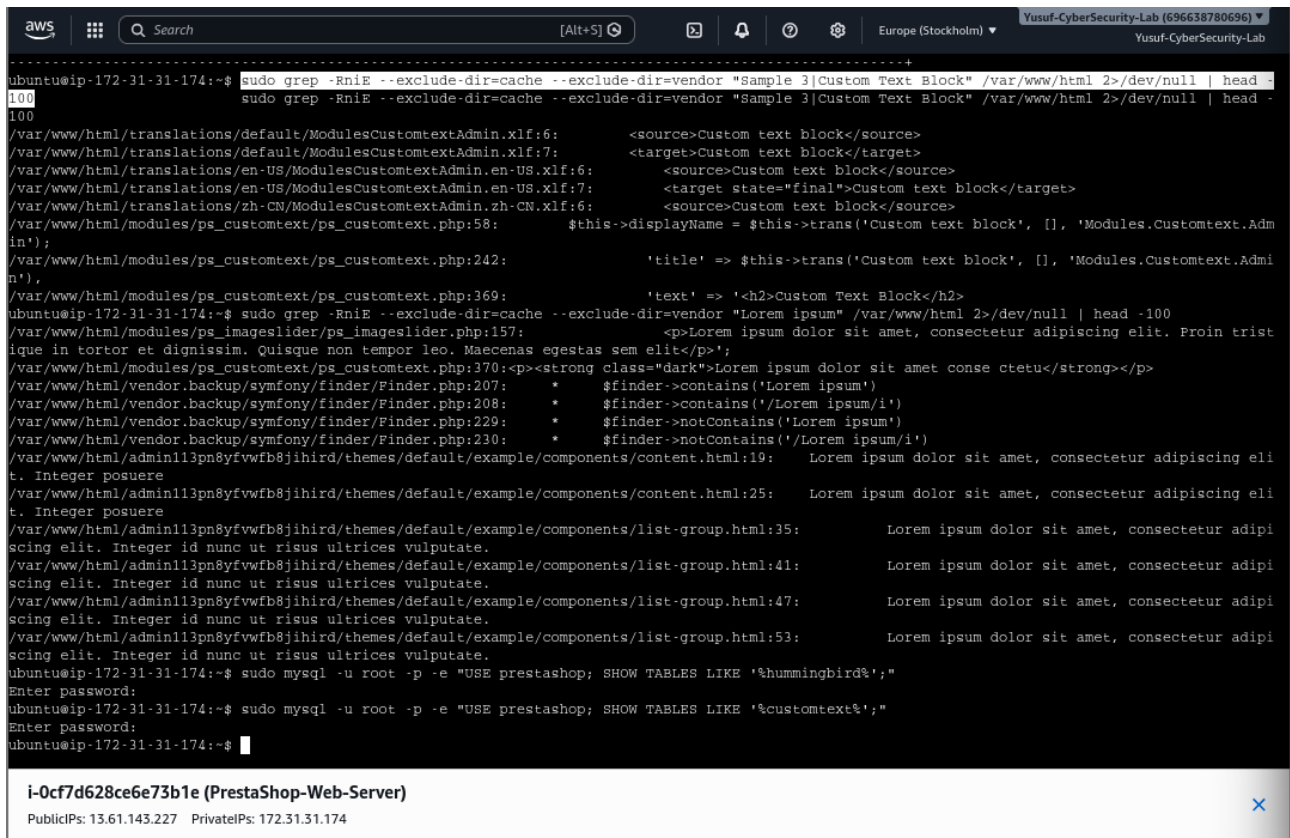
ubuntu@ip-172-31-31-174:~$ sudo grep -RniE "Sample 3|Custom Text Block|Lorem ipsum" /var/www/html/themes/hummingbird /var/www/html/themes/classic 2>/dev/null | head -80
sudo grep -RniE "Sample 3|Custom Text Block|Lorem ipsum" /var/www/html/themes/hummingbird /var/www/html/themes/classic 2>/dev/null | head -80
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT id_cms,id_lang,meta_title FROM ps_cms_lang WHERE content LIKE '%Lorem ipsum%' OR content LIKE '%Sample 3%' OR content LIKE '%Custom Text Block%';"
Enter password:
+-----+-----+-----+
| id_cms | id_lang | meta_title |
+-----+-----+-----+
| 3 | 1 | Terms and conditions of use |
| 4 | 1 | About us |
+-----+-----+-----+
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT c.id_category,cl.id_lang,cl.name,c.active FROM ps_category c JOIN ps_category_lang cl ON c.id_category=cl.id_category ORDER BY c.id_category;"
Enter password:
+-----+-----+-----+
| id_category | id_lang | name | active |
+-----+-----+-----+
| 1 | 1 | Root | 1 |
| 2 | 1 | Home | 1 |
+-----+-----+-----+
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT id_product,active,date_add FROM ps_product;"
Enter password:
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

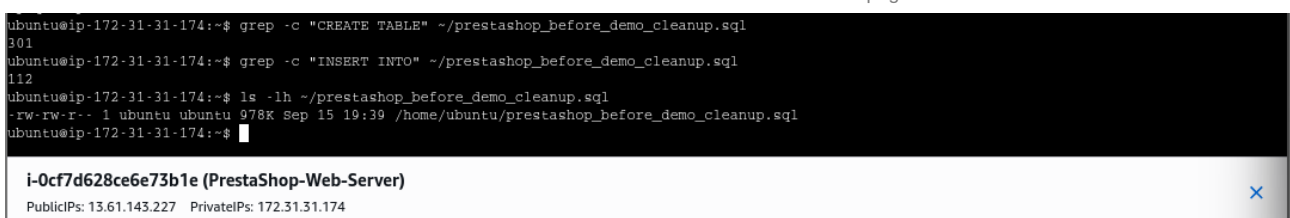
Archive screenshot: Screenshot 2026-09-15 202713.png



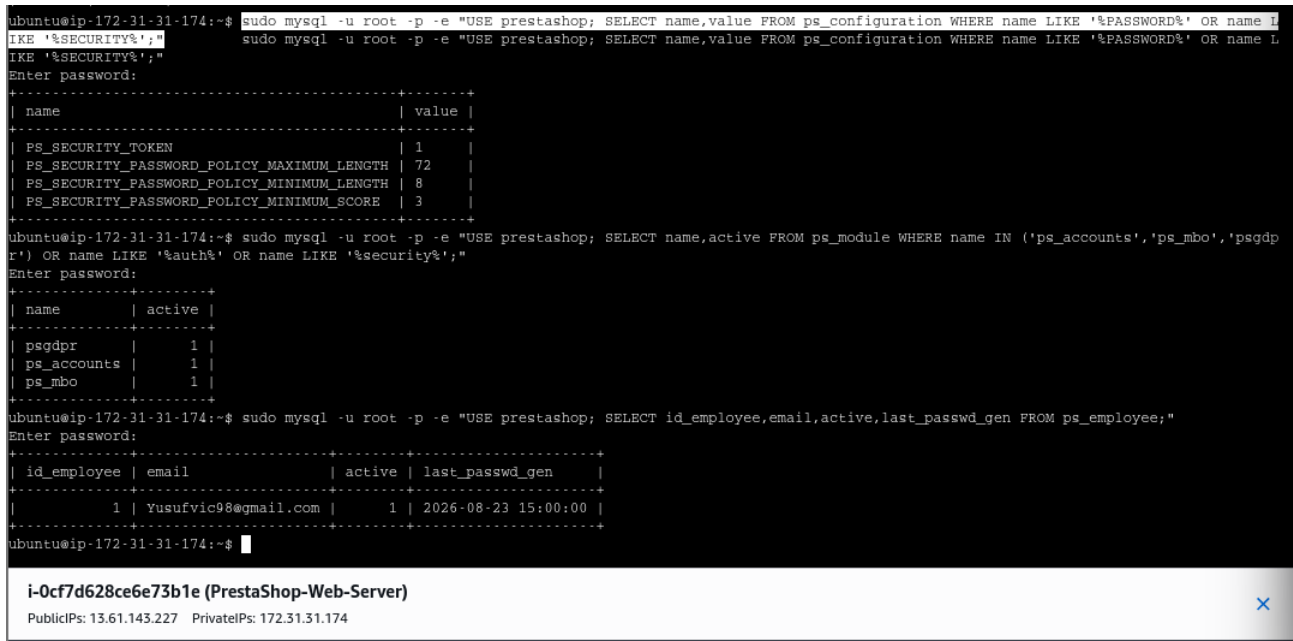
Archive screenshot: Screenshot 2026-09-15 203607.png



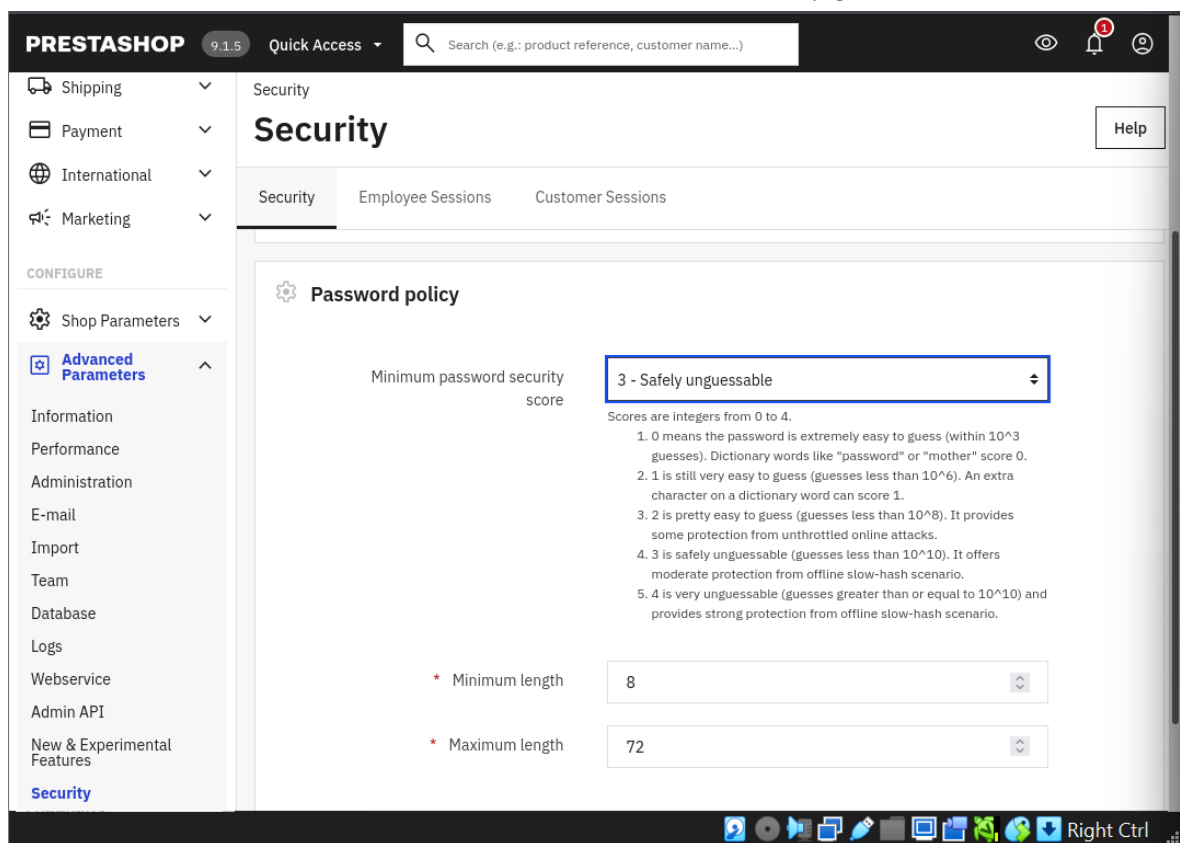
Archive screenshot: Screenshot 2026-09-15 204055.png



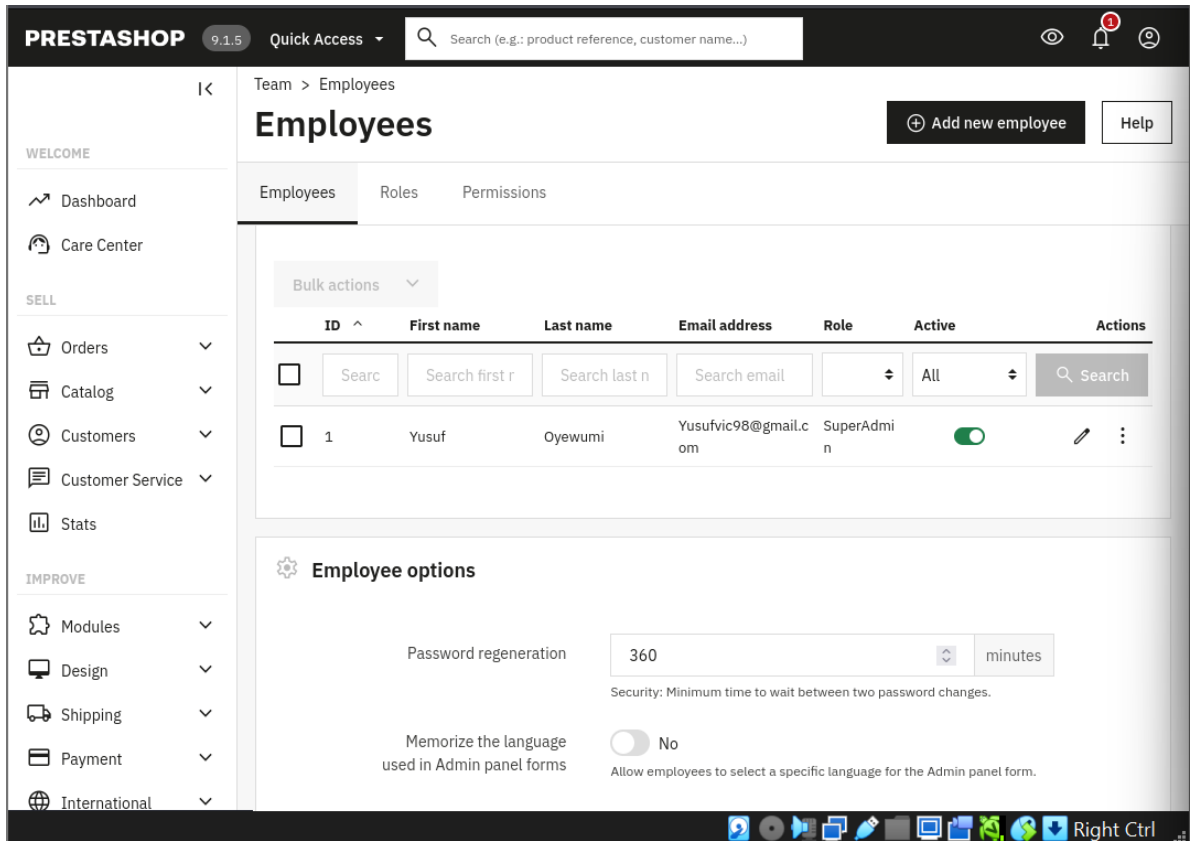
Archive screenshot: Screenshot 2026-09-15 204348.png



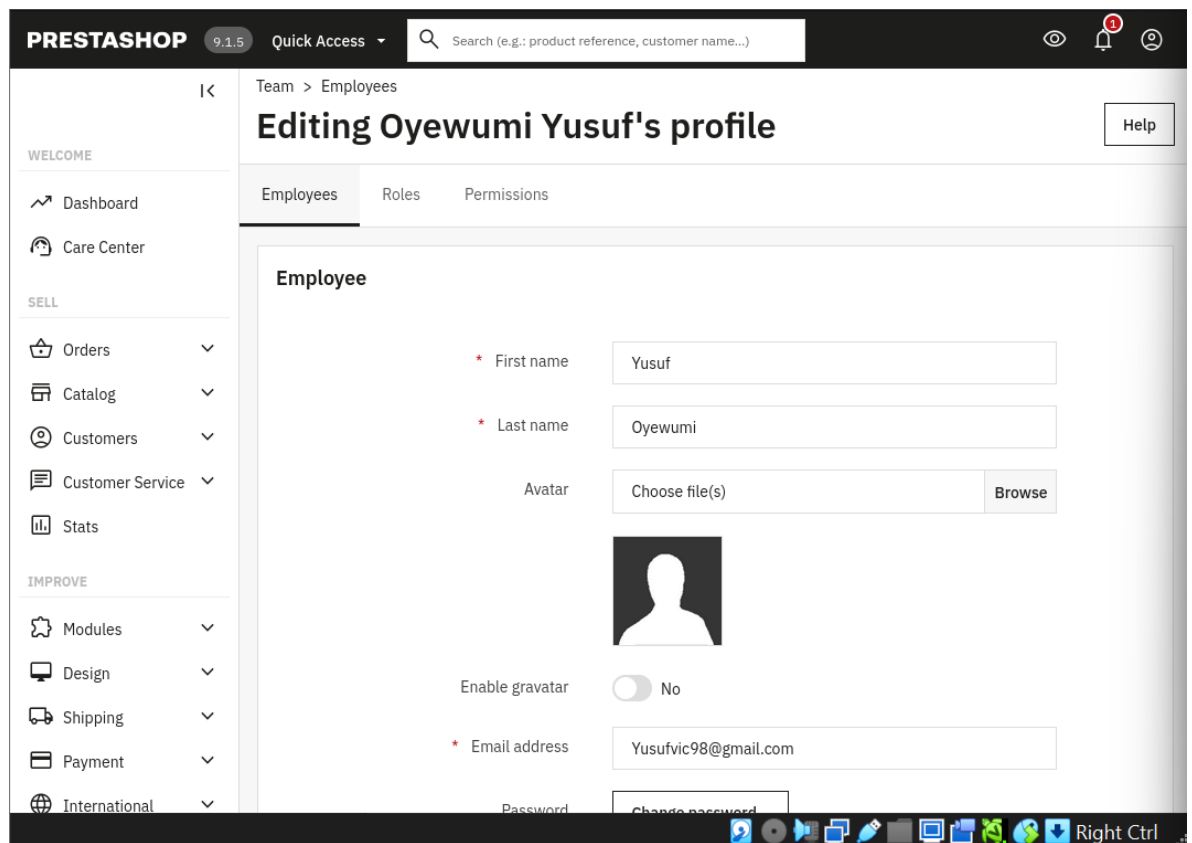
Archive screenshot: Screenshot 2026-09-15 205209.png



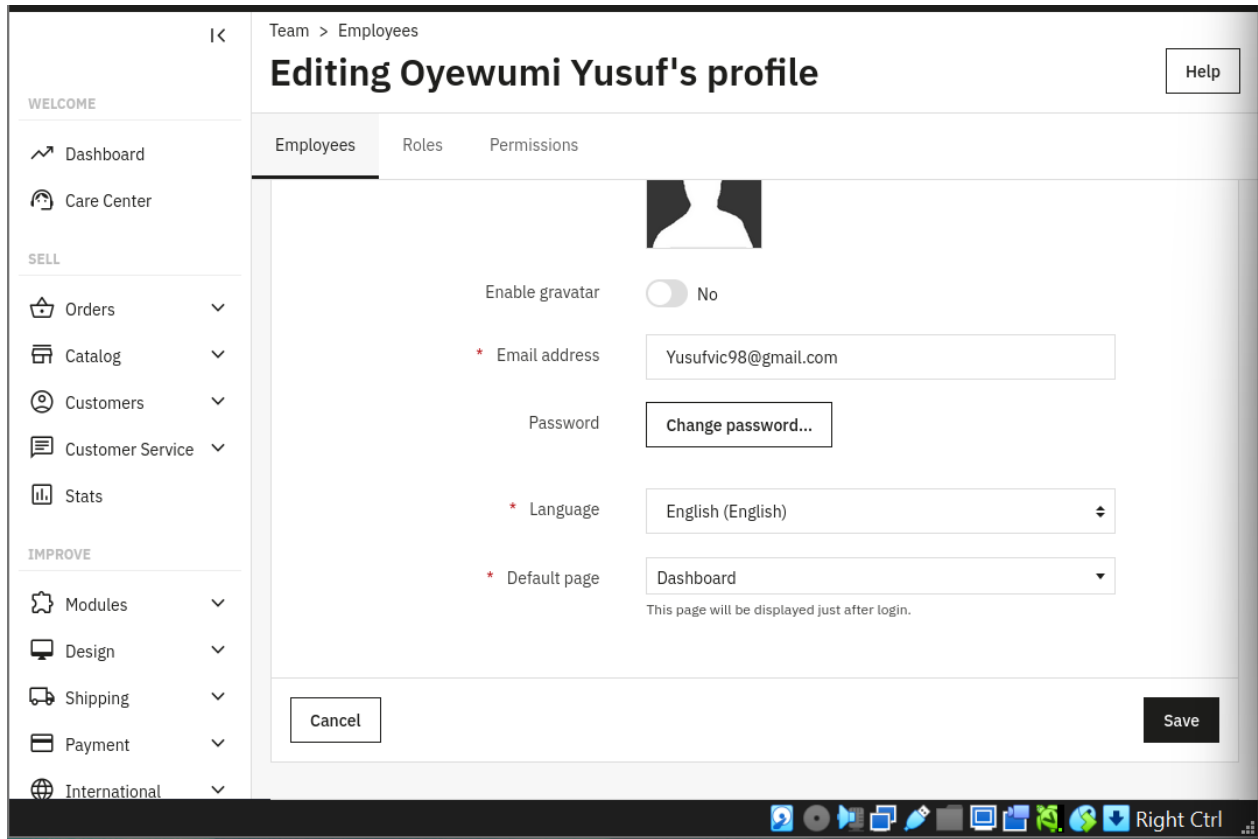
Archive screenshot: Screenshot 2026-09-15 205656.png



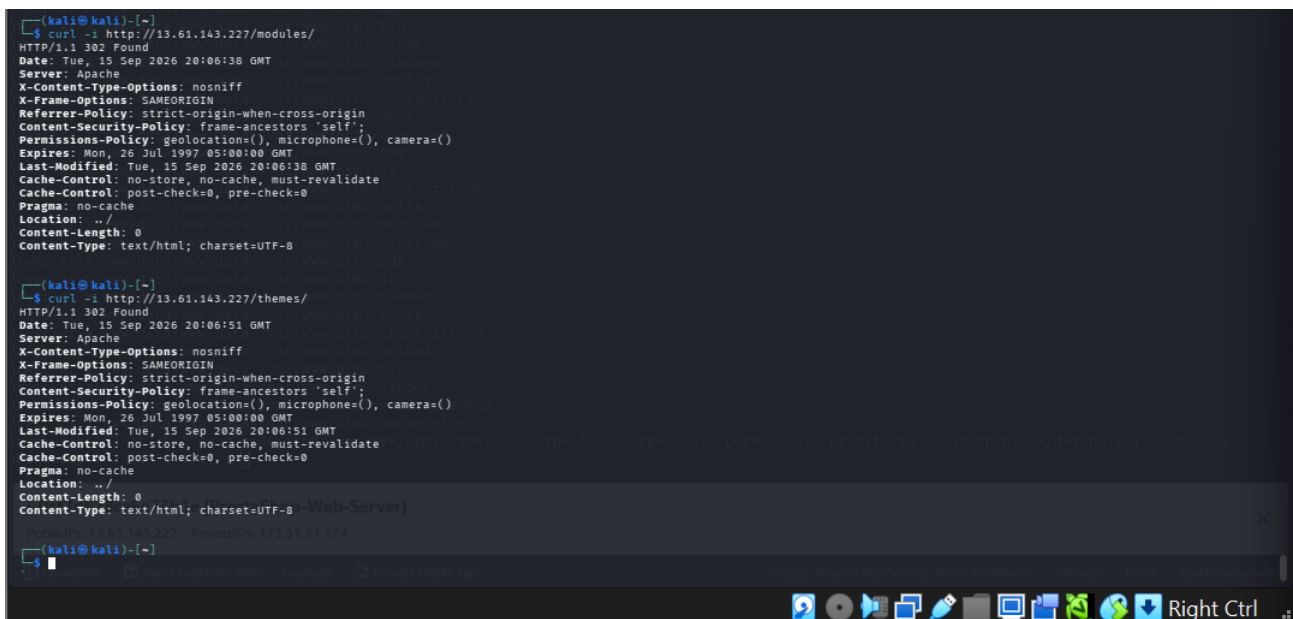
Archive screenshot: Screenshot 2026-09-15 210021.png



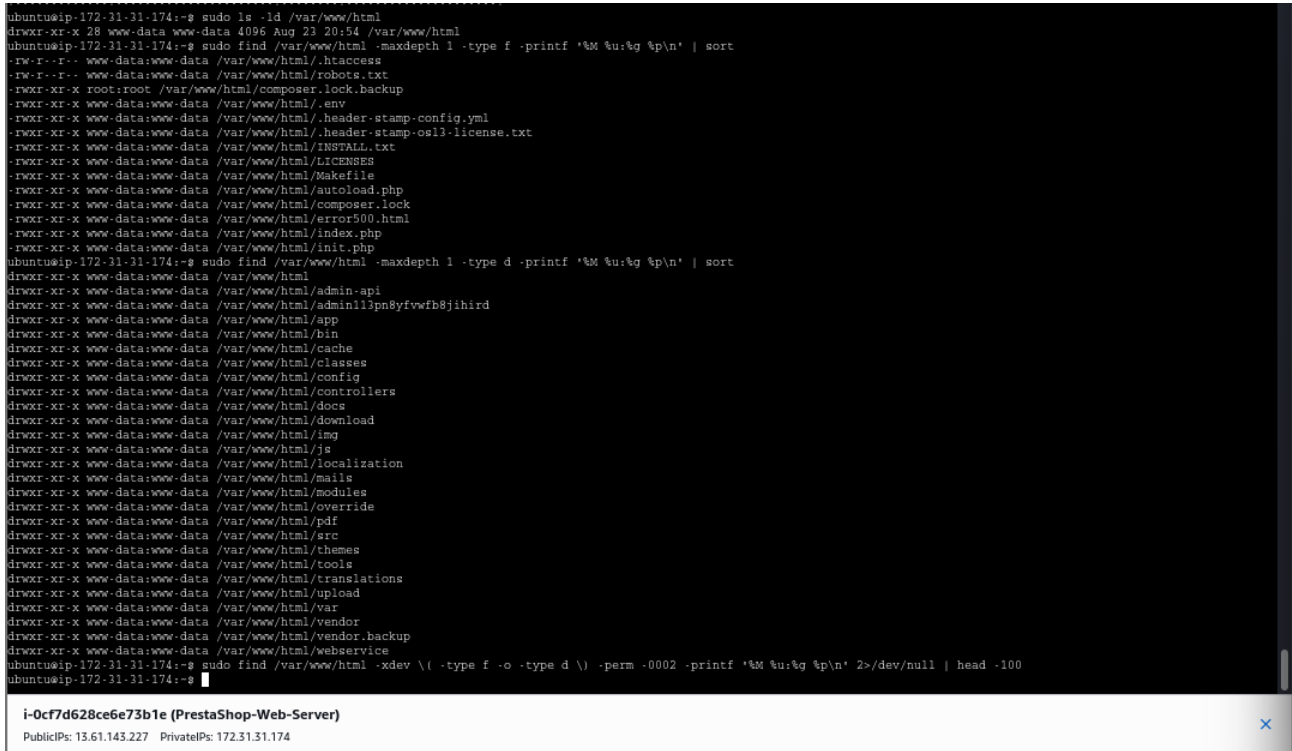
Archive screenshot: Screenshot 2026-09-15 210240.png



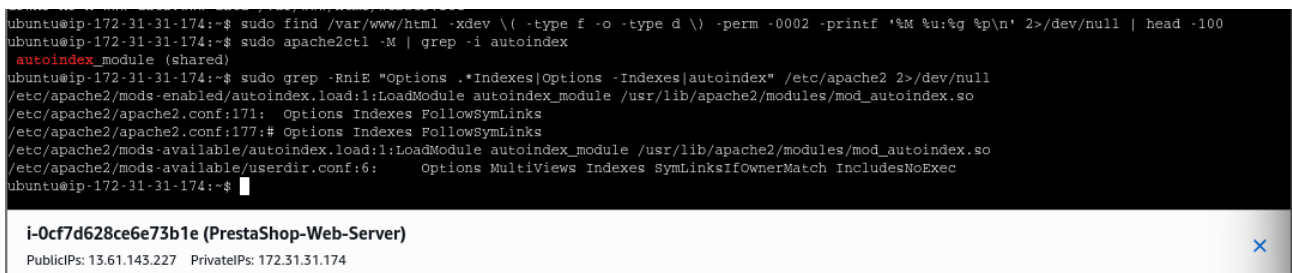
Archive screenshot: Screenshot 2026-09-15 210355.png



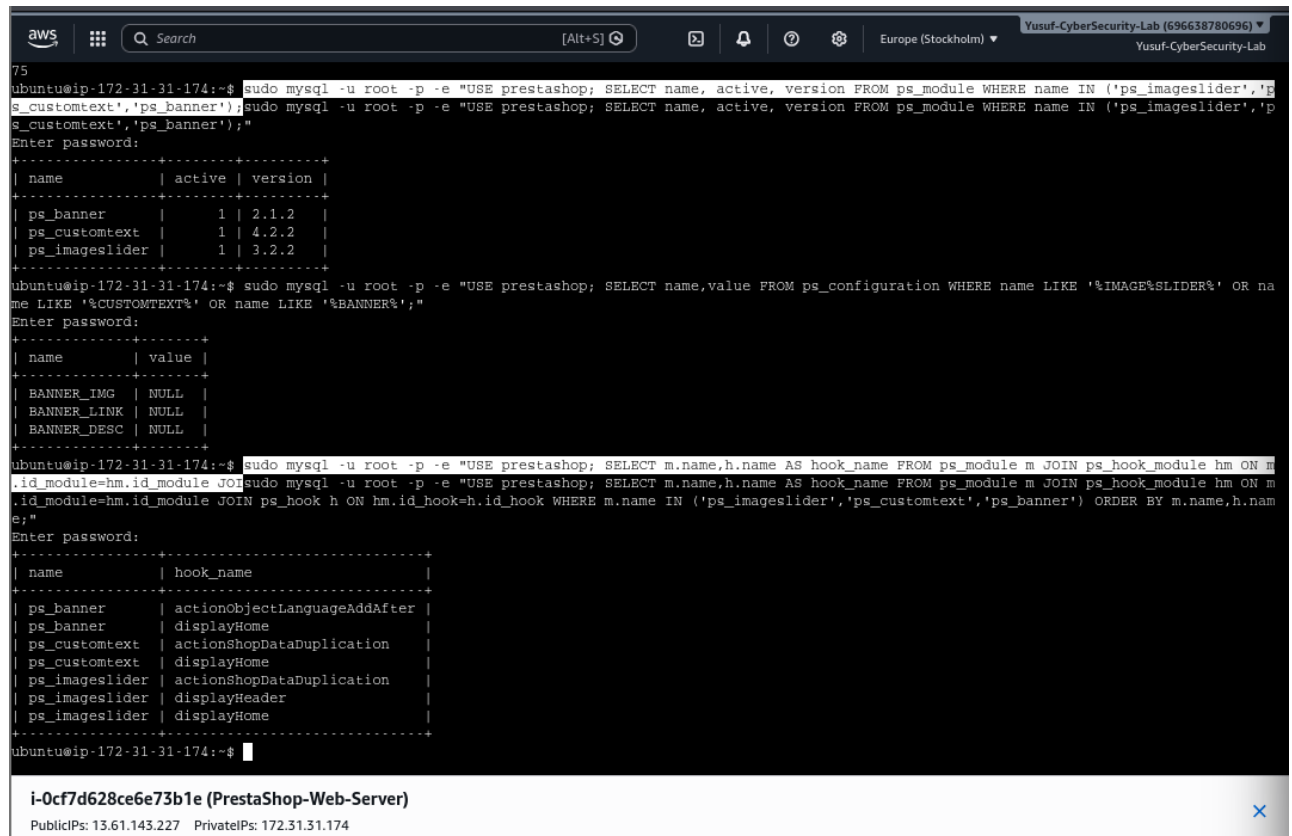
Archive screenshot: Screenshot 2026-09-15 210711.png



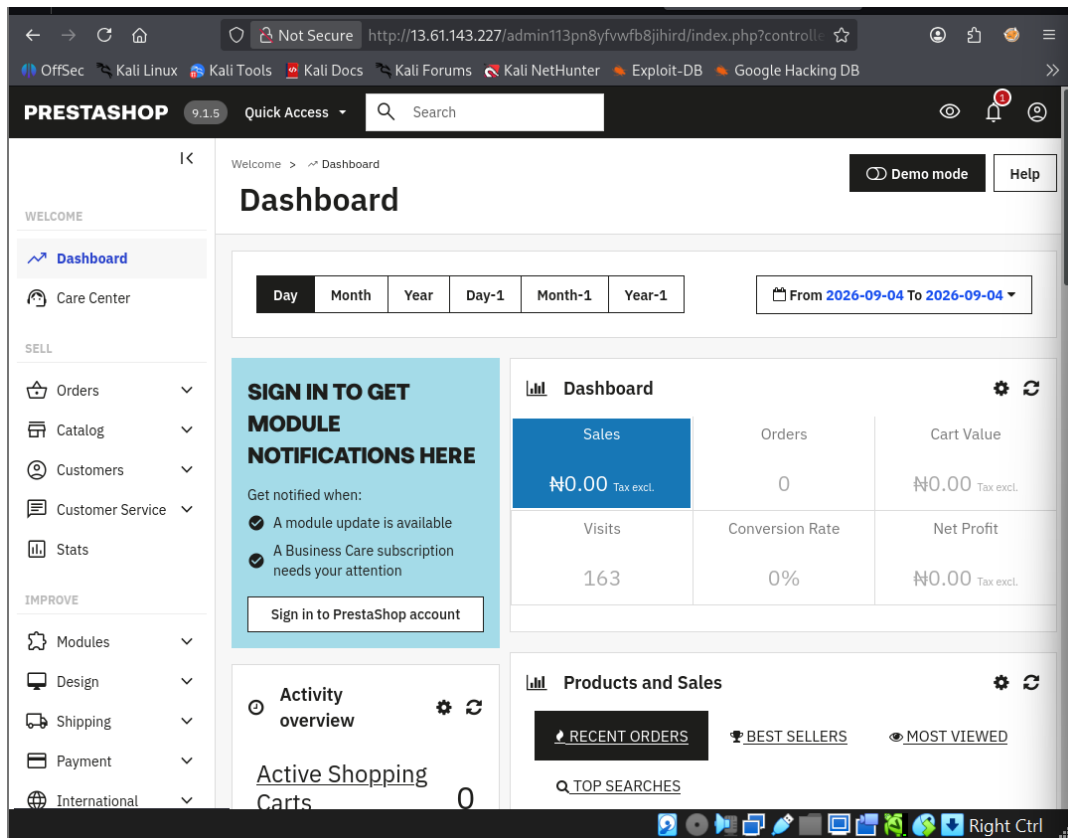
Archive screenshot: Screenshot 2026-09-15 210752.png



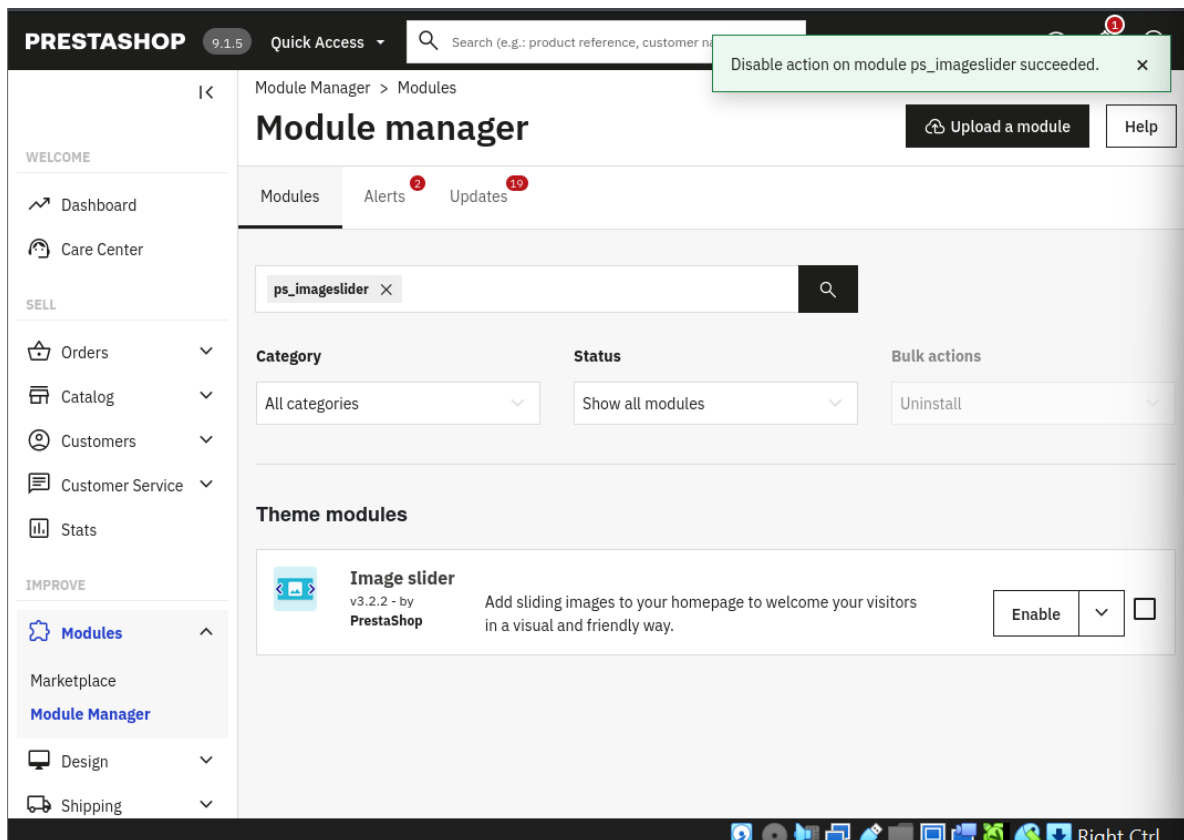
Archive screenshot: Screenshot 2026-09-15 211031.png



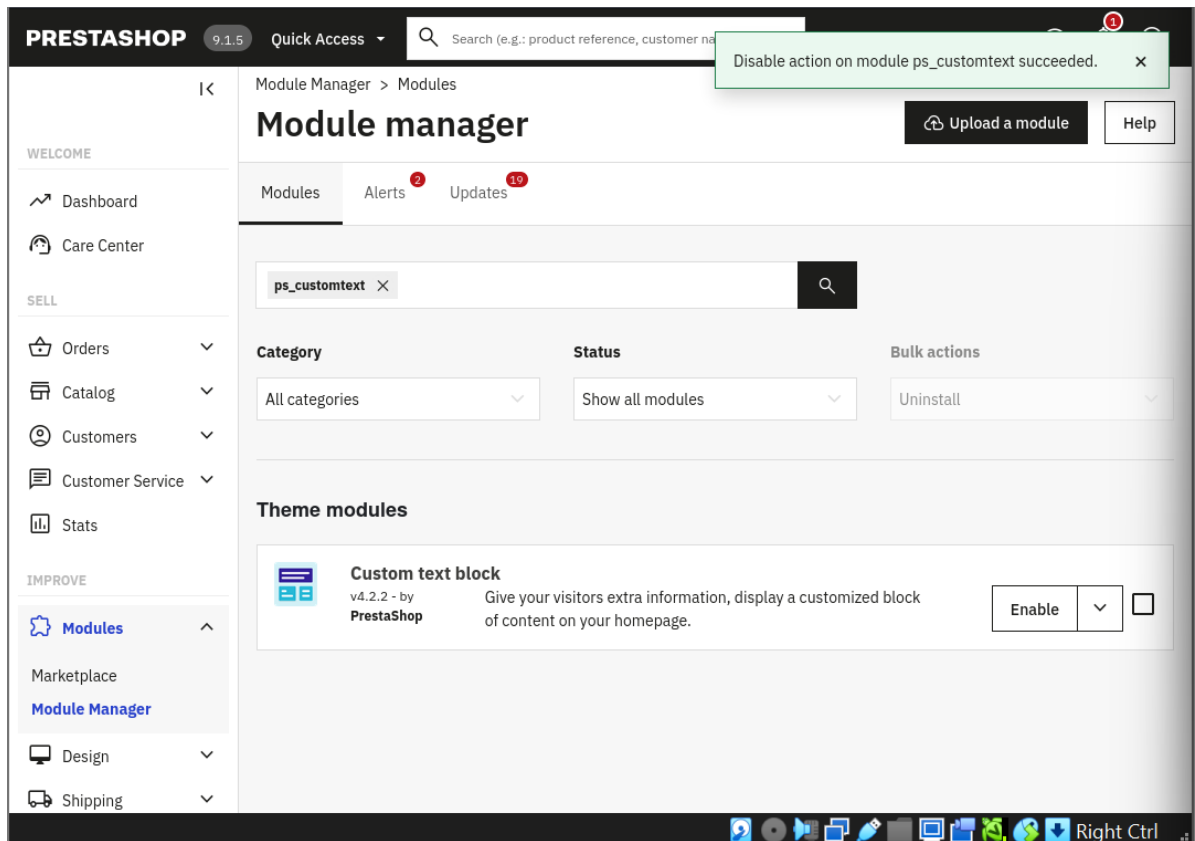
Archive screenshot: Screenshot 2026-09-15 211638.png



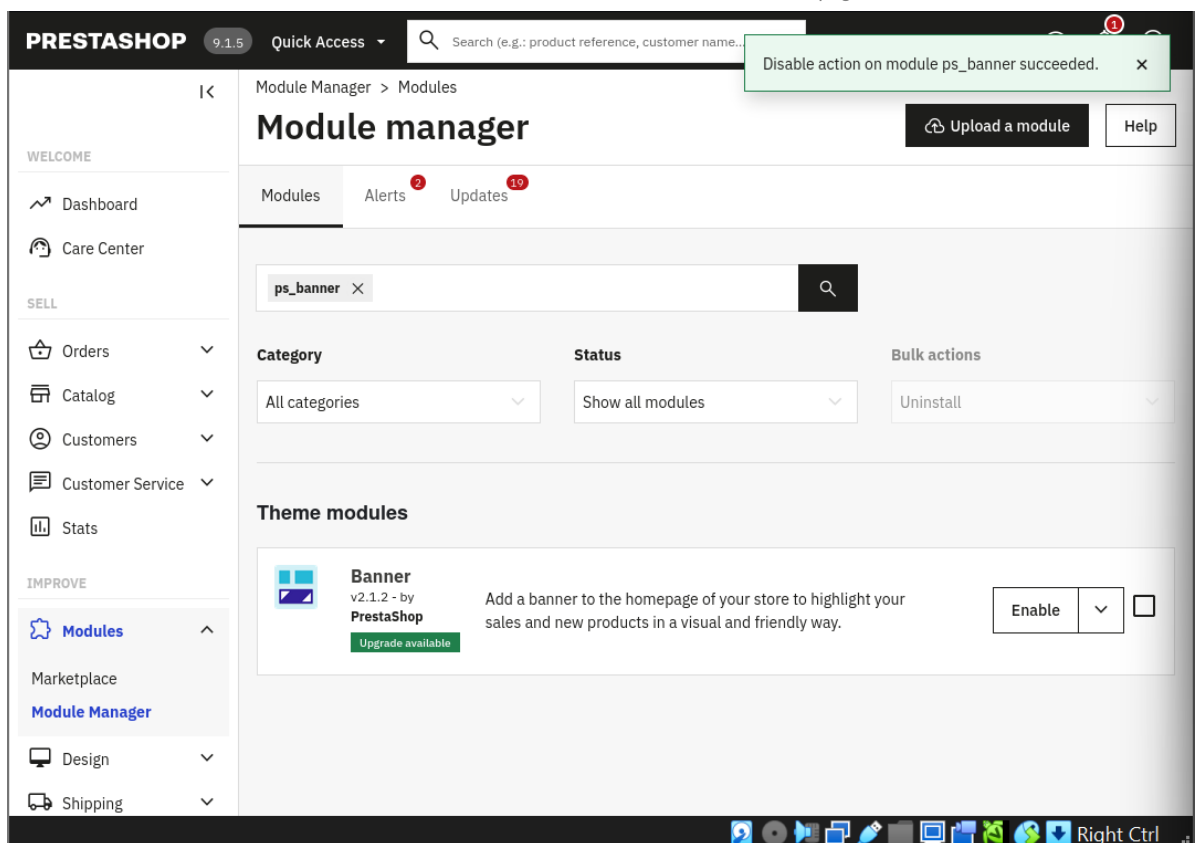
Archive screenshot: Screenshot 2026-09-15 212036.png



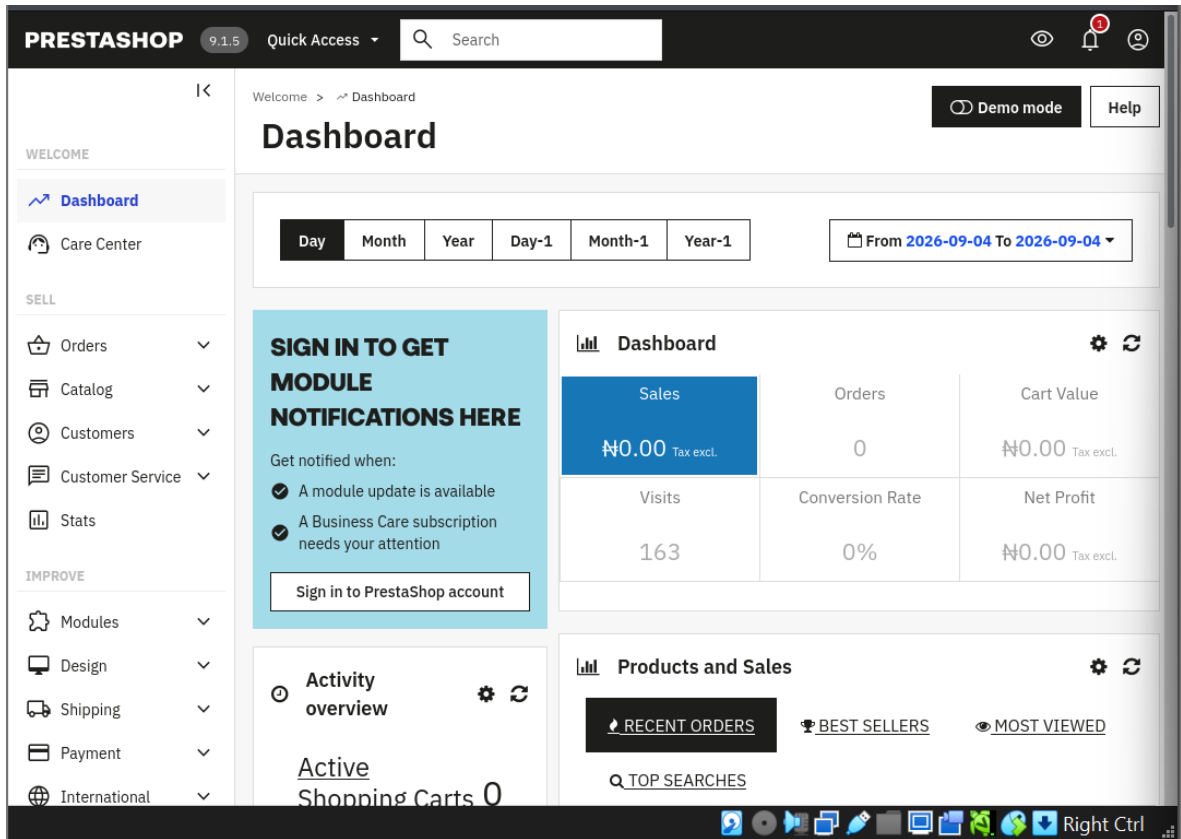
Archive screenshot: Screenshot 2026-09-15 212330.png



Archive screenshot: Screenshot 2026-09-15 212449.png



Archive screenshot: Screenshot 2026-09-15 212556.png



Archive screenshot: Screenshot 2026-09-15 212642.png

```
ubuntu@ip-172-31-31-174:~$ sudo mysql -u root -p -e "USE prestashop; SELECT name,active,version FROM ps_module WHERE name IN ('ps_imageslider','ps_customtext','ps_banner');"
Enter password:
+-----+-----+-----+
| name      | active | version |
+-----+-----+-----+
| ps_banner | 0      | 2.1.2   |
| ps_customtext | 0    | 4.2.2   |
| ps_imageslider | 0    | 3.2.2   |
+-----+-----+-----+
ubuntu@ip-172-31-31-174:~$
```

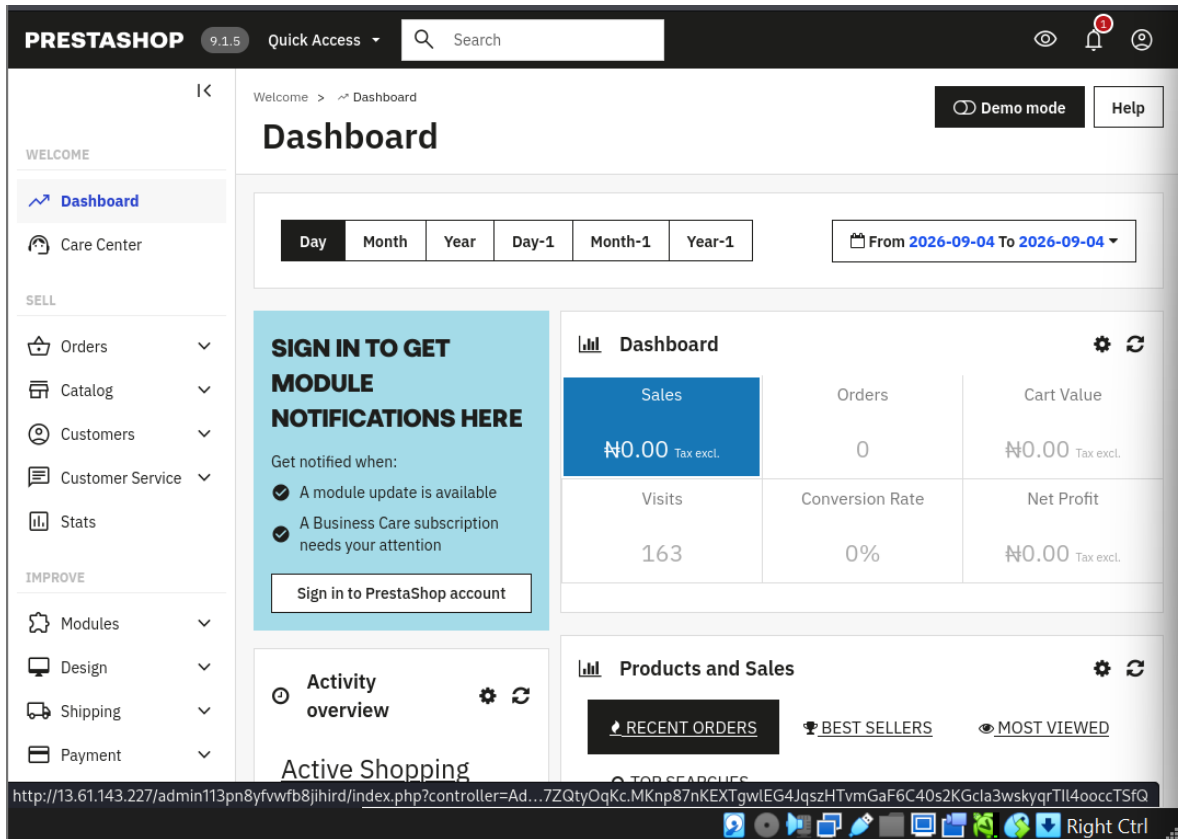
i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

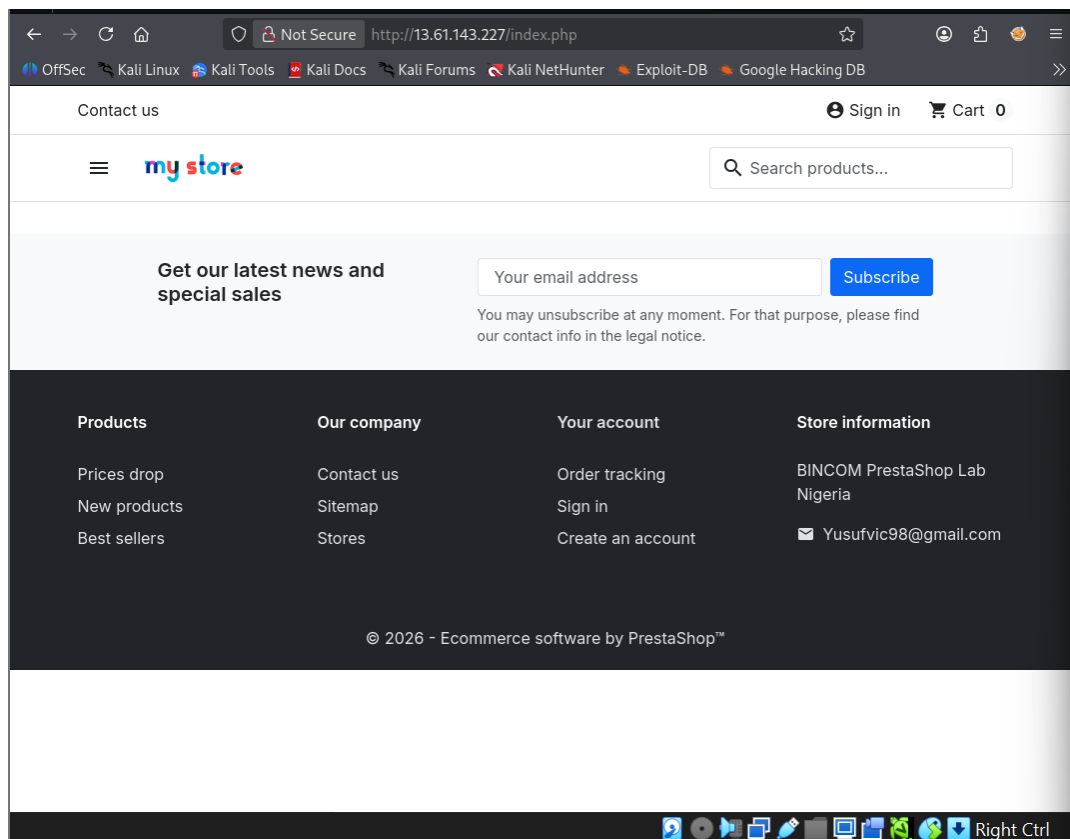
Archive screenshot: Screenshot 2026-09-15 212734.png

```
(kali@kali)-[~]
└─$ curl -s http://13.61.143.227/ | grep -iE "Sample|Custom Text|Lorem|sale70" | head -30
(kali@kali)-[~]
└─$ curl -s http://13.61.143.227/ | grep -iE "Sample 1|Sample 2|Sample 3|Custom Text Block|Lorem ipsum|sale70" | head -30
(kali@kali)-[~]
└─$ curl -s http://13.61.143.227/ | grep -oiE "Sample [123]|Custom Text Block|Lorem ipsum|sale70["\`<"]*" | sort -u
(kali@kali)-[~]
└─$
```

Archive screenshot: Screenshot 2026-09-15 213043.png



Archive screenshot: Screenshot 2026-09-15 213128.png



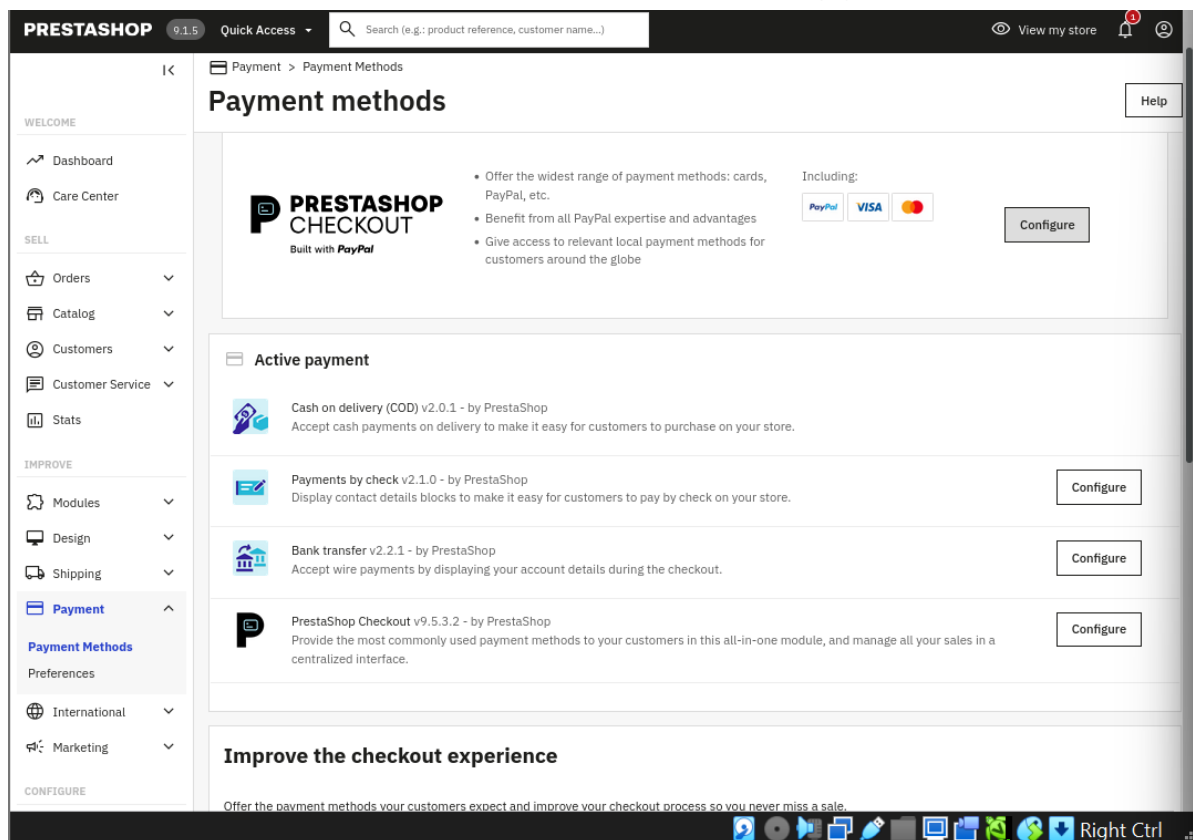
Archive screenshot: Screenshot 2026-09-15 213306.png

```
(kali@kali)-[~]
└─$ curl -i http://13.61.143.227/api/
HTTP/1.1 404 Not Found
Date: Tue, 15 Sep 2026 20:35:07 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Length: 236
Content-Type: text/html; charset=iso-8859-1

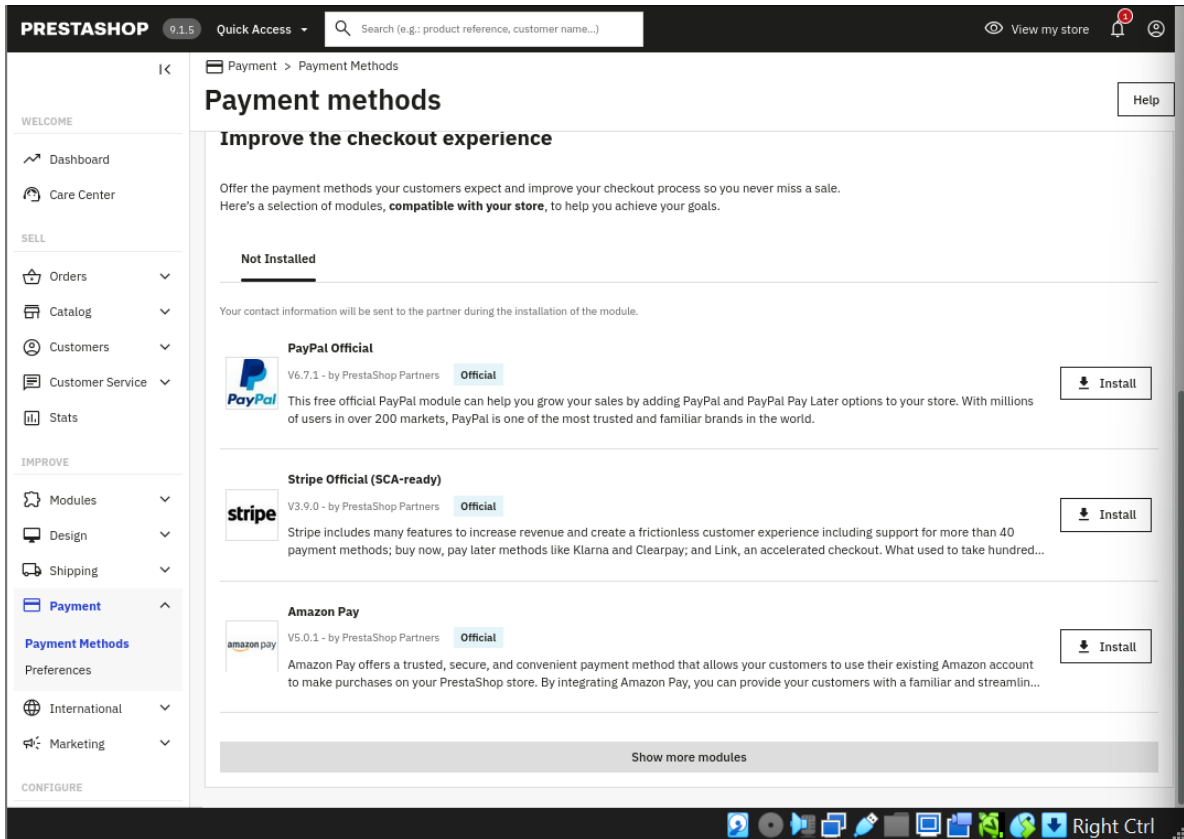
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
</body></html>

(kali@kali)-[~]
└─$ curl -i http://13.61.143.227/webService/
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 20:35:16 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Expires: Mon, 26 Jul 1997 05:00:00 GMT
Last-Modified: Tue, 15 Sep 2026 20:35:16 GMT
Cache-Control: no-store, no-cache, must-revalidate
Cache-Control: post-check=0, pre-check=0
Pragma: no-cache
Location: ../
Content-Length: 0
Content-Type: text/html; charset=UTF-8
PublicIP: 13.61.143.227 PrivateIP: 172.31.31.174
```

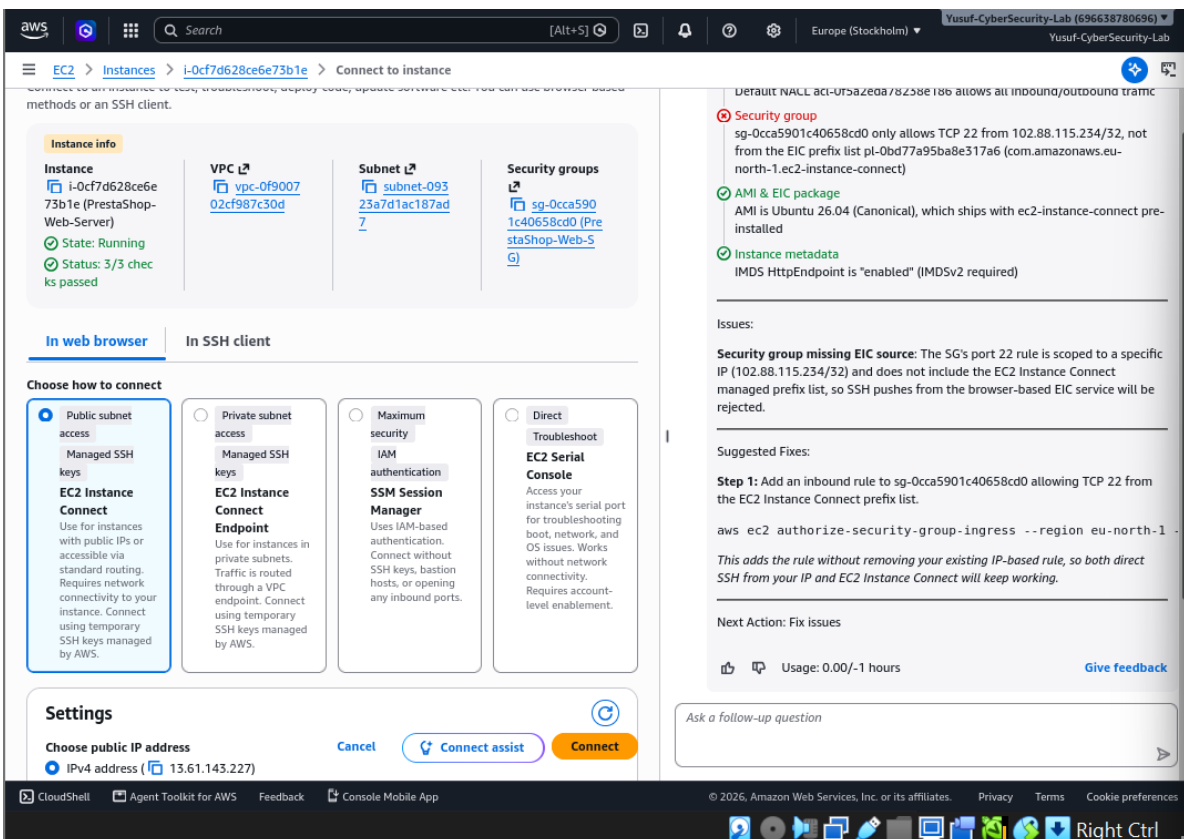
Archive screenshot: Screenshot 2026-09-15 213532.png



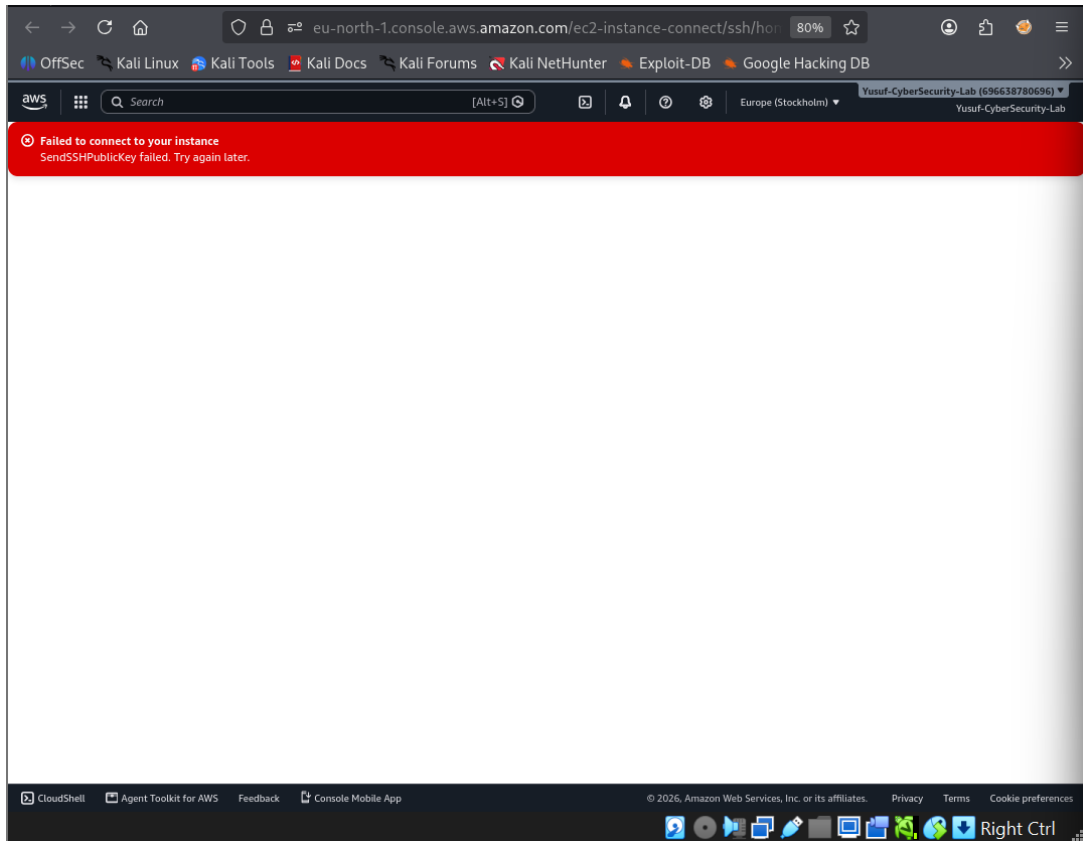
Archive screenshot: Screenshot 2026-09-15 214106.png



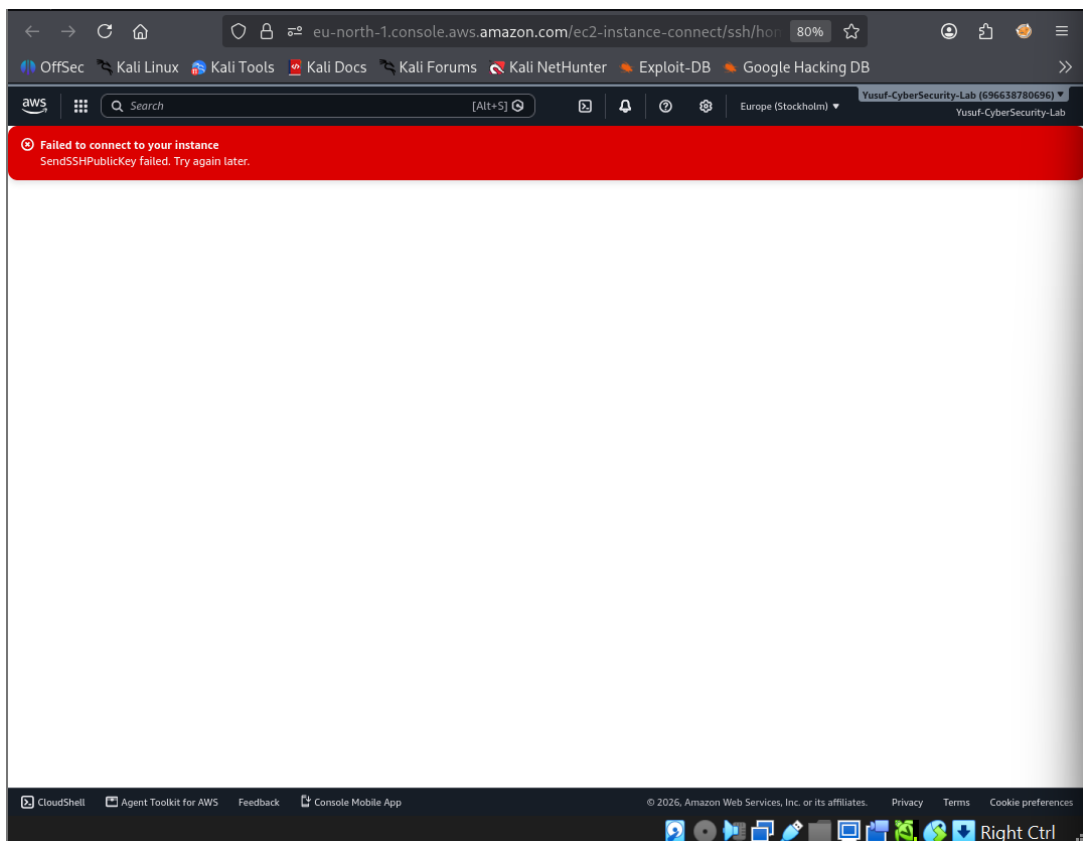
Archive screenshot: Screenshot 2026-09-15 214128.png



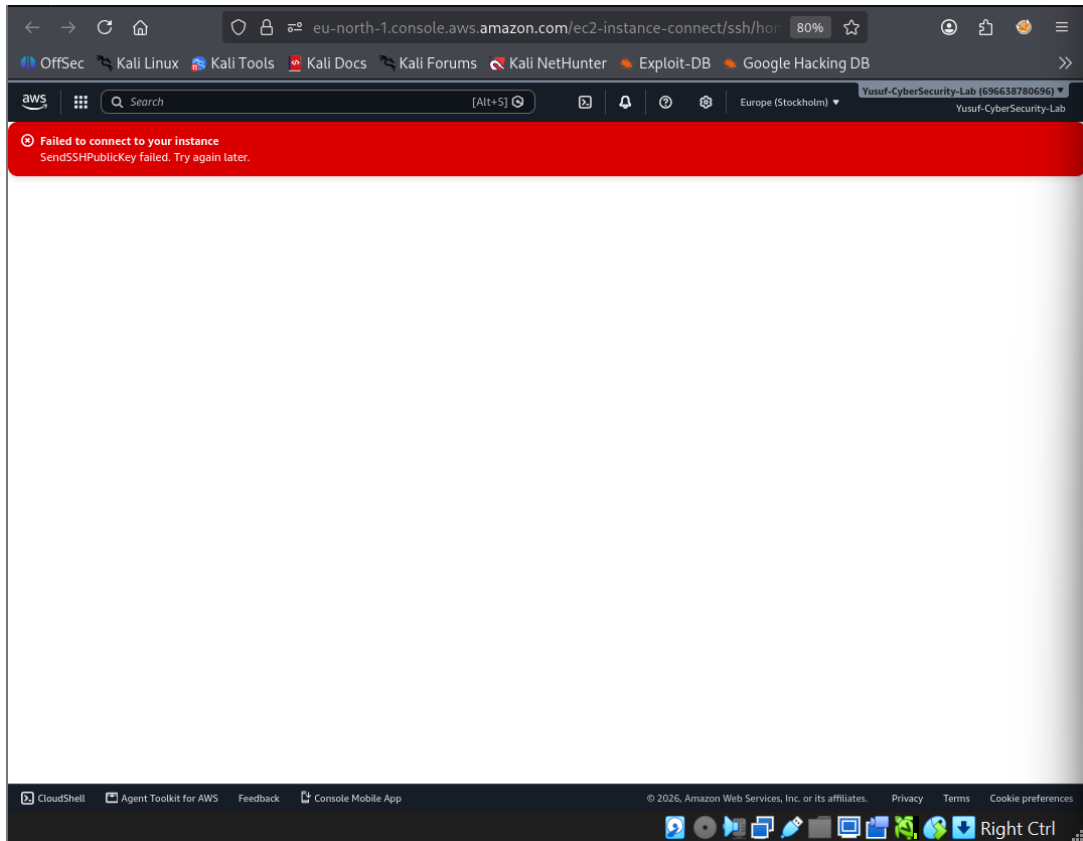
Archive screenshot: Screenshot 2026-09-15 215816.png



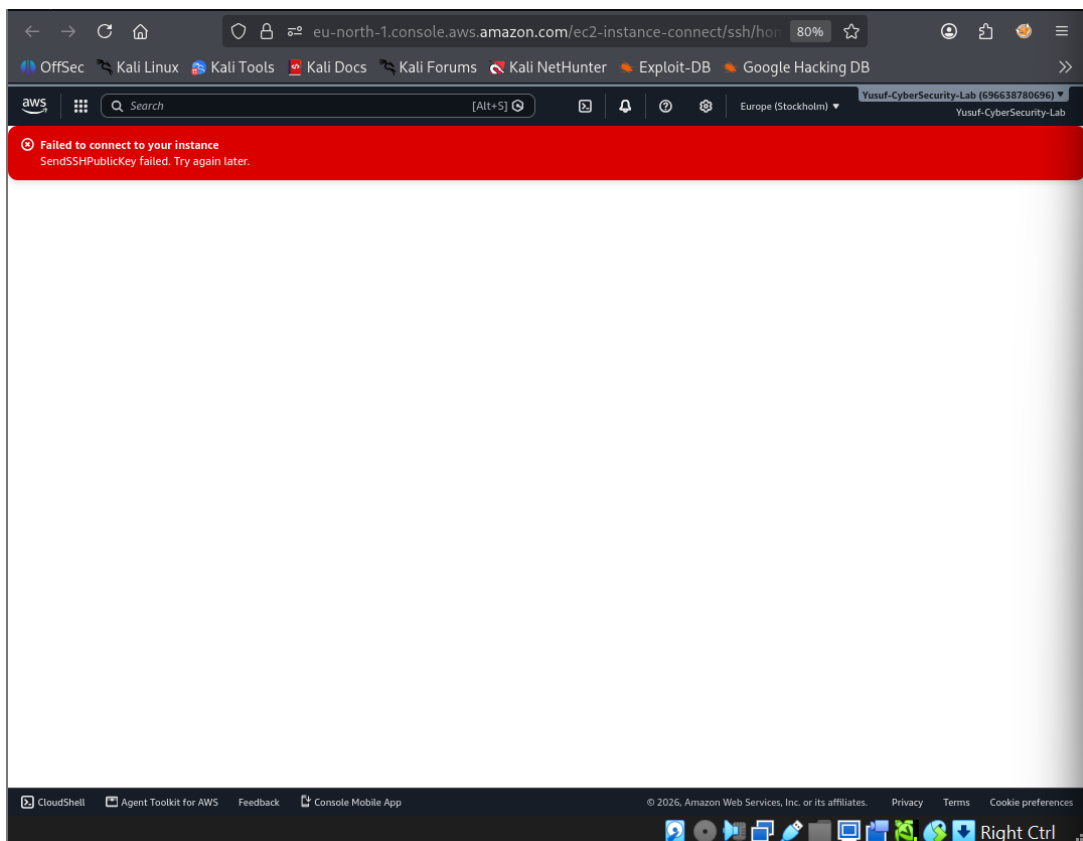
Archive screenshot: Screenshot 2026-09-15 215854(1).png



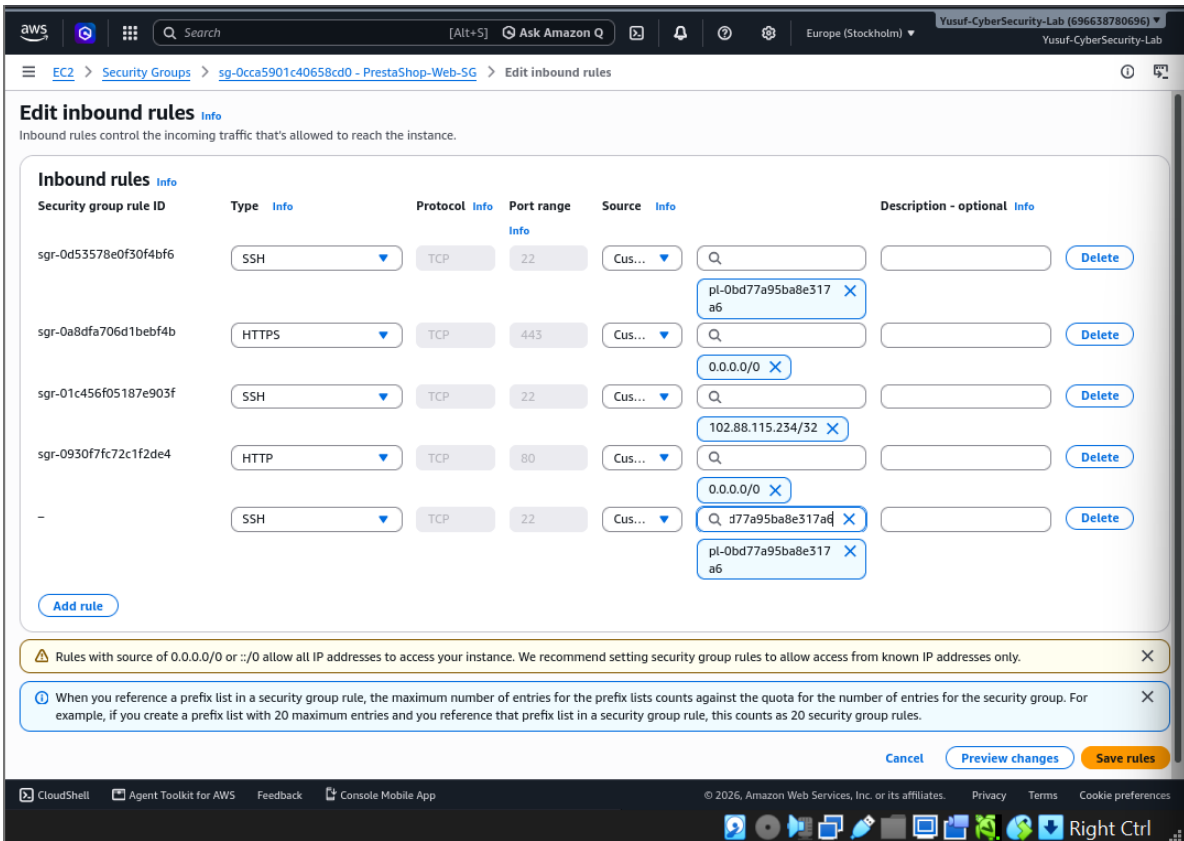
Archive screenshot: Screenshot 2026-09-15 215854(2).png



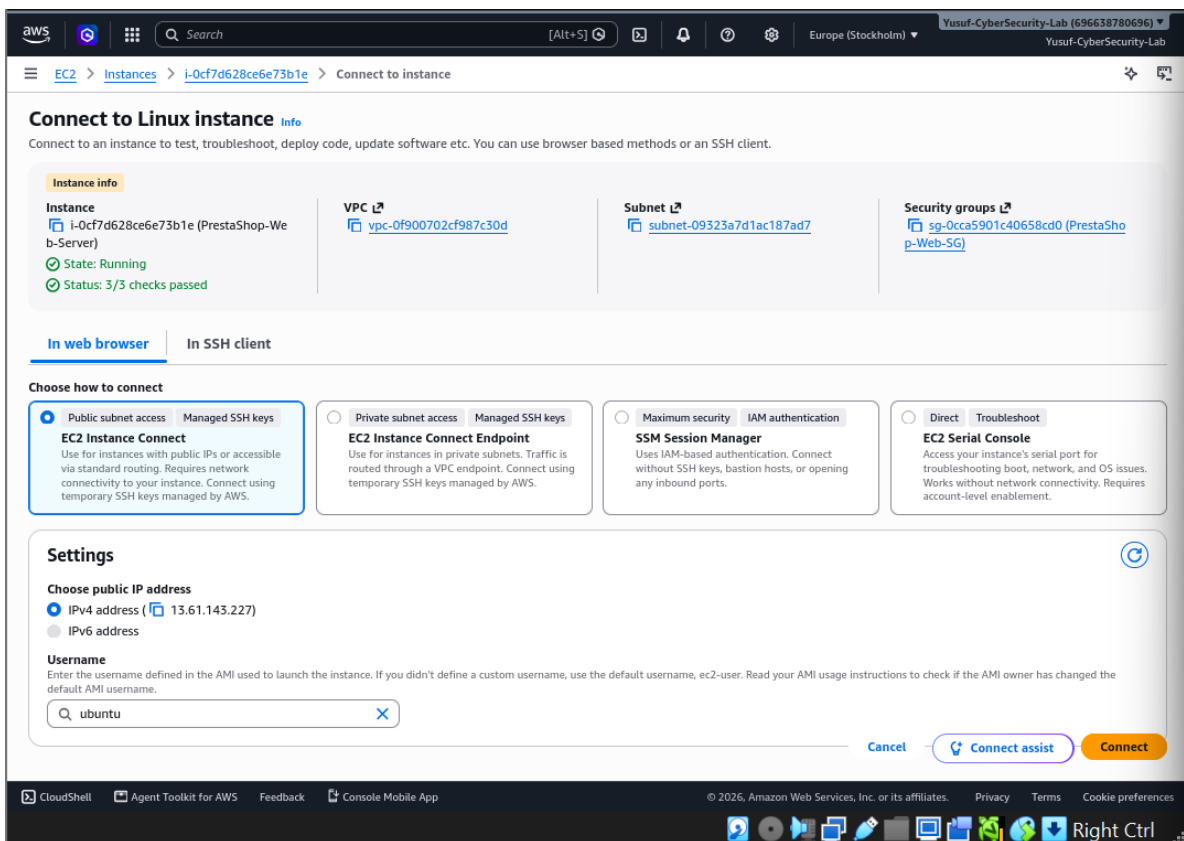
Archive screenshot: Screenshot 2026-09-15 215854(3).png



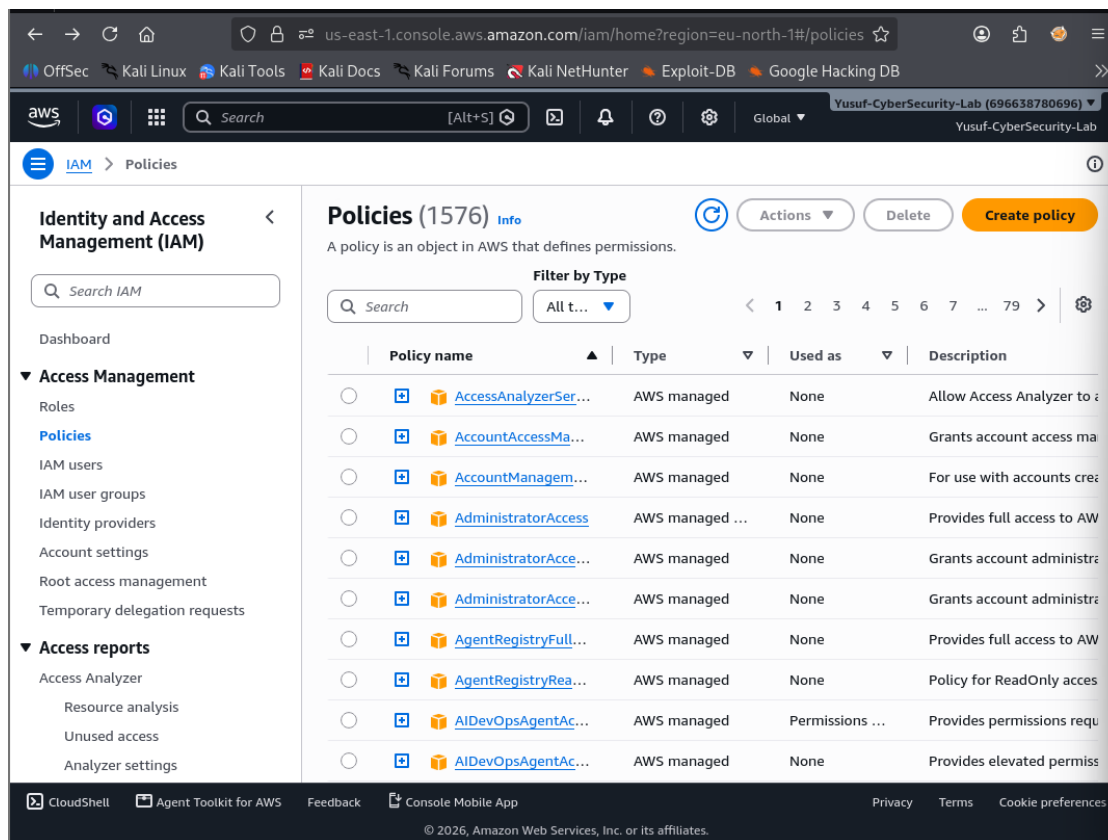
Archive screenshot: Screenshot 2026-09-15 215854.png



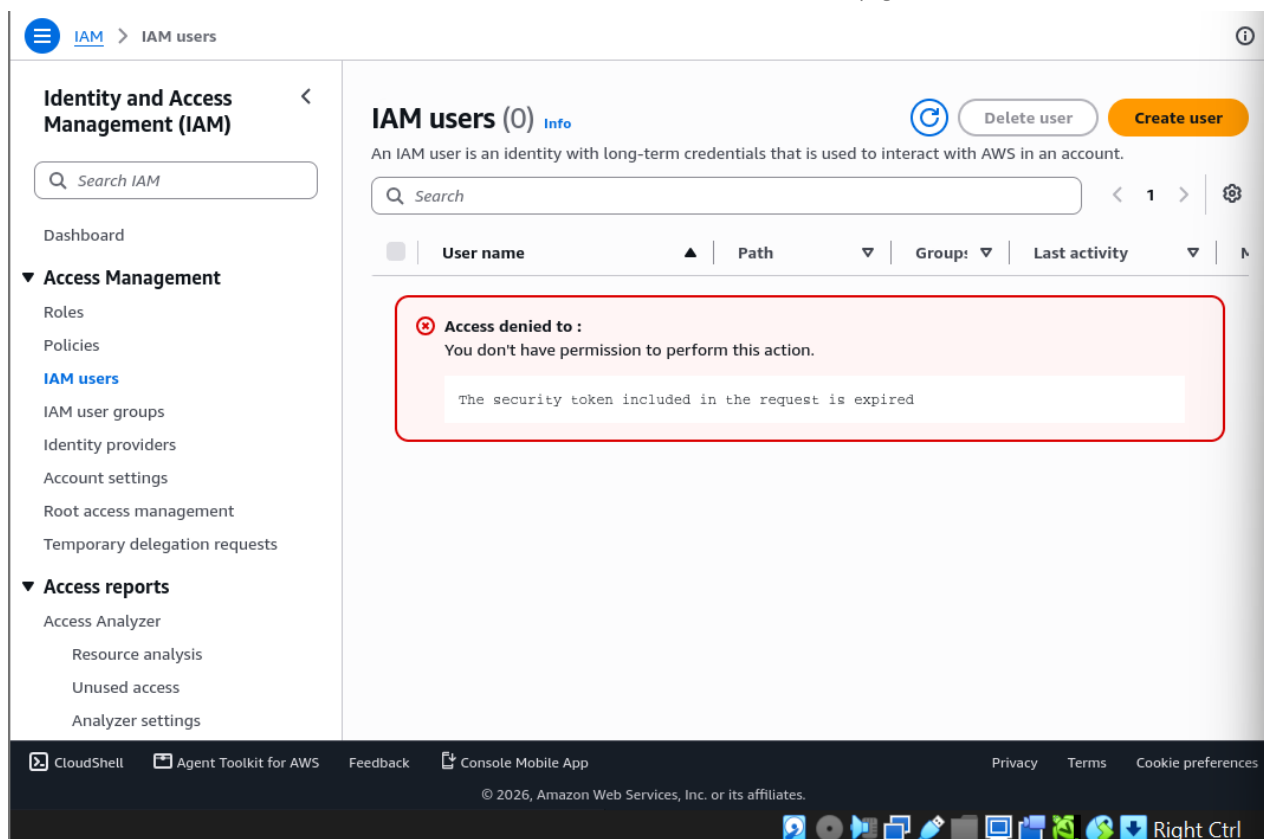
Archive screenshot: Screenshot 2026-09-15 220722.png



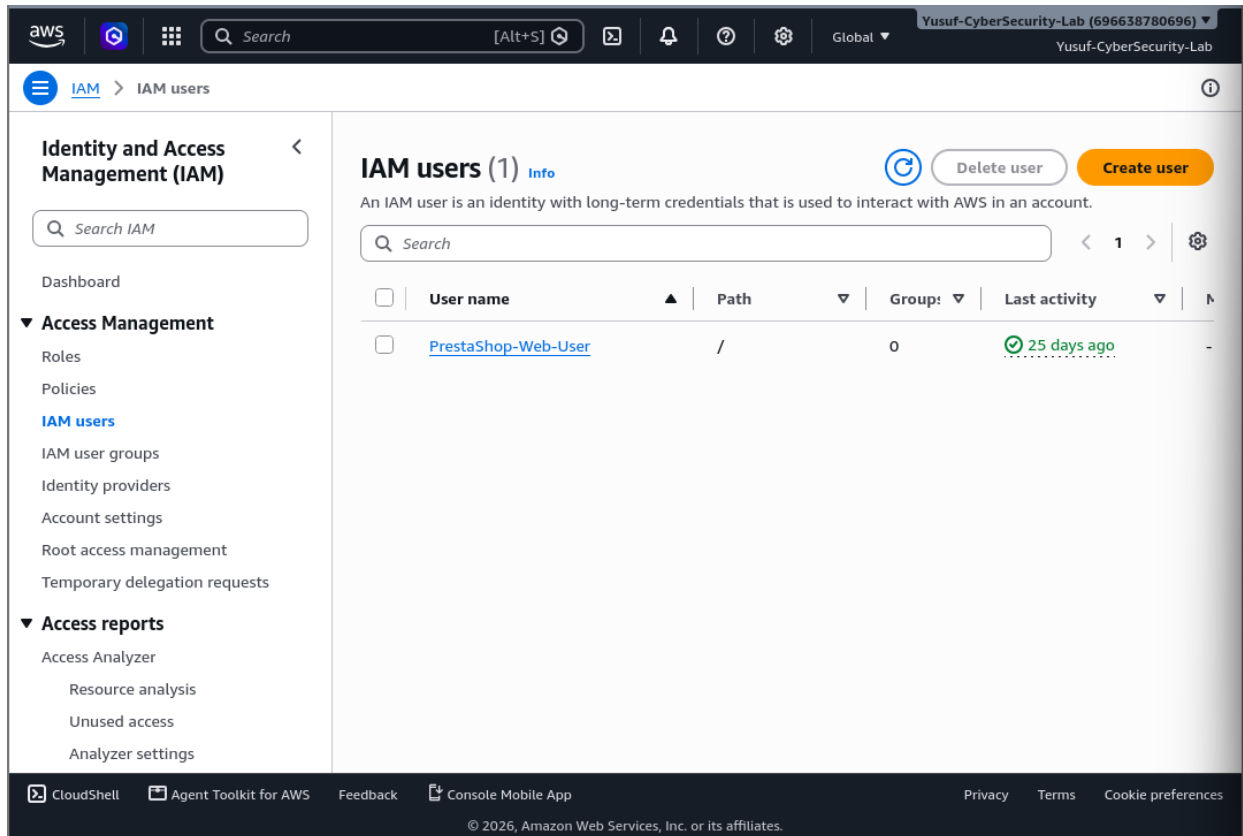
Archive screenshot: Screenshot 2026-09-15 221059.png



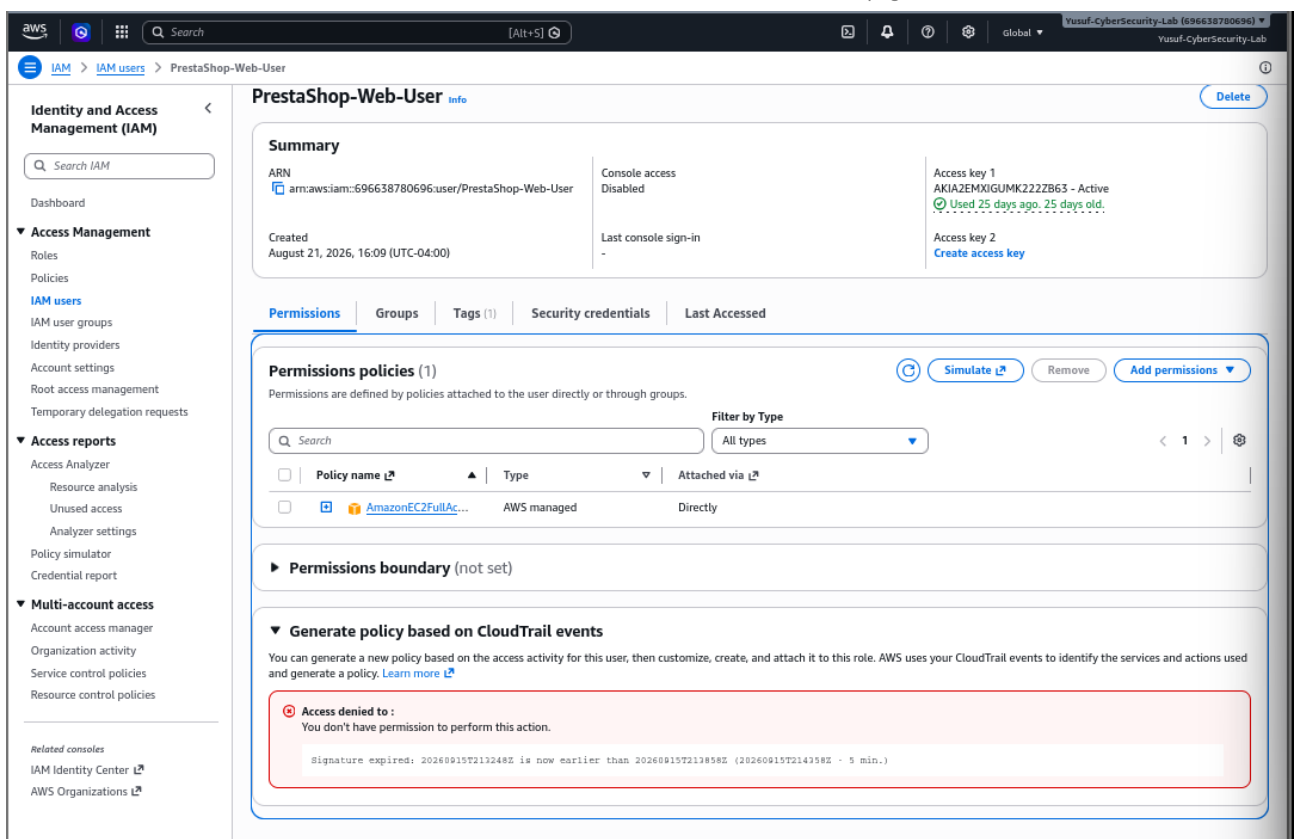
Archive screenshot: Screenshot 2026-09-15 223525.png



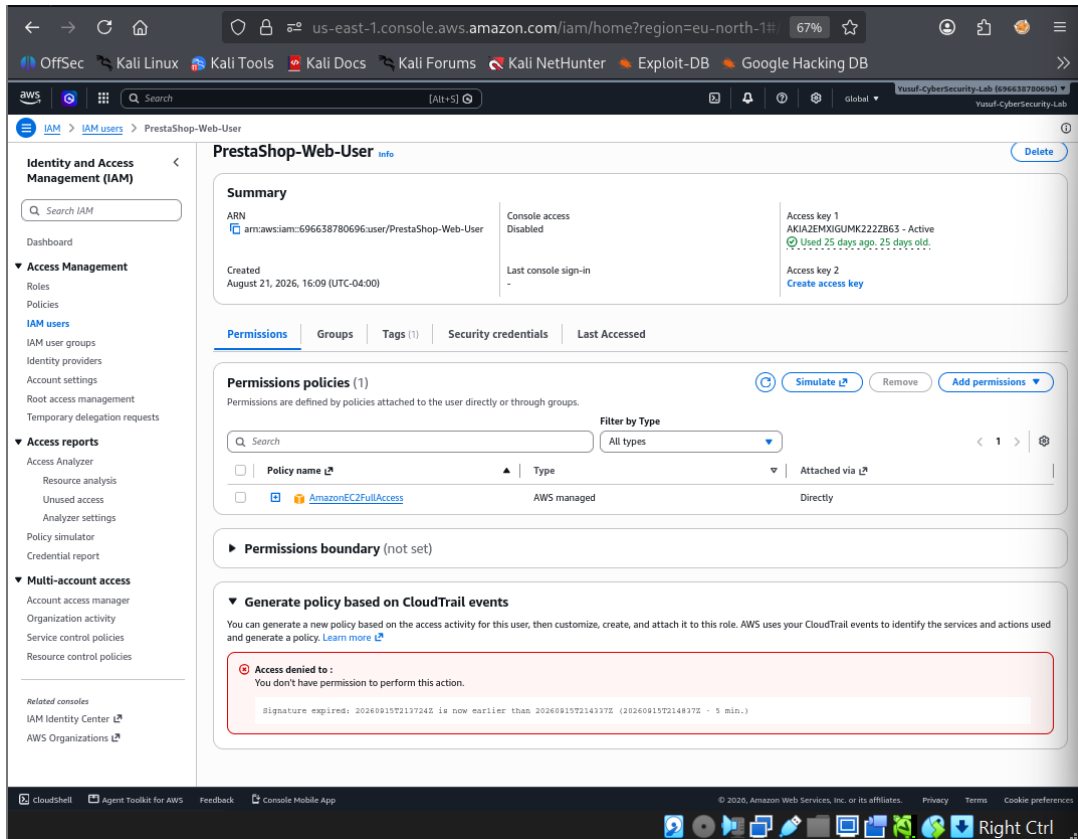
Archive screenshot: Screenshot 2026-09-15 223643.png



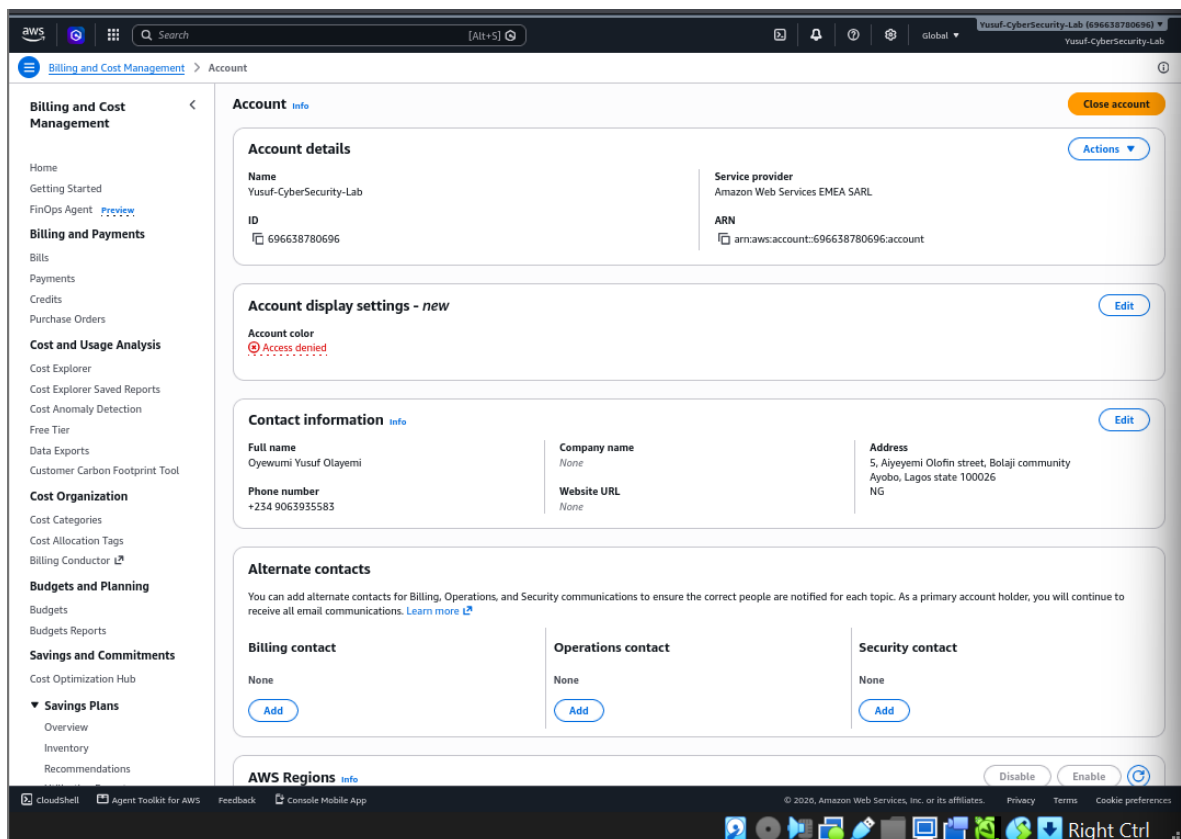
Archive screenshot: Screenshot 2026-09-15 224054.png



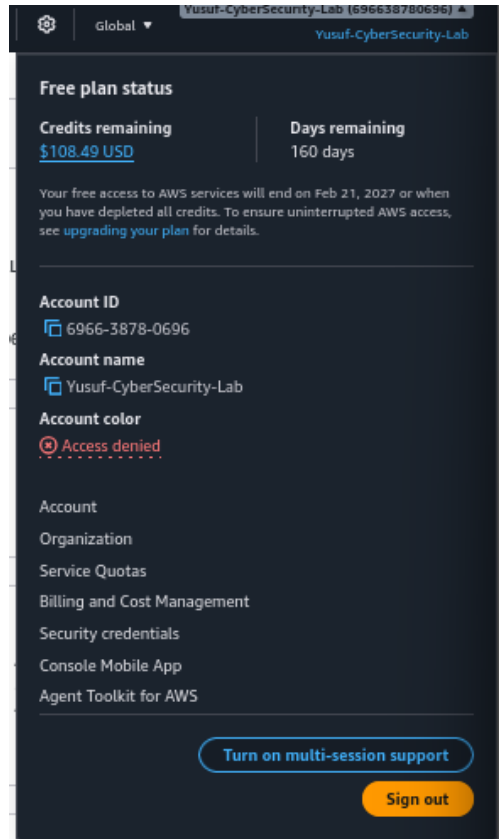
Archive screenshot: Screenshot 2026-09-15 224500.png



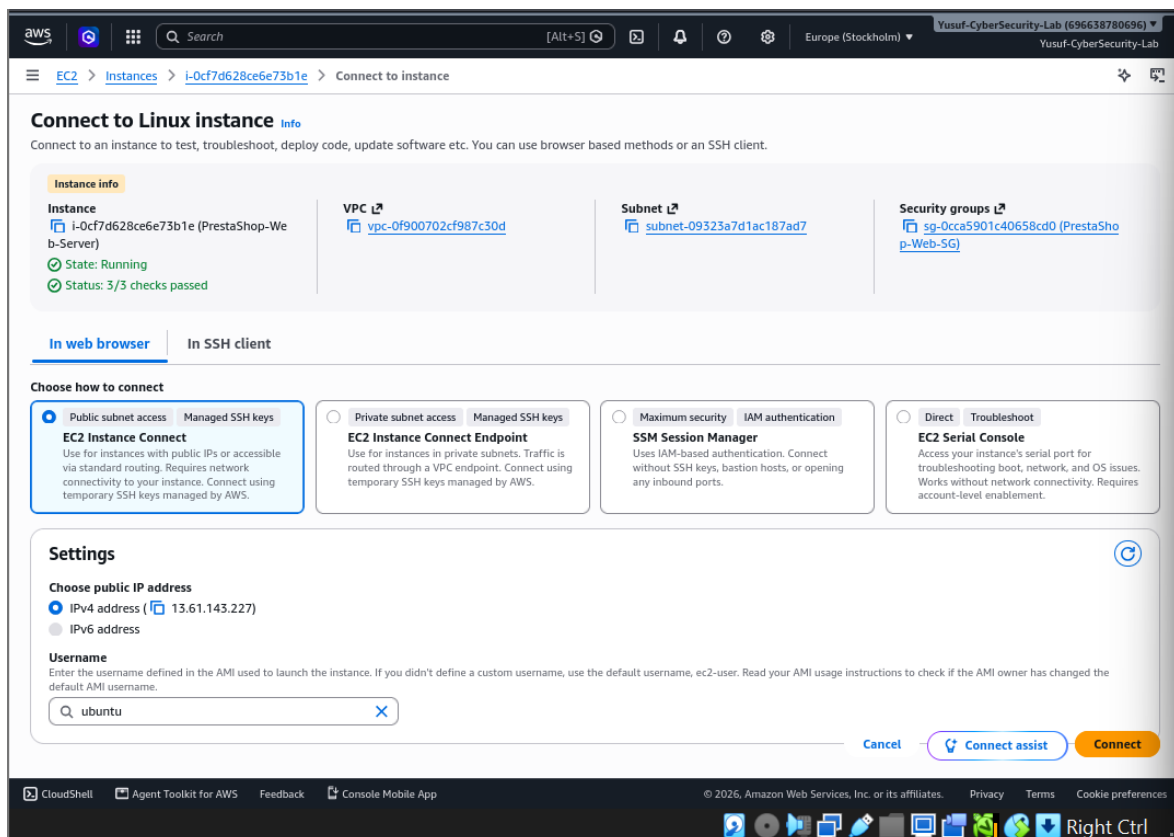
Archive screenshot: Screenshot 2026-09-15 225007.png



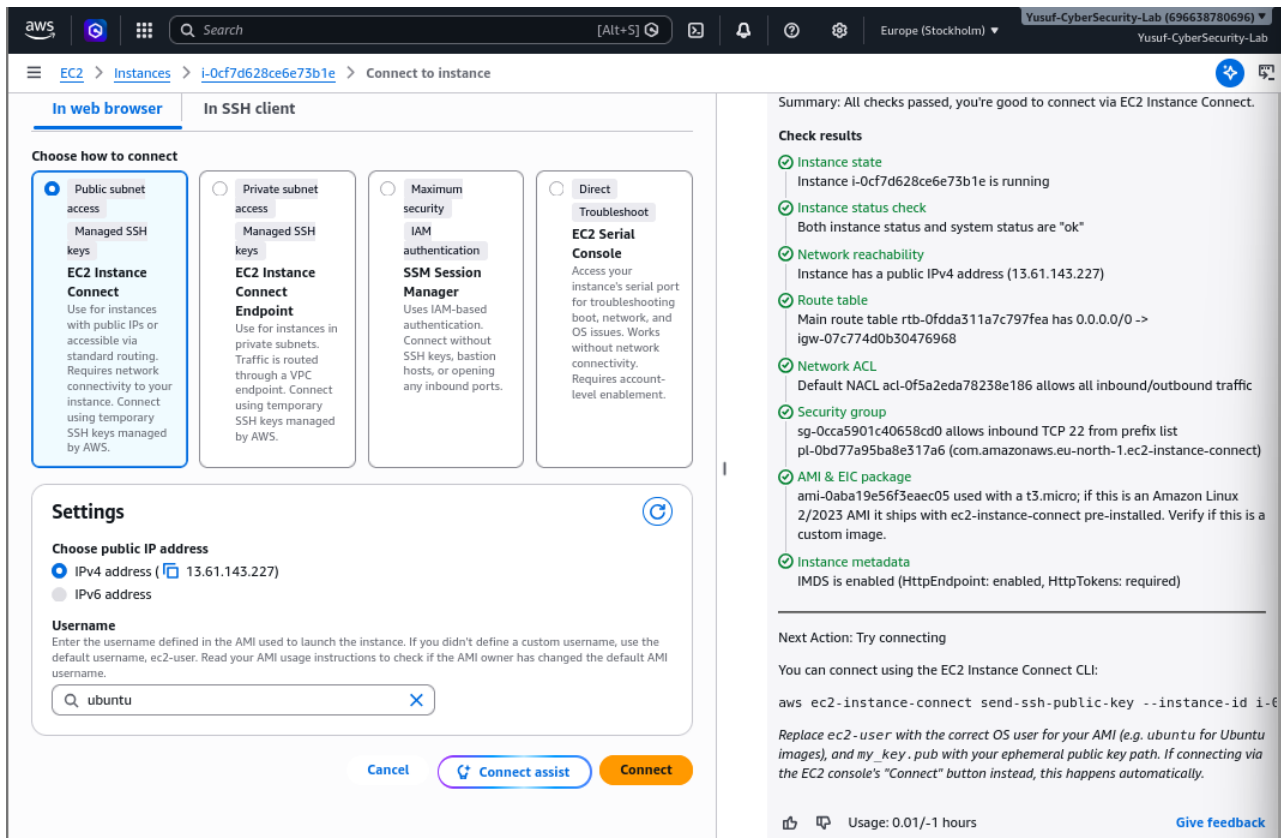
Archive screenshot: Screenshot 2026-09-15 230402.png



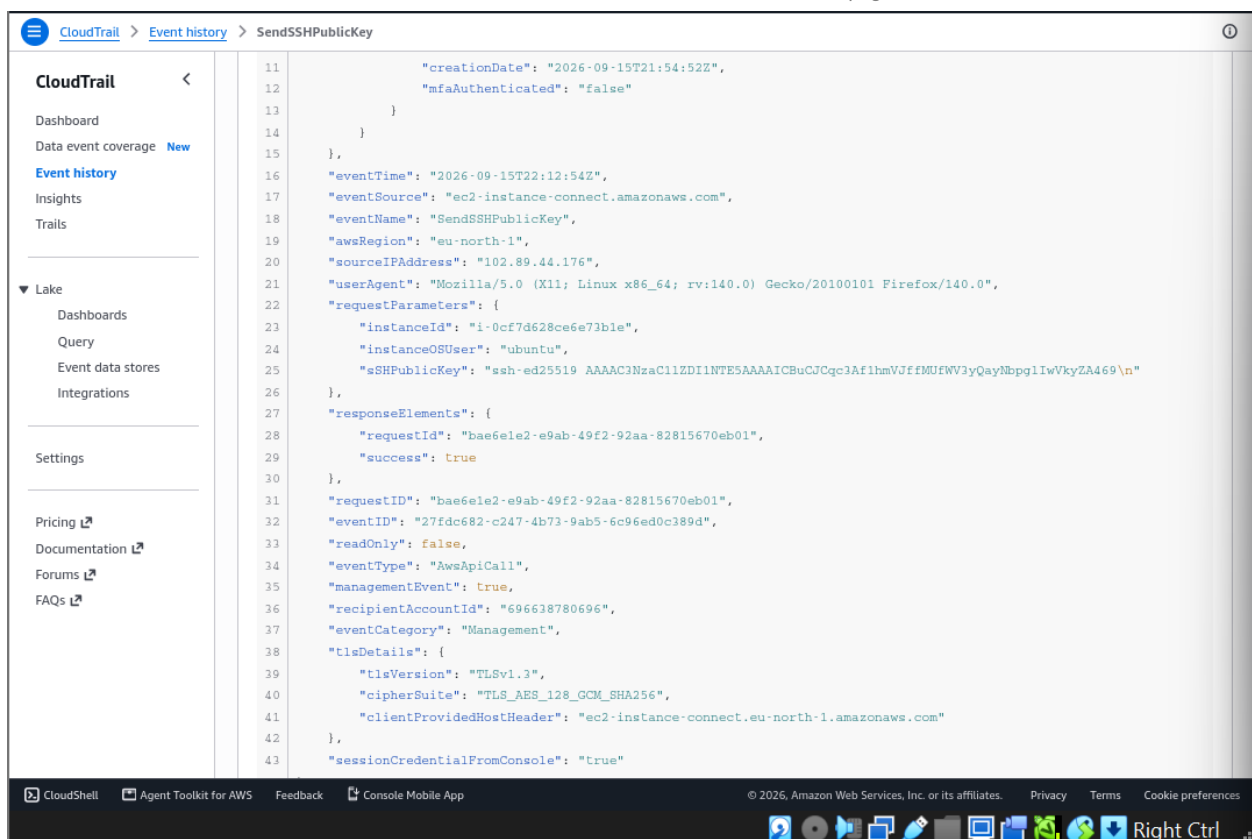
Archive screenshot: Screenshot 2026-09-15 230620.png



Archive screenshot: Screenshot 2026-09-15 230715.png



Archive screenshot: Screenshot 2026-09-15 231126.png



Archive screenshot: Screenshot 2026-09-15 232133.png

EC2

Instances

i-0cf7d628ce6e73b1e

Connect to instance

Instance info

Instance

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

State: Running

Status: 3/3 checks passed

VPC

vpc-0f900702cf987c30d

Subnet

subnet-09323a7d1ac187ad7

Security groups

sg-0cca5901c40658cd0 (PrestaShop-Web-SG)

In web browser

In SSH client

SSH info

SSH key pair assigned at launch

PrestaShop-Lab-Key

Public DNS

ec2-13-61-143-227.eu-north-1.compute.amazonaws.com

Username

ubuntu

Connect assist

How to connect

1. Open an SSH client

2. Locate or download your private key file

3. Ensure your key is not publicly viewable

4. Connect to your instance using its Public DNS

```
chmod 400 "PrestaShop-Lab-Key.pem"
```

```
ssh -i "PrestaShop-Lab-Key.pem" ubuntu@ec2-13-61-143-227.eu-north-1.compute.amazonaws.com
```

Summary: All checks passed, you're good to connect via EC2 Instance Connect.

Check results

Instance state

Instance i-0cf7d628ce6e73b1e is running

Instance status check

Both instance status and system status are "ok"

Network reachability

Instance has a public IPv4 address (13.61.143.227)

Route table

Main route table rtb-0fdda311a7c797fea has 0.0.0.0/0 -> igw-07c774d0b30476968

Network ACL

Default NACL acl-0f5a2eda78238e186 allows all inbound/outbound traffic

Security group

sg-0cca5901c40658cd0 allows inbound TCP 22 from prefix list pl-0bd77a95ba8e317a6 (com.amazonaws.eu-north-1.ec2-instance-connect)

AMI & EIC package

ami-0aba19e56f3eae05 used with a t3.micro; if this is an Amazon Linux 2/2023 AMI it ships with ec2-instance-connect pre-installed. Verify if this is a custom image.

Instance metadata

IMDS is enabled (HttpEndpoint: enabled, HttpTokens: required)

Next Action: Try connecting

You can connect using the EC2 Instance Connect CLI:

```
aws ec2-instance-connect send-ssh-public-key --instance-id i-0cf7d628ce6e73b1e --availability-zone eu-north-1 --public-key-path my-key.pub --ec2-user ubuntu
```

Replace `ec2-user` with the correct OS user for your AMI (e.g. `ubuntu` for Ubuntu images), and `my_key.pub` with your ephemeral public key path. If connecting via the EC2 console's "Connect" button instead, this happens automatically.

Usage: 0.01/-1 hours

Give feedback

Archive screenshot: Screenshot 2026-09-15 232350.png

EC2

Instances

i-0cf7d628ce6e73b1e

Connect to instance

In web browser

In SSH client

Choose how to connect

Public subnet access

Managed SSH keys

EC2 Instance Connect

Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.

Private subnet access

Managed SSH keys

EC2 Instance Connect Endpoint

Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.

Maximum security

IAM authentication

SSM Session Manager

Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.

Direct

Troubleshoot

EC2 Serial Console

Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.

Serial port info

Serial port

ttyS0

This account is not authorized to use the EC2 serial console

To connect to this instance using the EC2 serial console, the account must be authorized in the EC2 account settings.

Manage access

Cancel

Connect

Summary: All checks passed, you're good to connect via EC2 Instance Connect.

Check results

Instance state

Instance i-0cf7d628ce6e73b1e is running

Instance status check

Both instance status and system status are "ok"

Network reachability

Instance has a public IPv4 address (13.61.143.227)

Route table

Main route table rtb-0fdda311a7c797fea has 0.0.0.0/0 -> igw-07c774d0b30476968

Network ACL

Default NACL acl-0f5a2eda78238e186 allows all inbound/outbound traffic

Security group

sg-0cca5901c40658cd0 allows inbound TCP 22 from prefix list pl-0bd77a95ba8e317a6 (com.amazonaws.eu-north-1.ec2-instance-connect)

AMI & EIC package

ami-0aba19e56f3eae05 used with a t3.micro; if this is an Amazon Linux 2/2023 AMI it ships with ec2-instance-connect pre-installed. Verify if this is a custom image.

Instance metadata

IMDS is enabled (HttpEndpoint: enabled, HttpTokens: required)

Next Action: Try connecting

You can connect using the EC2 Instance Connect CLI:

```
aws ec2-instance-connect send-ssh-public-key --instance-id i-0cf7d628ce6e73b1e --availability-zone eu-north-1 --public-key-path my-key.pub --ec2-user ubuntu
```

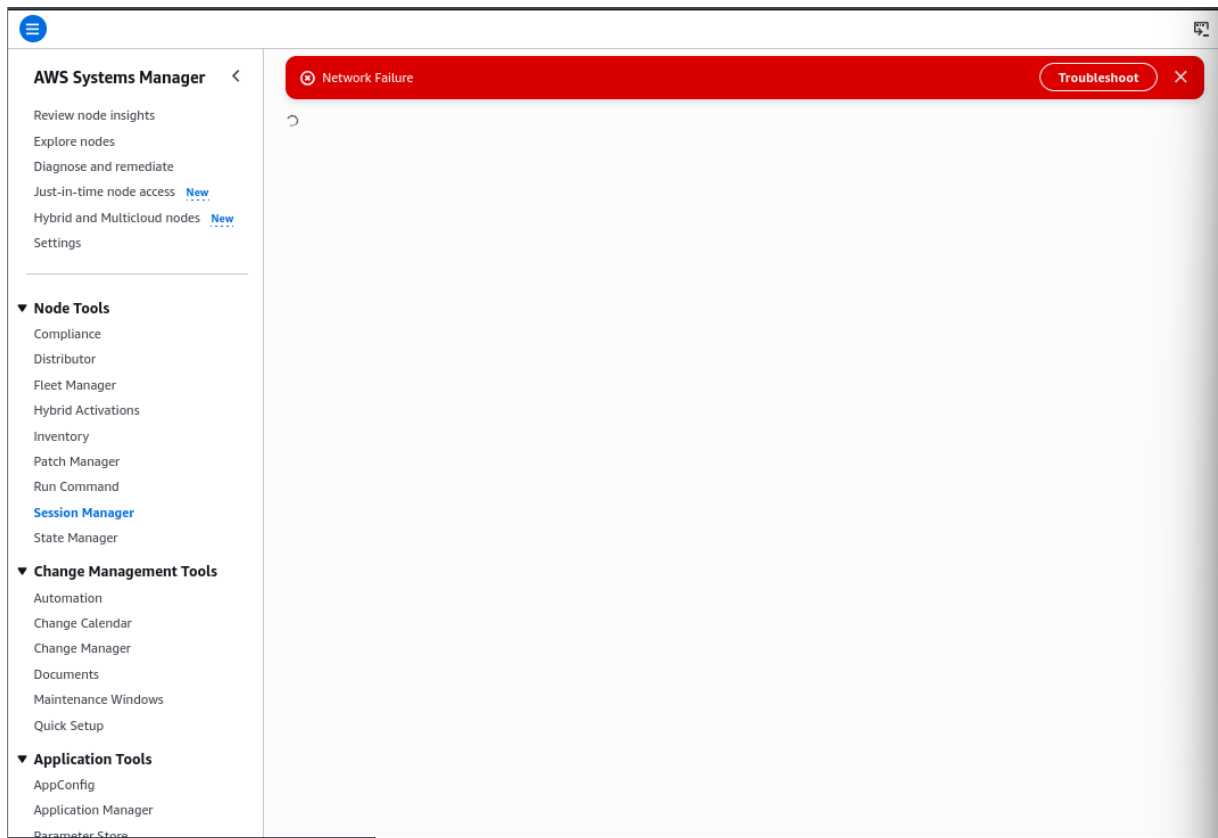
Replace `ec2-user` with the correct OS user for your AMI (e.g. `ubuntu` for Ubuntu images), and `my_key.pub` with your ephemeral public key path. If connecting via the EC2 console's "Connect" button instead, this happens automatically.

Usage: 0.01/-1 hours

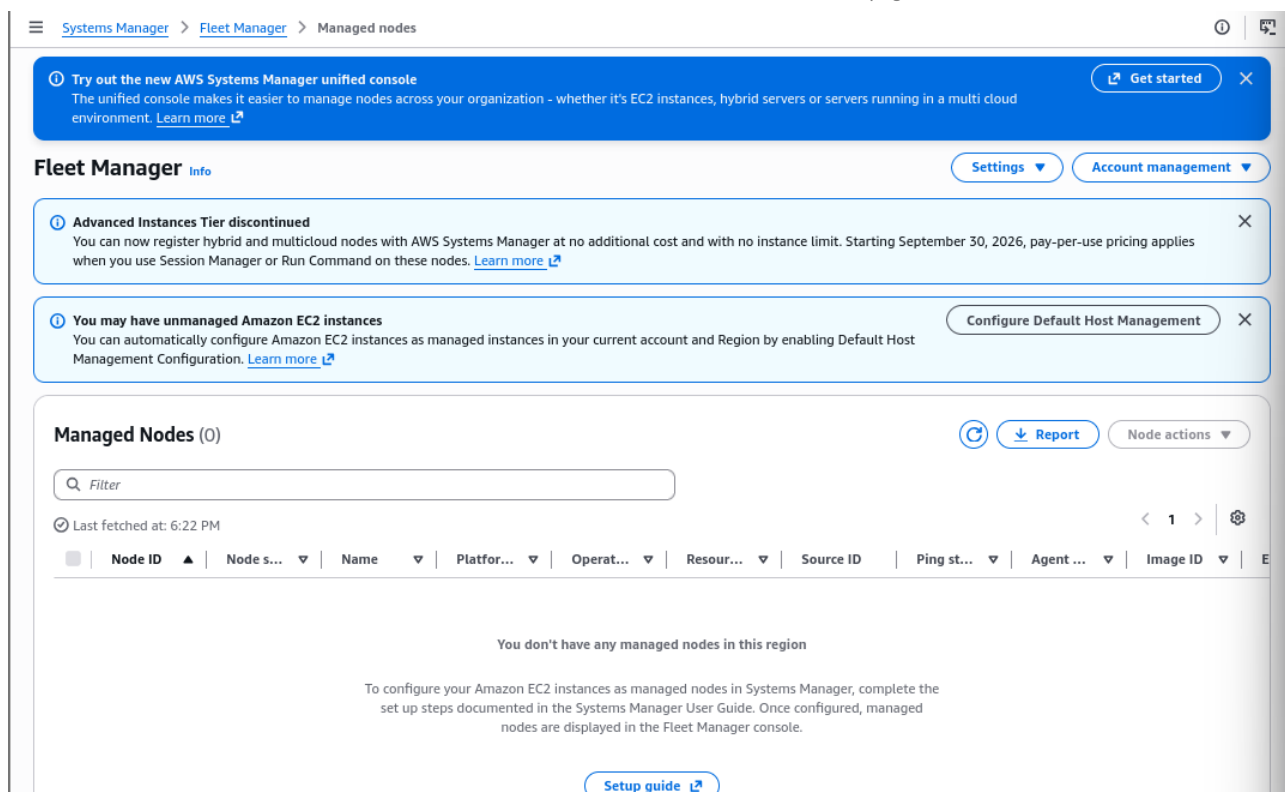
Give feedback

Archive screenshot: Screenshot 2026-09-15 232934.png

BINCOM Assessment 3 — Complete Technical Report | Page 65



Archive screenshot: Screenshot 2026-09-15 233159.png



Archive screenshot: Screenshot 2026-09-15 233424.png

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

IAM role

-

IMDSv2

Required

Operator

-

Subnet ID

 subnet-09323a7d1ac187ad7 

Instance ARN

 arn:aws:ec2:eu-north-1:696638780696:instance/i-0cf7d628ce6e73b1e

Auto Scaling Group name

-

Managed

false

Archive screenshot: Screenshot 2026-09-15 233659.png

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

▼ Instance details [Info](#)

AMI ID

 ami-0aba19e56f3eac05

AMI name

 ubuntu/images/hvm-ssd-gp3/ubuntu-resolut
e-26.04-amd64-server-20260604

Stop protection

Monitoring

disabled

Allowed image

-

Launch time

Platform details

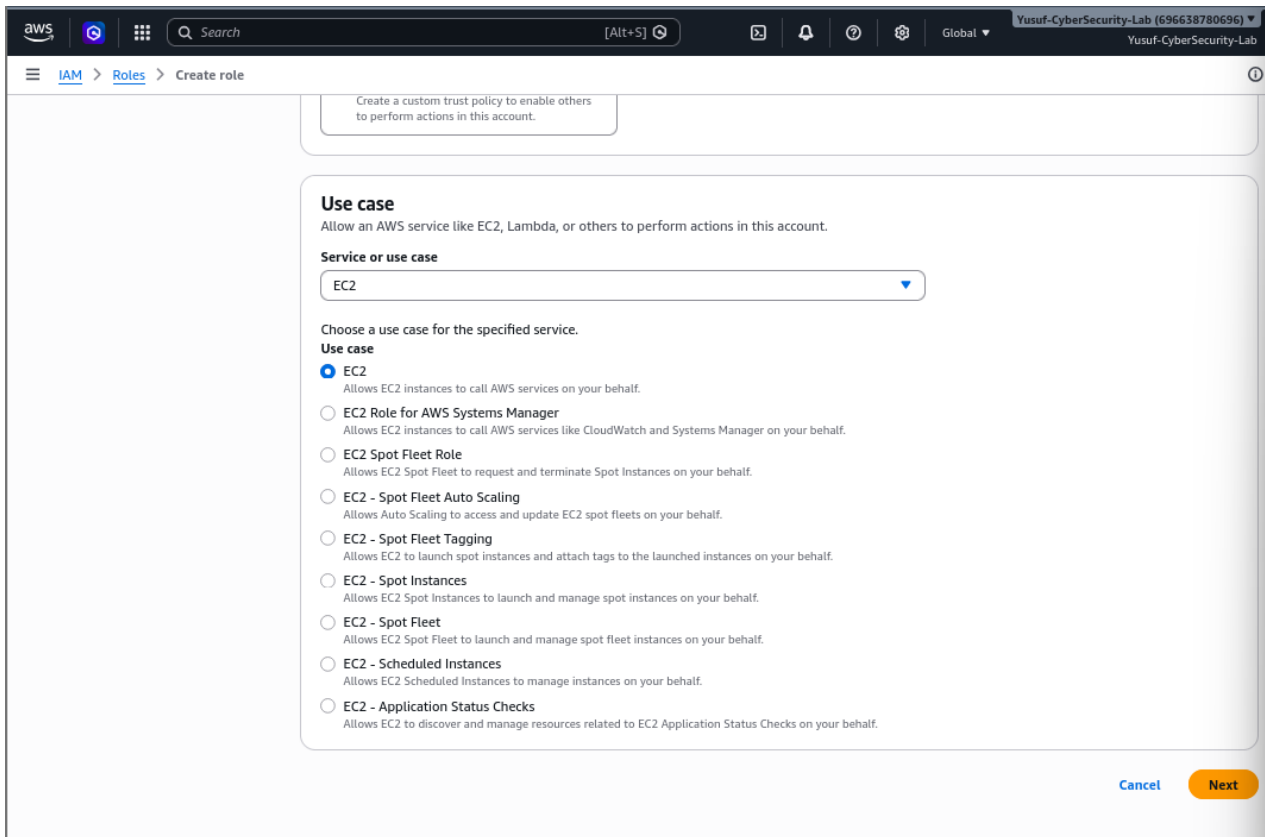
 Linux/UNIX

Termination protection

Disabled

AMI location

Archive screenshot: Screenshot 2026-09-15 233731.png



aws    Search [Alt+S]    Global Yusuf-CyberSecurity-Lab (696638780696) Yusuf-CyberSecurity-Lab

[IAM](#) > [Roles](#) > Create role

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

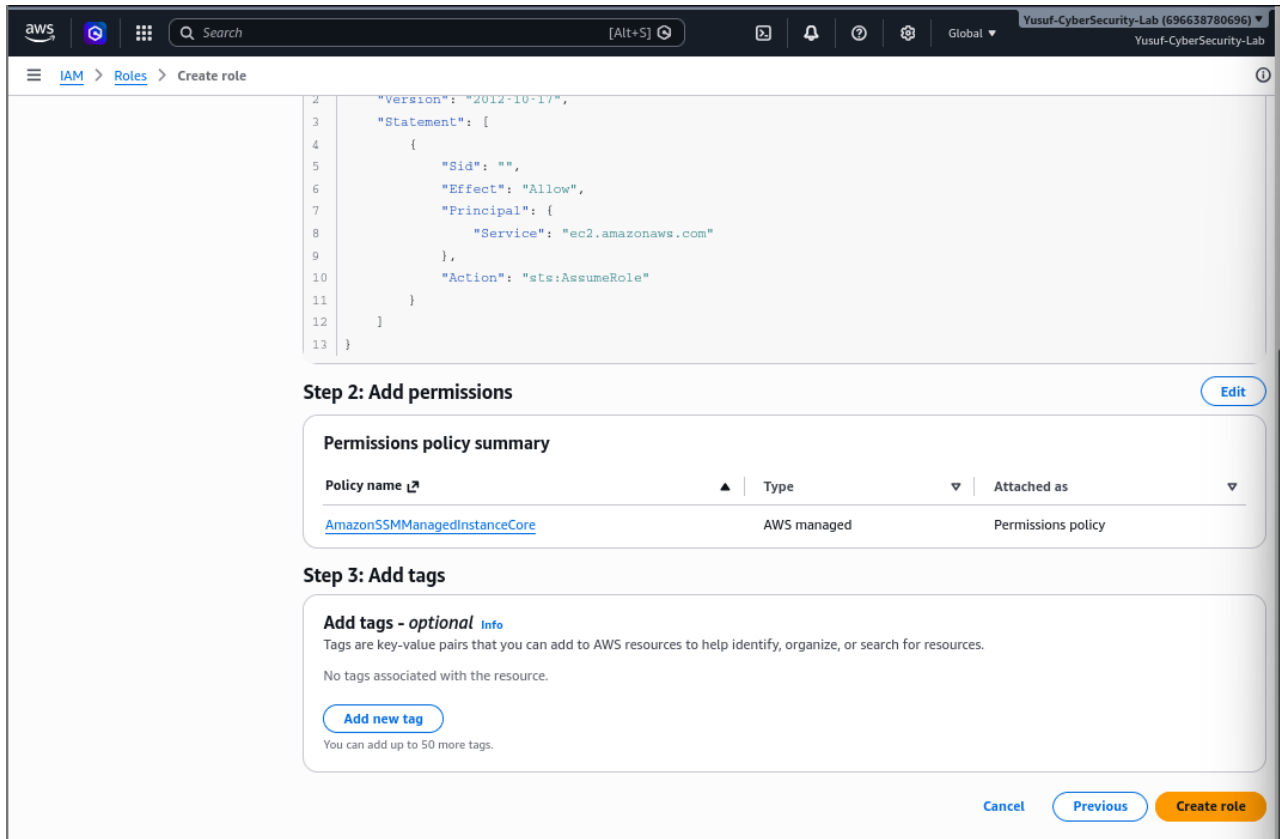
Choose a use case for the specified service.

Use case

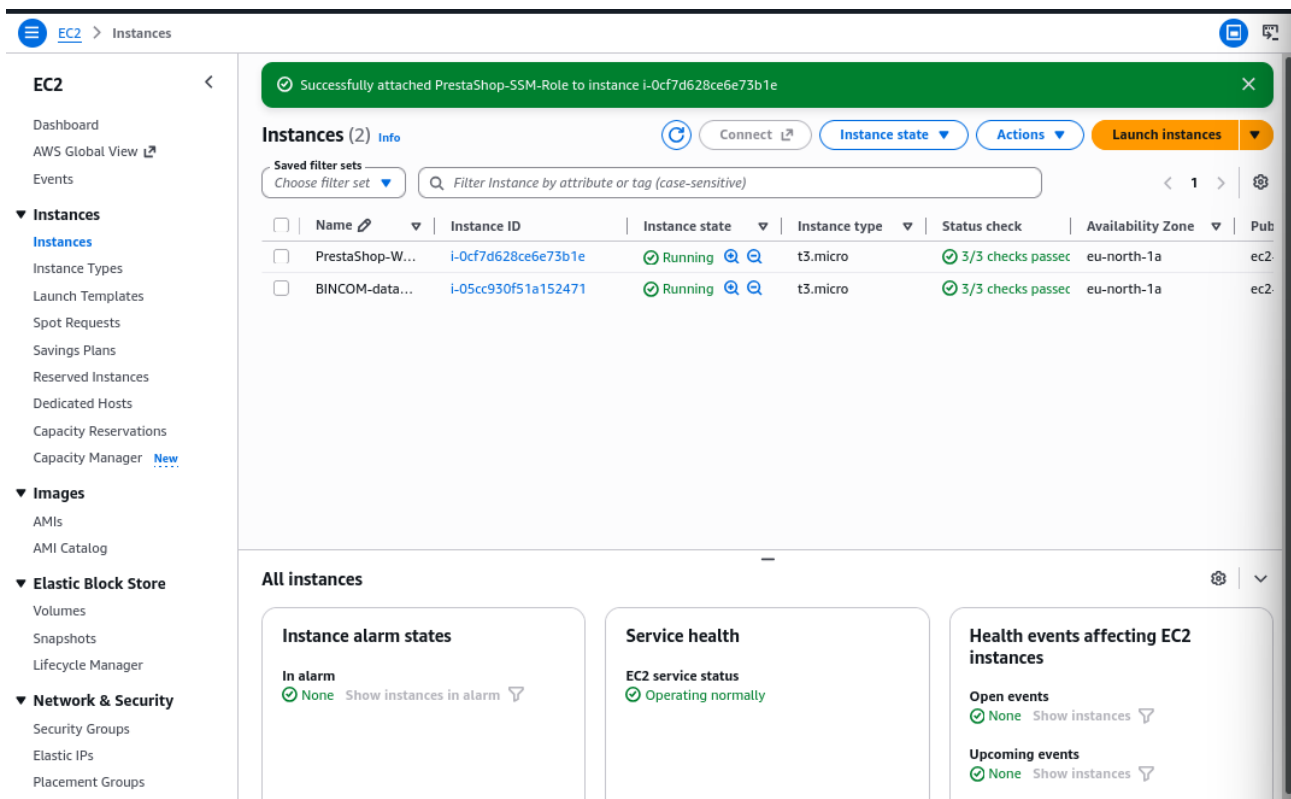
- ☒ EC2
Allows EC2 instances to call AWS services on your behalf.
- ☐ EC2 Role for AWS Systems Manager
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.
- ☐ EC2 Spot Fleet Role
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.
- ☐ EC2 - Spot Fleet Auto Scaling
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.
- ☐ EC2 - Spot Fleet Tagging
Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.
- ☐ EC2 - Spot Instances
Allows EC2 Spot Instances to launch and manage spot instances on your behalf.
- ☐ EC2 - Spot Fleet
Allows EC2 Spot Fleet to launch and manage spot fleet instances on your behalf.
- ☐ EC2 - Scheduled Instances
Allows EC2 Scheduled Instances to manage instances on your behalf.
- ☐ EC2 - Application Status Checks
Allows EC2 to discover and manage resources related to EC2 Application Status Checks on your behalf.

[Cancel](#) [Next](#)

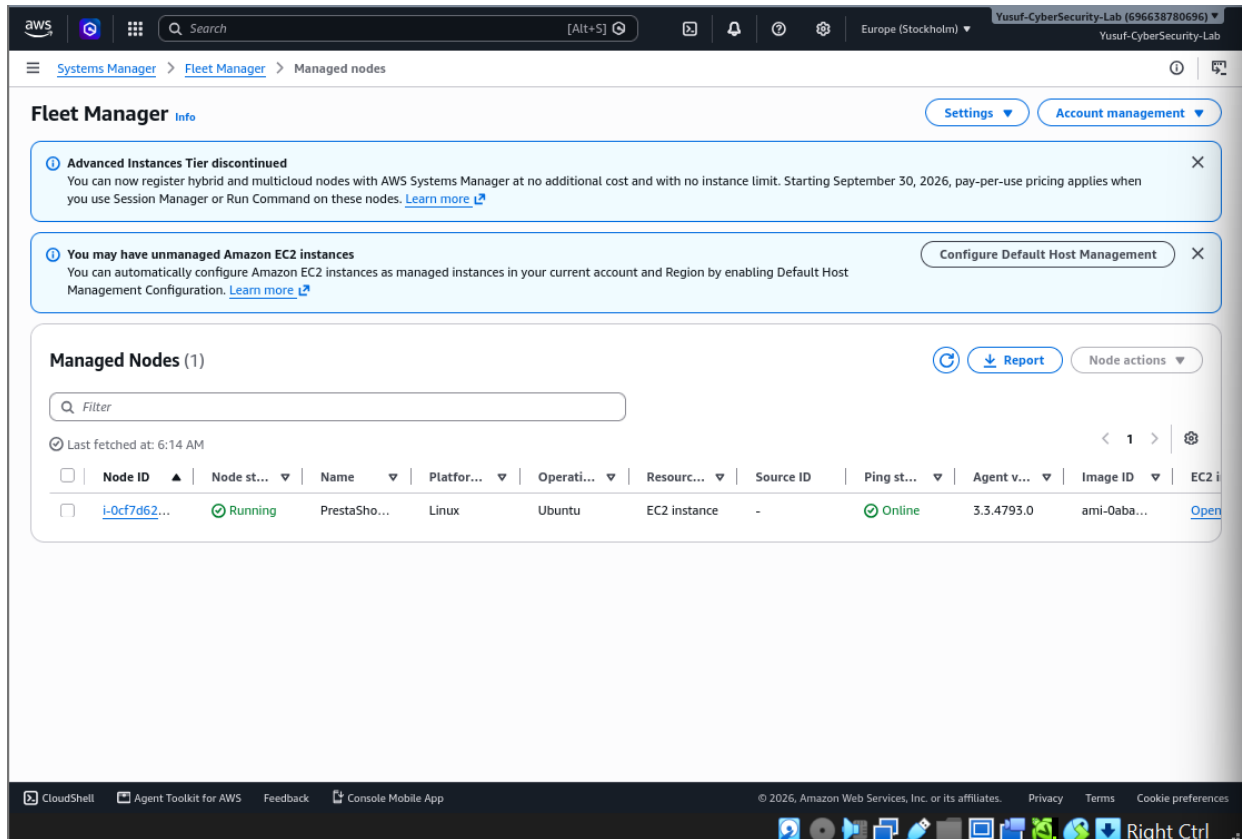
Archive screenshot: Screenshot 2026-09-15 234349.png



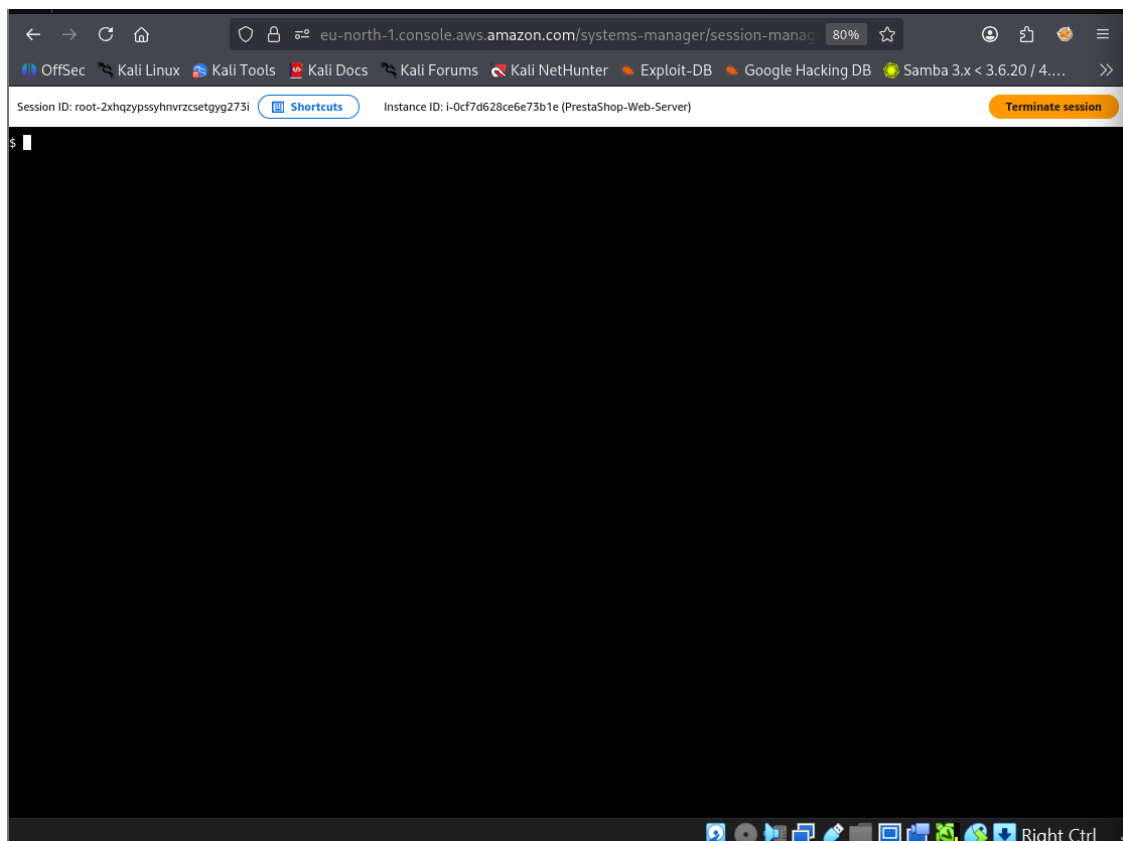
Archive screenshot: Screenshot 2026-09-15 234625.png



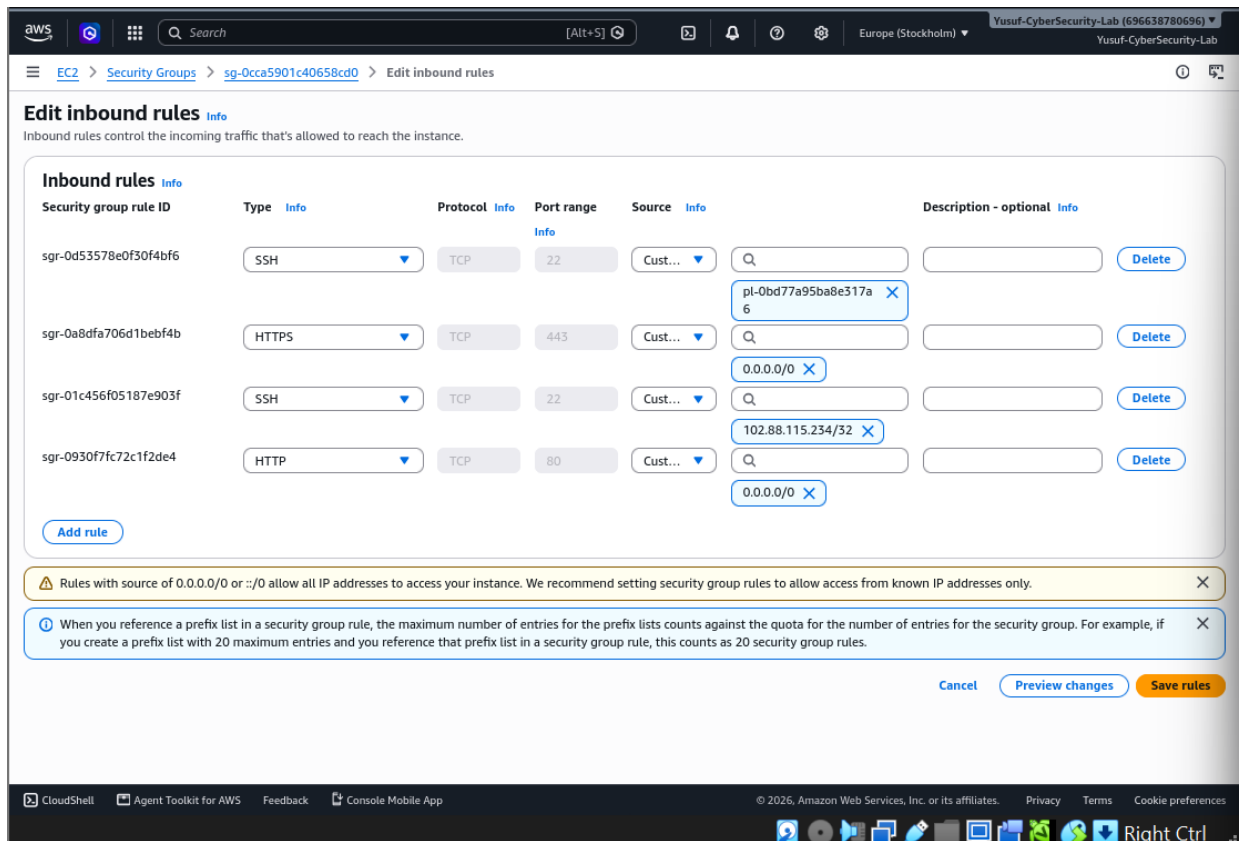
Archive screenshot: Screenshot 2026-09-15 235025.png



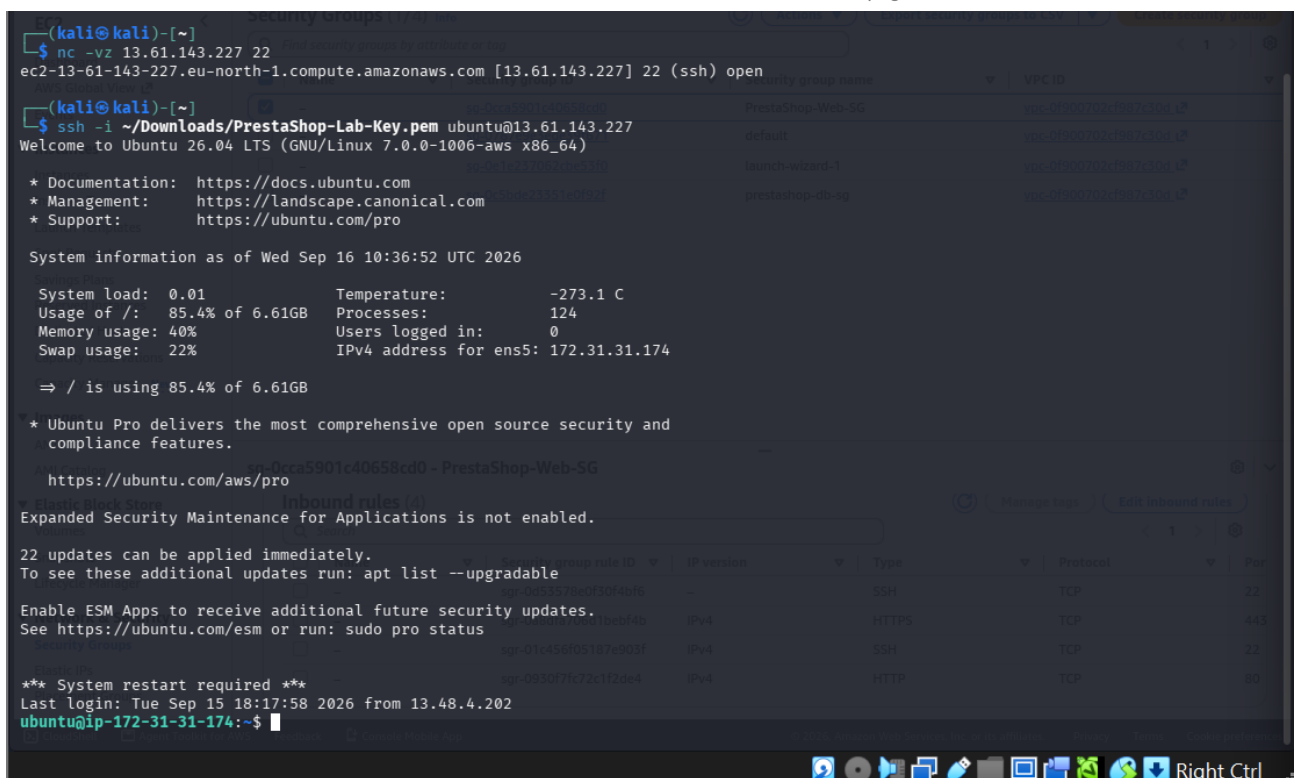
Archive screenshot: Screenshot 2026-09-16 111642.png



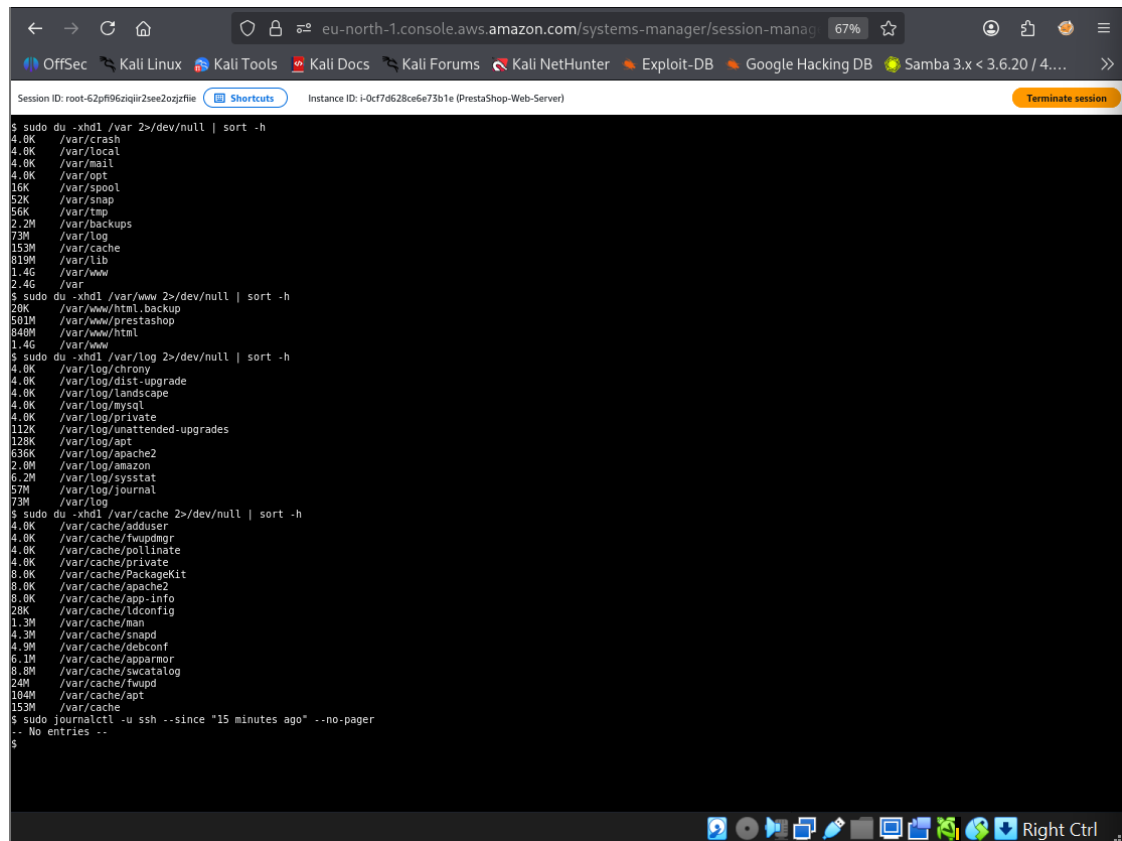
Archive screenshot: Screenshot 2026-09-16 111657.png



Archive screenshot: Screenshot 2026-09-16 113338.png



Archive screenshot: Screenshot 2026-09-16 113808.png



Archive screenshot: Screenshot 2026-09-16 120425.png



Archive screenshot: Screenshot 2026-09-16 123854.png



Archive screenshot: Screenshot 2026-09-16 124835.png

```
(kali@kali)-[~]
└─$ curl -I http://13.61.143.227/
HTTP/1.1 200 OK
Date: Wed, 16 Sep 2026 11:47:35 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Set-Cookie: PHPSESSID=b043bffd29ef76f33343cfa1e6f883c1; expires=Mon, 21 Jun 2083 23:35:10 GMT; Max-Age=1791287255; path=/; HttpOnly; SameSite=Lax
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Set-Cookie: PrestaShop-e0f449b036c2189050860c8fa7d8b50d=def50200e89ac93415859d36b4935df5c96ada2d2041eb8a78fa72f8230208c235832c65498ea21e
b9bd87bbe75114a52cddfc9c6d8a2c70367a7bffcfc8cf76239e30e1aeef60b8c1a916abb807efb5fde4647c4f82bebd8fb5e14fb03a32d8c4ac5b4a72e1ae9c87ae5
0f0363f2f3a0bb64b1d45784fbb946af25ca80f4ef20c5adb9b7042d7af7d8a2a9529e265a63b21709d63d5aa8cd2f8b94dcd16d0903afac4f29e1bb7f7ba85f293d80b8b
cbc59058b2a8ee720c55a7fa0c37a66179ad0cb18bbcc4dcae5e6c36cc3629a60fb66a; expires=Tue, 06 Oct 2026 11:47:35 GMT; Max-Age=1728000; path=/;
HttpOnly; SameSite=Lax
Set-Cookie: PrestaShop-e0f449b036c2189050860c8fa7d8b50d=def502009a28bfb5670d6c9001c0af6d8a99121453eb4493e10fb7c8f330d3dc24817b70dfe58727
7e12ac2eddfcfceae4c1cf587165e35422247d5c5ad9176666188a81dd39809d71ac0ea0f89f57af659d6eaa833a3df19b8a51b8e17a67b4dced9829046093f2649431921
73f666c55a21cf1cc5b240707974a7ae46cdac09375aac1afbbe251bca1216c88f45770bb18200ab90b716279064dc81e7f03d8b06868aa6178458d94c4b491ac83351f89
9427cb0cf350d9630a31b7feb99afbb601d3e55057cf3335861e67d40f7000ff8a2a1d826e1a8eafddbcd1e8487393ef8190ccce2d3ac53cafdbb9c94a0138c8002a4
f1dc77; expires=Tue, 06 Oct 2026 11:47:35 GMT; Max-Age=1728000; path=/; HttpOnly; SameSite=Lax
Content-Type: text/html; charset=utf-8

(kali@kali)-[~]
└─$ curl -s http://13.61.143.227/ | grep -Ei 'PrestaShop|Hummingbird|Sample|Lorem|Custom Text' | head

(kali@kali)-[~]
└─$
```

Archive screenshot: Screenshot 2026-09-16 124851.png

```
(kali@kali)-[~]
└─$ curl -I http://13.61.143.227/
HTTP/1.1 200 OK
Date: Wed, 16 Sep 2026 11:53:29 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Set-Cookie: PHPSESSID=25f306d844d91f0459c180b25f65fef1; expires=Mon, 21 Jun 2083 23:46:58 GMT; Max-Age=1791287609; path=/; HttpOnly; SameSite=Lax
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Set-Cookie: PrestaShop-e0f449b036c2189050860c8fa7d8b50d=def502003d9711178af1afcaad4645c933ea59ff7760c17c03d76abc59f5a655d081c0b9bf2432a1d
bd624ad68be6943bfff3fe08a32c0b6713b491cf2782b33aa27538ab24cfedaff77f3651eafb52c93fbeca6d40347a48fc219bc7ee22731b7eba5840fdd5ae5343b45dc053
ad3b07c8347cce202625dacdb4bf4bfa1f9b93c2b8fd8ec9036a4938607ca161e7a2aa768efcf9531d63c3a74bc50c0fa553184b8889ca4b261be3833c3c0a0c6d8fed9f
b3875028a8129910d455b0fe68b8d9c16662298a1e2abaab84c7402a354fce54965a916; expires=Tue, 06 Oct 2026 11:53:29 GMT; Max-Age=1728000; path=/;
HttpOnly; SameSite=Lax
Set-Cookie: PrestaShop-e0f449b036c2189050860c8fa7d8b50d=def50200d89d215139b251f19580193c7559335ff1374435c40b7d4605c98167563f6ade3d2b530f0f
1c1cab4589726f5dc1f1ca366352cb41a5fa478cc492469ff04559ae91827be446e085d74b1c7c5eb4d11723f9b9dac6befeab0aede63a0d879607fa1003bfb057bfd45a
c2927b85d5e9023cb2e3420533b27c676193b4b18fbf308d067ac0fe2611ce0c6d162d7f352f0bac3e9610930d83e3df2661d918447b803ea19e79a77b0fd3e12bea3d306
fa3c1542fe0680f36f1c4f2ab67e08d864e494d80bf57595ff765742c41d542086fa3be35237f16eed25b57a1f95239d1a6494c27e2e5375447aba90dbbd020295d7274f0
03d889; expires=Tue, 06 Oct 2026 11:53:29 GMT; Max-Age=1728000; path=/; HttpOnly; SameSite=Lax
Content-Type: text/html; charset=utf-8

(kali@kali)-[~]
└─$ sudo apache2ctl -t
[sudo] password for kali:
Syntax OK

(kali@kali)-[~]
└─$
```

Archive screenshot: Screenshot 2026-09-16 125405.png

```
*** System restart required ***
Last login: Wed Sep 16 10:36:53 2026 from 102.89.47.185
ubuntu@ip-172-31-31-174:~$ sudo ss -ltnp | grep -E ':(80|443)\b'
LISTEN 0      511          *      *      *:80      *:*      users: ((("apache2",pid=231212,fid=4), ("apache2",pid=223970,fid=4), ("apache2",pid=223969,fid=4), ("apache2",pid=197235,fid=4)))
ubuntu@ip-172-31-31-174:~$ sudo apache2ctl -s
VirtualHost configuration:
*:80               ip:172-31-31-174.eu-north-1.compute.internal (/etc/apache2/sites-enabled/000-default.conf:1)
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex watchdog-callback: using_defaults
Mutex rewrite-map: using_defaults
Mutex default: dir="/run/apache2/" mechanism=default
Mutex mpm-accept: using_defaults
PidFile: "/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
Define: DUMP_RUN_CFG
User: name="www-data" id=33
Group: name="www-data" id=33
ubuntu@ip-172-31-31-174:~$ curl -I http://13.61.143.227/
HTTP/1.1 200 OK
Date: Wed, 16 Sep 2026 12:07:48 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Set-Cookie: PHPSESSID=6dab4d6668865c18508f523a6a0dc; expires=Tue, 22 Jun 2083 00:15:36 GMT; Max-Age=1791288468; path=/; HttpOnly; SameSite=Lax
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Set-Cookie: PrestaShop-e0f449b036c2189050860c8fa7d8b50d-def502001da300e92be7e273acd422d7b45a4ff6d3e9a61a062dac19f9fc26861a3fa2b532e76044d33b46e466952599cfd397b5b5f3fe394fc0a3222f3f05497b011129c4dbb495258a8a6649d84d97f36a1d4ba10bca73880a04b77c9fbc70e6260b2c6e2826613b0eeb28802213220d101de4c4989b9535c6104f7d8a6dc3cbc9f572826551f30e3d143bf3c5551189ae7cd11e6182103e7b1f22e2d82f3b78a5770a80f7c1343c356a30fa78edc2c59329cb0ff61f6ca2e4734587bb257199bee7fa341b70af35bb045c78aeee90eeec8; expires=Tue, 06 Oct 2026 12:07:48 GMT; Max-Age=1728000; path=/; HttpOnly; SameSite=Lax
Set-Cookie: PrestaShop-e0f449b036c2189050860c8fa7d8b50d-def5020068e74eab9277c8ea80del711dcbae348e11f9e341310afac20eee9eeffefc95043210b6af66777cac261ba3566e9ed2c22204d06b1069e52aa2d7b2e04a079a7b33fd3435c6301a8fa87500aa0276a5c6feef5b10953d45c918729d6c15f544c6ee54221c513c64c5e84ca8f2a23a03272f211be4bd170fa57c258b1c9a9c09989cb9748ecdcc95aed952d26320fef2755541d50aalc4b389578cca212abd3945e84c3a8c4c6b7523837491c35ff1bc50708eac8de0289e4150b6b1d15e93b2c78671433b4ba3202896f04d8fc13c446004fccc0ff5b7b01b77a7bd9c84401fd01f3579be9142c3345313591ceb4d19ee5; expires=Tue, 06 Oct 2026 12:07:48 GMT; Max-Age=1728000; path=/; HttpOnly; SameSite=Lax
Content-Type: text/html; charset=utf-8
ubuntu@ip-172-31-31-174:~$
```

i-0c7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-16 130910.png

```
kali@kali: ~
Session Actions Edit View Help
kali@kali ~$ curl -I http://13.61.143.227/admin113pn8yfwfb8jhiird/
HTTP/1.1 404 Not Found
Date: Wed, 16 Sep 2026 12:11:37 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Set-Cookie: PrestaShop-69479823f3242ba2c04a3860126e6cd6-def502008f793d027525c731b340292e4c61284f451626a806b23438bb7c05fcddeb1bb2edb68010a08a15c696e0064e3200f5e97970188025272b6d020fa2b90c099e92cd00b62d7ad017e9d5cd789e4ba4e537034d80f9d01c0760c4f757d0b78ece4b50d8281999e28fa8f0c10a170456dad429fa10054ad92d3b623bb561cf1bfbbbf4ca6f1e6daf0e5a104f51ce7dcb6f3176646826442d00cbe3555c0b294cb79854620191504245d03276f13a5bfc4843af68b81cfccf3cf9b868eba6b429d7ca26bd1a5dd9573006c6a8cce1ec2211ae; expires=Tue, 06 Oct 2026 12:11:37 GMT; Max-Age=1727999; path=/; HttpOnly; SameSite=Lax
Cache-Control: no-cache, private
Content-Type: text/html; charset=UTF-8

kali@kali ~$ curl -I http://13.61.143.227/Login.php
HTTP/1.1 404 Not Found
Date: Wed, 16 Sep 2026 12:11:48 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Content-Type: text/html; charset=iso-8859-1

kali@kali ~$ curl -I http://13.61.143.227/admin113pn8yfwfb8jhiird/index.php
HTTP/1.1 404 Not Found
Date: Wed, 16 Sep 2026 12:12:00 GMT
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referer-Policy: strict-origin-when-cross-origin
Content-Security-Policy: frame-ancestors 'self';
Permissions-Policy: geolocation=(), microphone=(), camera=()
Set-Cookie: PrestaShop-69479823f3242ba2c04a3860126e6cd6-def50200f6f97099125a76e1f6abb75f6ed861b1ad7fbf347b7c7ef20cb053dd29f1dabdcde1c5e76c13e71bb108804c122d3dfc4c77fc95d1defa24c033234a761605e02af87d68a6452e0138007214e832ebc92739b9cf062bc57e8ed087bb557631f8442c32b3e70c9b0dcc88ba14ecff4936fed83007836f0caddd87f2c3dfa881c8fac84130cc70aa4b82585dd77a4282c5e5c73d397d036155d22e5c25df08265e5adc87aa19d8527f8d1ce0f86fd1bb133dfcf82b517a0e7bb3d3f6ea8d710502d478948ee74af93d1a4445fbf986344fbcabd8929; expires=Tue, 06 Oct 2026 12:12:00 GMT; Max-Age=1728000; path=/; HttpOnly; SameSite=Lax
Cache-Control: no-cache, private
Content-Type: text/html; charset=UTF-8
```

Archive screenshot: Screenshot 2026-09-16 131238.png

```

ubuntu@ip-172-31-31-174:~$ sudo apache2ctl -M | grep -i security
ubuntu@ip-172-31-31-174:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-11 06:36:07 UTC; 5 days ago
     Invocation: dc51dd67530f409799fb7e03af40ae93
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 197235 (apache2)
      Status: "Total requests: 2449, Idle/Busy workers 100/0,Requests/sec: 0.0051, Bytes served/sec: 22 B/sec"
        Tasks: 4 (limit: 627)
       Memory: 93.6M (peak: 584.2M, swap: 51.3M, swap peak: 90.7M)
          CPU: 2min 17.677s
      CGroup: /system.slice/apache2.service
              └─197235 /usr/sbin/apache2 -k start -DFOREGROUND
                 └─223969 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─223970 /usr/sbin/apache2 -k start -DFOREGROUND
                       └─231212 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 13 00:29:54 ip-172-31-31-174 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 13 00:29:54 ip-172-31-31-174 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 14 00:36:01 ip-172-31-31-174 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 14 00:36:01 ip-172-31-31-174 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 00:43:48 ip-172-31-31-174 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 00:43:48 ip-172-31-31-174 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 18:47:19 ip-172-31-31-174 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 18:47:19 ip-172-31-31-174 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 16 00:36:42 ip-172-31-31-174 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 16 00:36:42 ip-172-31-31-174 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-31-174:~$ sudo ls -lah /etc/modsecurity
ls: cannot access '/etc/modsecurity': No such file or directory
ubuntu@ip-172-31-31-174:~$ sudo ls -lah /etc/apache2/mods-enabled/ | grep -i security
ubuntu@ip-172-31-31-174:~$ sudo grep -Rni "SecRuleEngine" /etc/modsecurity /etc/apache2 2>/dev/null | head -20
ubuntu@ip-172-31-31-174:~$

```



Archive screenshot: Screenshot 2026-09-16 132403.png

```

ubuntu@ip-172-31-31-174:~$ sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [613 kB]
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [147 kB]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [293 kB]
Get:8 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [93.3 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [477 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [112 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [173 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [54.5 kB]
Fetched 2237 kB in 1s (4413 kB/s)
27 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-31-174:~$ apt-cache policy libapache2-mod-security2
libapache2-mod-security2:
  Installed: (none)
  Candidate: 2.9.12-2build1
  Version table:
     2.9.12-2build1 500
        500 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages
ubuntu@ip-172-31-31-174:~$

```



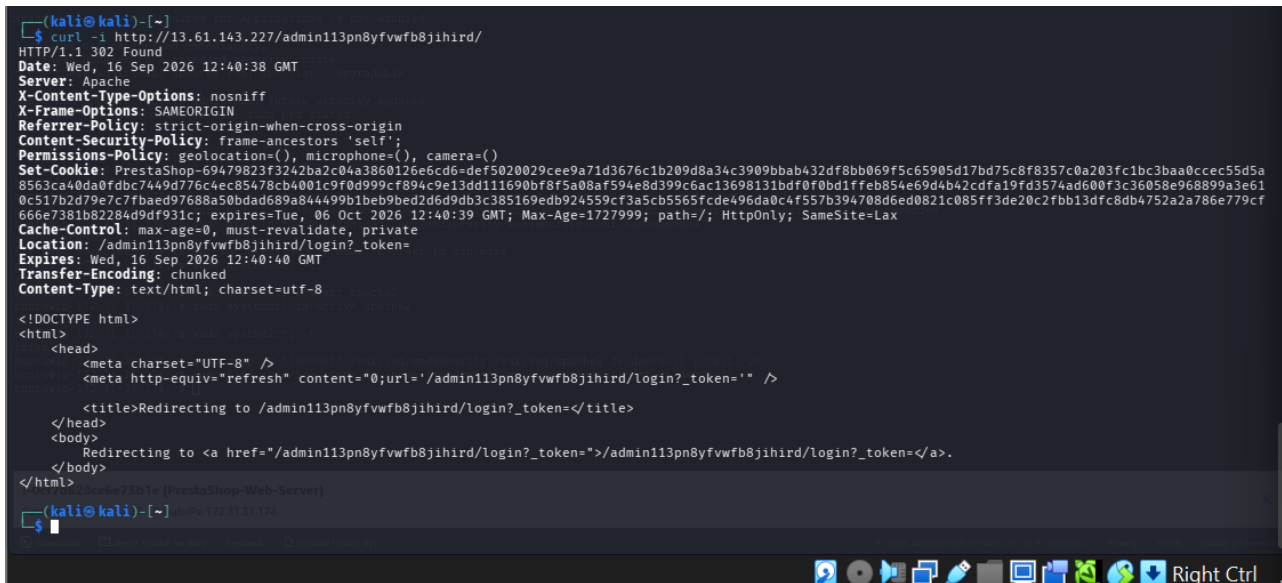
Archive screenshot: Screenshot 2026-09-16 132703.png

```

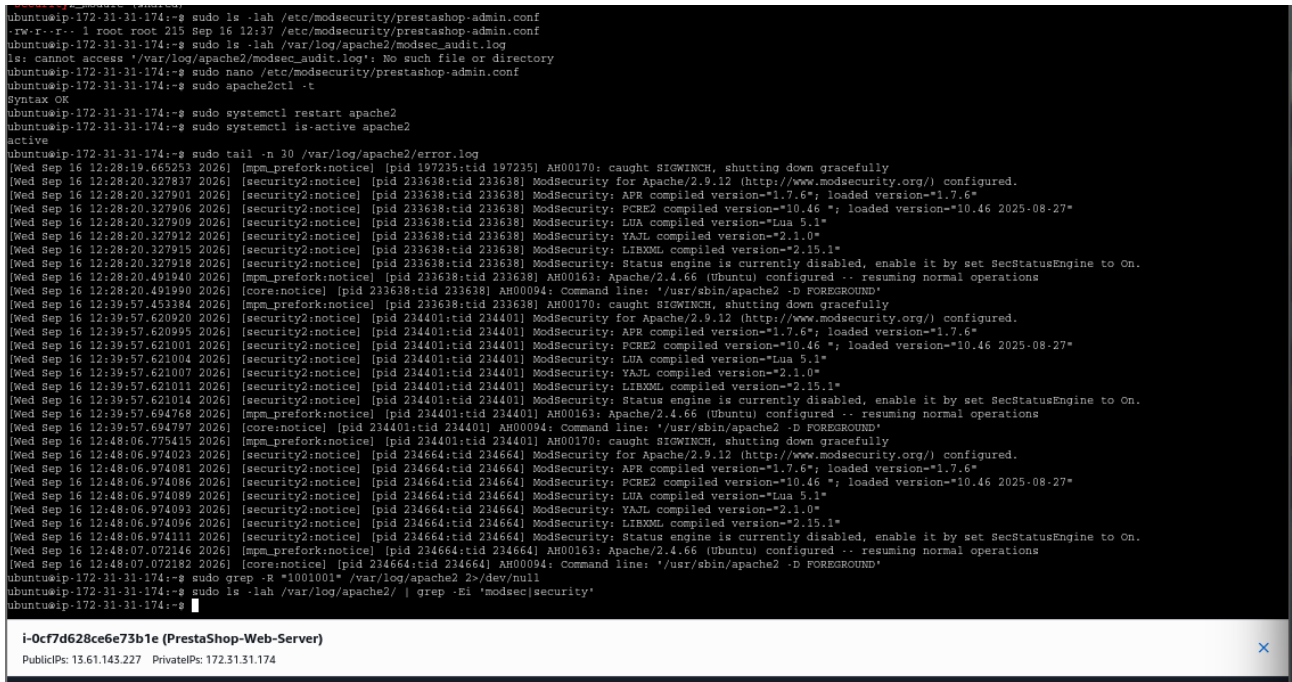
ubuntu@ip-172-31-31-174:~$ sudo apache2ctl -t
Syntax OK
ubuntu@ip-172-31-31-174:~$ sudo systemctl restart apache2
ubuntu@ip-172-31-31-174:~$ sudo systemctl is-active apache2
active
ubuntu@ip-172-31-31-174:~$ sudo apache2ctl -t
Syntax OK
ubuntu@ip-172-31-31-174:~$ sudo grep -R "1001001" /var/log/modsecurity /var/log/apache2 2>/dev/null | tail -20
ubuntu@ip-172-31-31-174:~$ sudo find /var/log -type f -iname **modsec** -o -iname **security** 2>/dev/null
ubuntu@ip-172-31-31-174:~$

```

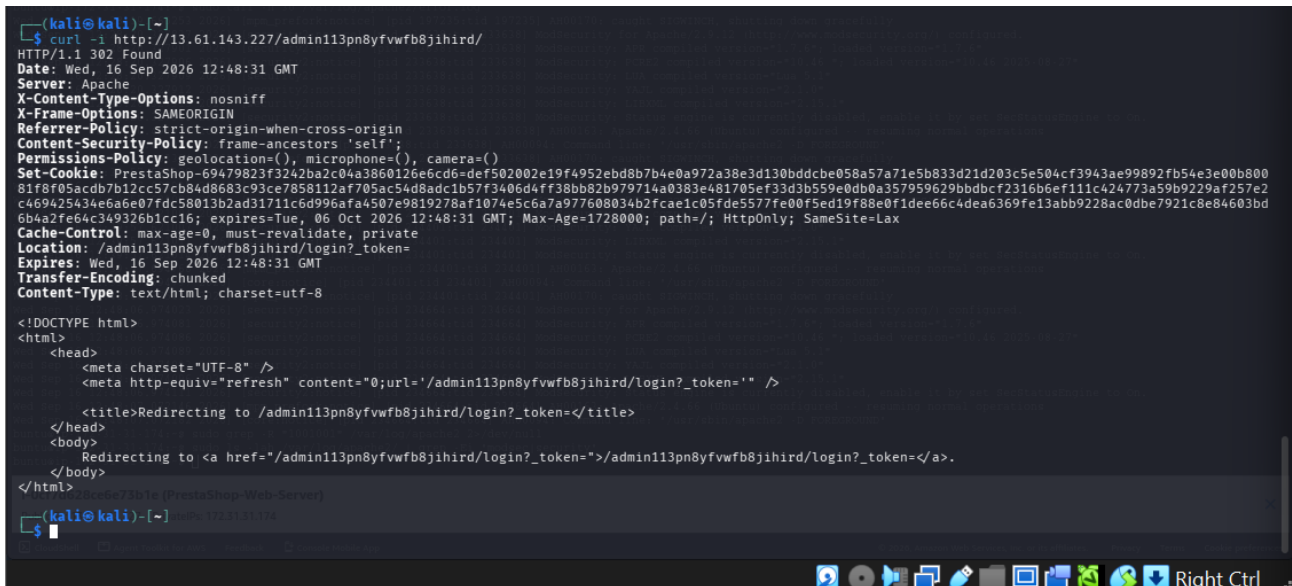
Archive screenshot: Screenshot 2026-09-16 134201.png



Archive screenshot: Screenshot 2026-09-16 134214.png



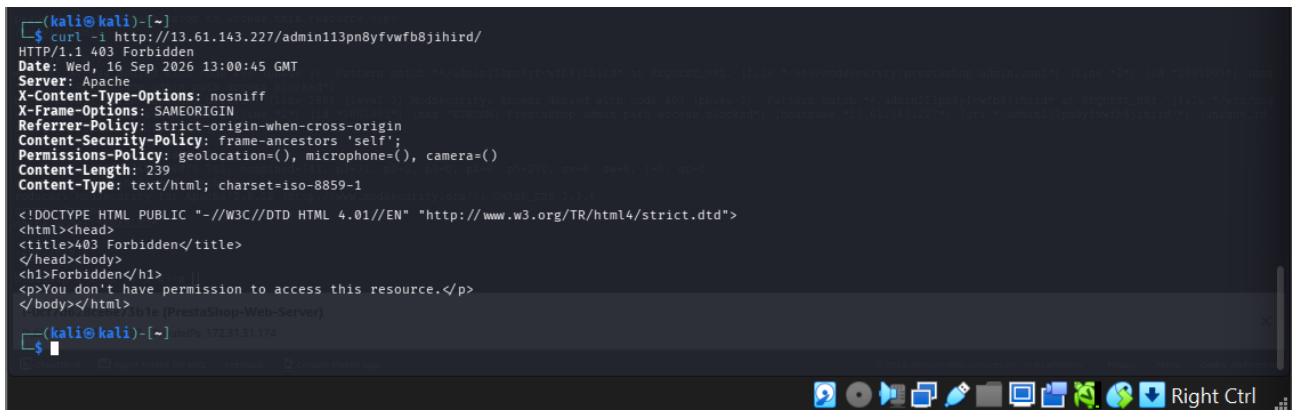
Archive screenshot: Screenshot 2026-09-16 134942.png



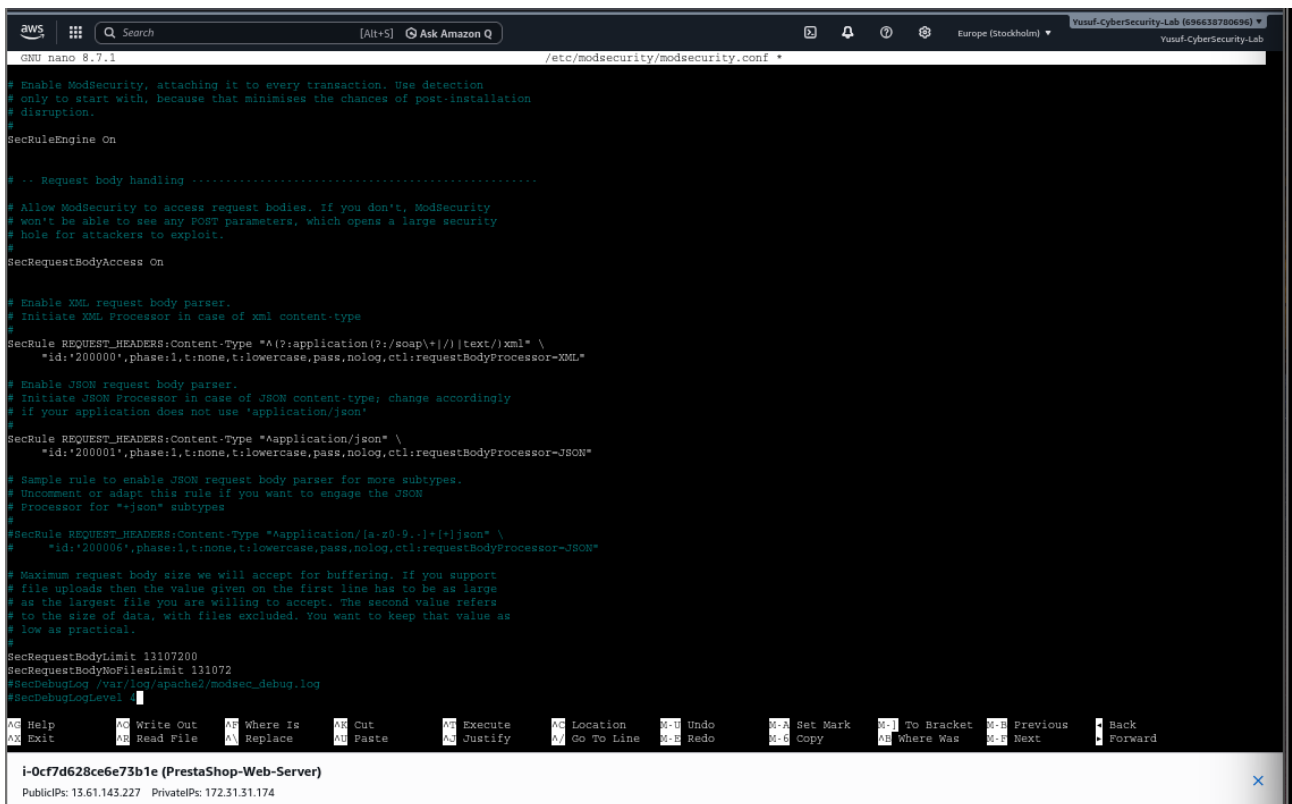
Archive screenshot: Screenshot 2026-09-16 134954.png



Archive screenshot: Screenshot 2026-09-16 140132.png



Archive screenshot: Screenshot 2026-09-16 140147.png



Archive screenshot: Screenshot 2026-09-16 141000.png

```
ubuntu@ubuntu-172-31-174:~$ sudo apt-get install -y syntaxd
Syntax OK
ubuntu@ubuntu-172-31-174:~$ sudo systemctl reload aptdaemon
ubuntu@ubuntu-172-31-174:~$ sudo systemctl is-active aptdaemon
active
ubuntu@ubuntu-172-31-174:~$ ls -lh -/prestashop_before_demo_cleanup.sql
-rw-rw-r-- 1 ubuntu:ubuntu 978K Sep 15 10:19 /home/ubuntu/prestashop_before_demo_cleanup.sql
ubuntu@ubuntu-172-31-174:~$ file -/prestashop_before_demo_cleanup.sql
-/prestashop_before_demo_cleanup.sql: ASCII text
ubuntu@ubuntu-172-31-174:~$ grep -c "CHANGESET TABLE" /prestashop_before_demo_cleanup.sql
361
ubuntu@ubuntu-172-31-174:~$ grep -c "INSERT INTO" /prestashop_before_demo_cleanup.sql
112
ubuntu@ubuntu-172-31-174:~$ sudo systemctl is-active mariadb
active
ubuntu@ubuntu-172-31-174:~$ sudo mysql -e "SHOW DATABASES;"
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| prestashop |
| sys |
| weapp |
+-----+
ubuntu@ubuntu-172-31-174:~$ head -n 15 /prestashop_before_demo_cleanup.sql
DROP SCHEMA IF EXISTS `prestashop` /*!40101 IF NOT EXISTS */;
MariaDB root@10.10.11.8.6-MariaDB: [debian-linux-gnu (x86_64)]
*****
Host: localhost   Database: prestashop
*****
Server version      11.8.6-MariaDB-Subuntu0.1 from Ubuntu

/*+01001 SET WOLD_CHARACTER_SET_CLIENT=WORLD_CHARACTER_SET_CONNECTION */;
/*+01001 SET WOLD_CHARACTER_SET_SERVER=latin1 CHARACTER SET utf8mb4 */;
/*+01001 SET WOLD_COLLATION_CONNECTION=utf8mb4 COLLATION utf8mb4_bin */;
/*+01001 SET NAMES utf8mb4 */;
/*+01001 SET TIME_ZONE='+00:00' */;
/*+01001 SET @@SESSION_CHARACTER_SET=latin1 COLLATE latin1_general_ci */;
/*+01001 SET @@SESSION_COLLATION=utf8mb4 COLLATE utf8mb4_bin CHECKED=0 */;
ubuntu@ubuntu-172-31-174:~$ tail -n 15 /prestashop_before_demo_cleanup.sql
/*+01000 ALTER TABLE `ps_shop` ENABLE KEYS */;
CHECK TABLES;
COMMIT;
/*+01000 ALTER DATABASE AUTO_INCREMENT */;
/*+01001 SET TIME_ZONE=WORLD_TIME_ZONE */;

/*+01001 SET @@WOLD_WOLD_WOLD_WOLD_WOLD */;
/*+01001 SET FOREIGN_KEY_CHECKS=WORLD_FOREIGN_KEY_CHECKS */;
/*+01001 SET UNIQUE_CHECKS=WOLD_UNIQUE_CHECKS */;
/*+01001 SET CHARACTER_SET_CLIENT=WOLD_CHARACTER_SET_CLIENT */;
/*+01001 SET CHARACTER_SET_RESULTS=WOLD_CHARACTER_SET_RESULTS */;
/*+01001 SET COLLATION_CONNECTION=WOLD_COLLATION_CONNECTION */;
UNLOCK TABLES;
/*+01001 SET HOST_VEROSITY=WOLD_HOST_VEROSITY */;

Date completed on 2026-09-15 19:30:28
ubuntu@ubuntu-172-31-174:~$
```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-16 142027.png

```

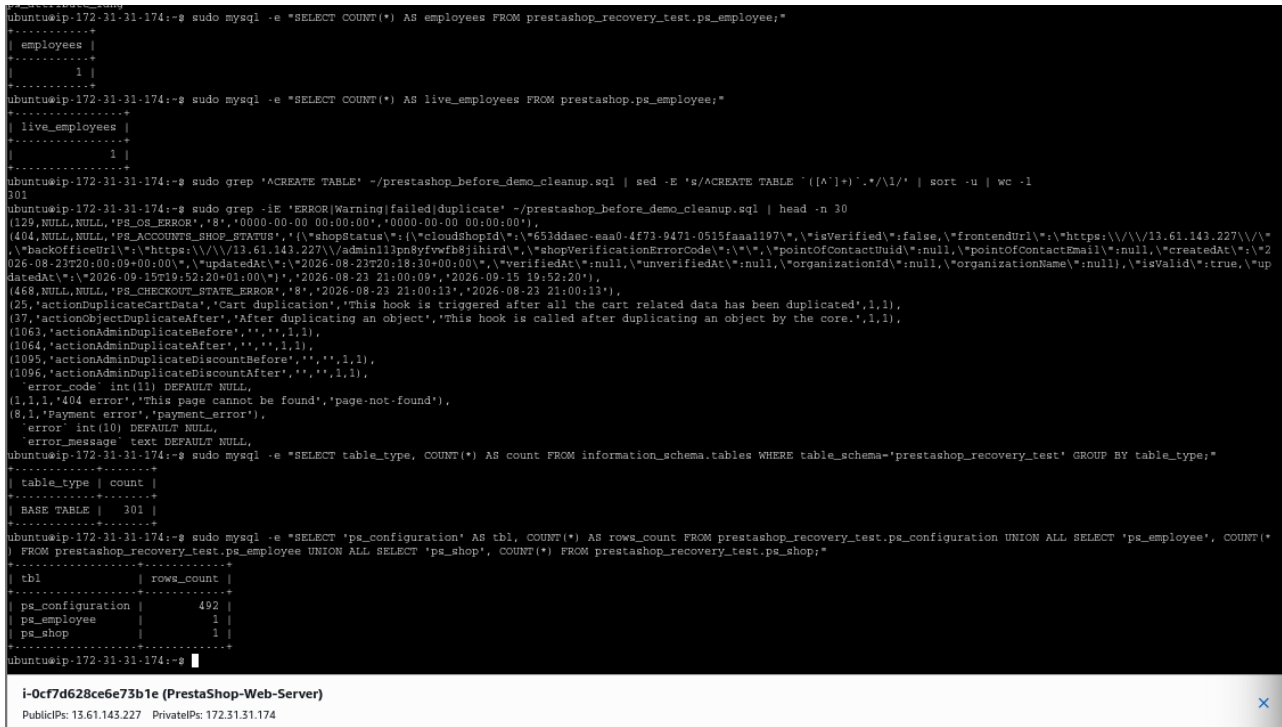
ubuntu@ip-172-31-31-174:~$ sudo mysql -e "CREATE DATABASE prestashop_recovery_test;"
ubuntu@ip-172-31-31-174:~$ sudo mysql -e "SHOW DATABASES LIKE 'prestashop_recovery_test';"
+-----+
| Database (prestashop_recovery_test) |
+-----+
| prestashop_recovery_test           |
+-----+
ubuntu@ip-172-31-31-174:~$ sudo mysql prestashop_recovery_test < ~/prestashop_before_demo_cleanup.sql
ubuntu@ip-172-31-31-174:~$ sudo mysql -e "SELECT COUNT(*) AS recovered_tables FROM information_schema.tables WHERE table_schema='prestashop_recovery_test';"
+-----+
| recovered_tables |
+-----+
|                201 |
+-----+
ubuntu@ip-172-31-31-174:~$ sudo mysql -e "SHOW TABLES FROM prestashop_recovery_test;" | head -n 15
tables_in_prestashop_recovery_test
ps_access
ps_accessory
ps_address
ps_address_format
ps_admin_filter
ps_alias
ps_appliance
ps_attachment
ps_attachment_lang
ps_attribute
ps_attribute_group
ps_attribute_group_lang
ps_attribute_group_shop
ps_attribute_lang
ubuntu@ip-172-31-31-174:~$ sudo mysql -e "SELECT COUNT(*) AS employees FROM prestashop_recovery_test.ps_employee;"
+-----+
| employees |
+-----+
|          1 |
+-----+
ubuntu@ip-172-31-31-174:~$ sudo mysql -e "SELECT COUNT(*) AS live_employees FROM prestashop.ps_employee;"
+-----+
| live_employees |
+-----+
|                1 |
+-----+
ubuntu@ip-172-31-31-174:~$

```

i-0cf7d628ce6e73b1e (PrestaShop-Web-Server)

PublicIPs: 13.61.143.227 PrivateIPs: 172.31.31.174

Archive screenshot: Screenshot 2026-09-16 145211.png



Archive screenshot: Screenshot 2026-09-16 145524.png



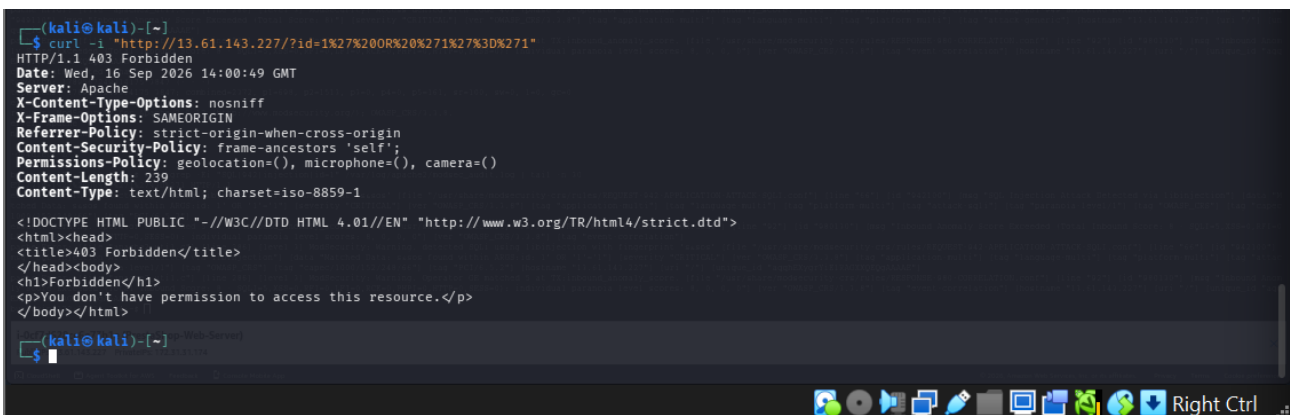
Archive screenshot: Screenshot 2026-09-16 145924.png



Archive screenshot: Screenshot 2026-09-16 150232.png



Archive screenshot: Screenshot 2026-09-16 150255.png



Archive screenshot: Screenshot 2026-09-16 150309.png

18. WAF Configuration Appendix

```
# BINCOM Assessment 3 - PrestaShop admin path protection
SecRule REQUEST_URI "@rx ^/admin113pn8yfwfwb8jhihird" "id:1001001,phase:1,deny,status:403,log,msg:'BINCOM: PrestaShop admin path access blocked'"
```

```
SecRuleEngine On
SecAuditEngine RelevantOnly
SecAuditLogRelevantStatus "^(?:5|4(?:04))"
SecAuditLog /var/log/apache2/modsec_audit.log
IncludeOptional /etc/modsecurity/*.conf
IncludeOptional /usr/share/modsecurity-crs/*.load
```

19. Source Materials

- PROJECT-GUIDELINES.txt — Assessment 3 brief and deliverables.
- BINCOM_PrestaShop_AWS_Deployment_Report.pdf — preliminary AWS deployment and hardening report.
- Project screenshots and terminal evidence collected during DVWA, ModSecurity, AWS and PrestaShop work.
- PrestaShop official documentation consulted during API/TLS review.